Distributional Learning of Multiple Context-Free Languages under Fixed Finite-Semigroup Typing: Local Groups and the Length of Characteristic Data

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Abstract

We study positive-data learning of multiple context-free languages under a fixed finite-semigroup typing: a morphism s from Σ^+ onto a finite semigroup S is fixed in advance, and nonempty factors and tuple components are compared only when their s-values agree. This continues the context-free case of a companion paper. Identification in the limit follows Yoshinaka's distributional framework; our subject is the total length of characteristic samples, measured against the size $\|G\|$ and the rule thickness $\bar{\theta}_G$ of a presentation G of the target. We show that local groups give polynomial characteristic data at every fan-out and are the exact pseudovariety boundary at fan-out one. If S is a local group, then for every fixed fan-out and rank bound, each target has characteristic data of length $O(\|G\|^2\bar{\theta}_G)$ for each of its reduced good presentations G of bounded rank, so these classes are identifiable in polynomial time and thick data. The factor $\|G\|\bar{\theta}_G$ is unavoidable: at fan-out f, over an alphabet of at least $4f+1$ letters (two at fan-out one) and under every typing, some finite targets with near-minimal presentations need data of length $\Omega(\|G\|\bar{\theta}_G)$, which matches Yoshinaka's upper bound for the trivial typing. At fan-out one and rank at least two, local groups are exactly the pseudovarieties of finite semigroups with uniformly polynomial characteristic data, and hence exactly those whose classes are identifiable in polynomial time and thick data; outside them, targets with presentations of size and rule thickness $O(n)$ need data of length exponential in n, for every set-driven reconstruction operator that has characteristic samples throughout the class. This witness family does not carry over to fan-out two: under any fixed typing, all but finitely many of its members fail to be tuple-substitutable there, and no target there keeps their long word as the only one of its type. Whether local groups remain the boundary at fan-out two is left open. The upper bound rests on a simultaneous shortening of interleaved tuple components that preserves ordered products in finite groups.

1 Introduction

Positive-data grammatical inference asks when a language can be identified in the limit from positive examples alone [1, 2]. Distributional learning treats words, or tuples of words, that occur in common contexts as candidate constituents. It began with substitutable CFGs [3, 12] and was extended to multiple context-free grammars (MCFGs), whose tuple-valued nonterminals express discontinuous and cross-serial dependencies [13, 14, 28, 29]. Yoshinaka's finite-signature framework [15] gives positive-data identification with characteristic samples of polynomially many words. Polynomially many words may still be exponentially long, so we measure a finite sample K by its total encoded length $\|K\|_+ = \sum_{w\in K}(|w|+1)$, following the thick-data perspective of [10, Section 2.5.2]. Qualitative identification is inherited from Yoshinaka's framework; our subject is the length of characteristic

samples relative to an MCFG presentation of the target, and the question is: *for which comparison biases is this length polynomial, and does the answer depend on the fan-out?*

The setting. Before any data are seen, we fix a fan-out bound f and a surjective finite-semigroup morphism $s : \Sigma^+ \twoheadrightarrow \mathsf{S}$, a *typing*. The reconstruction compares observed nonempty tuple components only when their s-values agree, and the target class $\mathcal{C}^{\mathrm{mcf}}_{f,s}$ consists of the f-MCFLs that are tuple-substitutable under this restriction (Definition 2.8). Already at fan-out two a group typing makes this class strictly larger than Yoshinaka's untyped class, with non-context-free witnesses on which the untyped reconstruction over-generalizes (Proposition 3.18). The typing is a class-level bias in the sense of [5], not chosen per target; other biases restrict derivations by control sets [6, 7] or restrict the evidence used to compare factors [9, 11]. Typical typings are prefix–suffix windows, values in a finite group, their products, and DFA transition morphisms. Group values compose in the order of the blocks, so for noncommutative groups such as S_3 the order of interleaved components matters. Local groups, finite semigroups T with $e\mathsf{T}e$ a group for every idempotent e, add group contributions of sliding factors: over $\{\mathtt{a}, \mathtt{b}\}$, recording the parity of the number of occurrences of $\mathtt{ab}$ is a local-group typing, but no prefix–suffix window combined with a group morphism determines it: for each window size and group morphism there are m, n such that $\mathtt{b}^m\mathtt{a}^n\mathtt{b}^m$ and $\mathtt{b}^{2m}$ agree on both but differ in that parity (Proposition B.6). Windows, groups and their products are local groups as well, but a DFA transition morphism need not be: if one letter acts as the identity and another as a constant map on at least two states, the two transformations form a copy of U_1, the obstruction of Theorem C below. For a reduced good presentation G we use its size $\|G\|$ and its rule thickness $\bar{\theta}_G$, the largest length of a shortest sentence using a given rule. Lower bounds are stated for every set-driven operator with characteristic samples throughout the class; upper bounds for Yoshinaka's reconstruction operator with typed links (Section 3).

Main results. Fix the alphabet, the fan-out f, a rule-rank bound $r_{\max} \geq 2$ and the typing s.

Theorem A (Local groups)**.** *If the positive image of s is a local group, every $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$ has characteristic data of total length $O(\|G\|^2\bar{\theta}_G)$ for every reduced good presentation G (Theorem 5.8).*

Theorem B (The linear factor)**.** (i) *If Σ has at least $4f + 1$ letters, or $f = 1$ and Σ has at least two letters, then for every typing s there are finite targets in $\mathcal{C}^{\mathrm{mcf}}_{f,s}$, with presentations within a constant factor of the smallest reduced good ones, whose characteristic samples all have length $\Omega(\|G\|\bar{\theta}_G)$ (Proposition 6.1).*

(ii) *For the trivial typing, Yoshinaka's bound $O(\|G\|\bar{\theta}_G)$ on characteristic data [14] is attained up to a constant factor: under the alphabet condition of* (i)*, by the targets of* (i) *(Corollary 6.2).*

Theorem C (The boundary at fan-out one)**.** *Let $f = 1$ and $r_{\max} \geq 2$.*

(i) *A pseudovariety of finite semigroups admits uniform polynomial characteristic-data length iff it is contained in* **LG** *(Theorem 7.3).*

(ii) *Outside* **LG** *the failure is exponential: for a typing s_U onto the two-element semilattice U_1 there are targets in $\mathcal{C}^{\mathrm{mcf}}_{1,s_U}$ with reduced good presentations of size and rule thickness $O(n)$ whose characteristic samples all have length at least $3 \cdot 2^n - 1$, for every set-driven operator with characteristic samples throughout the class (Theorem 7.3).*

In the terminology of identification in polynomial time and thick data [10,12], Theorems A and C say that for local-group typings the classes $\mathcal{C}^{\mathrm{mcf}}_{f,s} \cap \mathsf{MCFL}(f, r_{\max})$ are so identifiable at every fan-out,

and that at fan-out one **LG** is exactly the pseudovariety boundary of this property (Corollary 7.6). An MCFG of fan-out one is a context-free grammar, so Theorem C is a statement about context-free languages; it turns the sufficient condition of the companion manuscript [23] into a characterization.

The remaining question. Theorem A gives the *if* direction of Theorem C(i) at every fan-out. Whether **LG** is still the boundary at fan-out $f \geq 2$ is open (Open Problem 9.1). The witness family of Theorem C(ii) does not decide it: by a two-hole transfer of powers, under every fixed typing all but finitely many of its members, and their images under letter-disjoint morphisms, leave the class at fan-out two, and no member of the class keeps their long word as the only word of its type (Theorem 8.2). A typing that needs superpolynomial data at fan-out one but polynomial data at fan-out two would show that raising the fan-out can make learning strictly cheaper.

Approach. Refining the grammar by types and applying untyped estimates to the refinement can be expensive: for a fixed two-element typing there are presentations G_n of $\{\mathtt{a}, \mathtt{c}\}^+$ with $\|G_n\| = O(n)$ and rule thickness at most 2 whose full yield-type refinement has thickness at least 2^n (Proposition 4.9). We therefore bound characteristic data in terms of the original presentation, and we use distribution-equivalent representatives rather than generated witnesses (Corollary 4.10). To obtain short tuples of prescribed type, we replace long subderivations by short ones that need not preserve the group values of the replaced subtrees; every regime is an instance of one shortening principle (Lemma 5.1(ii)), and only the word-level replacement step changes. At fan-out one each replaced subtree occupies one contiguous block, and a pigeonhole on prefix products of the group values selects a replaceable interval. In an MCFG derivation of fan-out $f \geq 2$ a subtree contributes up to f blocks to different root components, interleaved with blocks of other subtrees, and for noncommutative groups the effect of simultaneous replacements depends on this order. The key step, with no fan-out-one counterpart, is an *owner-wise simultaneous selection* of replacements that preserves the ordered group product of every root component at once (Proposition 5.2). For local-group typings the group value is a product over sliding factors, which straddle block boundaries; selecting owners whose blocks lie far apart and enlarging each block by unchanged boundary letters reduces this case to the same selection (Lemma B.7). The lower bounds use marker families all of whose subsets stay in the class (Proposition 6.1) and rare types (Proposition 4.7). A second ingredient without a fan-out-one counterpart, a two-hole transfer of powers, excludes the exponential fan-out-one obstruction at fan-out two (Lemma 8.1, Theorem 8.2).

Relation to prior work. Yoshinaka bounded characteristic data of (k, ℓ)-substitutable CFGs with a thickness parameter [12], learned multidimensionally substitutable MCFGs from positive data [13, 14], and gave the finite-signature learning principle with finite witness cardinality [15]; the qualitative results and the Const/Link/Comp construction used here are his. The finite-semigroup structure theory and local-group representations are classical [33–35]. Our contribution is the encoded-length analysis and the boundary results above. Other mildly context-sensitive formalisms are learned from positive data in [19], and MCFGs with queries or dual context sets in [16, 18, 20]; the MCFL substitution lemma of [21] is a membership condition, not a learning criterion.

Relation to the companion CFG manuscript. The companion manuscript [23] treats fan-out one. In short, it proves polynomial data for typings in $\mathbf{LI} \vee \mathbf{Grp}$ as a sufficient condition and exponential data for one typing onto U_1; here the fan-out-one boundary is determined exactly (**LG**), and the upper bound for **LG** and the linear-factor lower bound hold at every fan-out. Table 1 separates its results from ours (numbers in the second column refer to the companion). No result of

the companion is used as a premise: instead of its bracketed doubling family [23, Proposition 6.4] we use a three-letter variant, proved in Appendix F, and its expressive results (counter and Dyck obstructions, comparison with congruential grammars) are not used.

Table 1: Results of the companion CFG manuscript [23] (fan-out one) and of the present paper.

Topic	Companion [23]	Here
Typed reconstruction, Gold learner, typed-thickness witness bound	Theorem 4.2, Theorems 5.2–5.3, Corollary 5.4; Lemma 6.2, Corollary 6.3	Tuple versions (Section 3, Theorem 4.3); qualitative results credited to Yoshinaka; not new
Window and window–group bounds	Proposition 3.5, Lemma 7.1, Theorem 7.2, Lemma 7.4, Corollary 7.5	Not new at $f = 1$; new for $f \geq 2$: owner-wise replacement across interleaved components (Proposition 5.2)
Local-group typings	Not treated (only $\mathbf{LI} \vee \mathbf{Grp}$; Lemma 7.6)	New for data length, at every f (Theorems B.4 and 5.8); the characterization of $\mathbf{LG}$ by U_1 and the semigroup algebra are classical [33]
Doubling family under a typing onto U_1	Proposition 6.4, Corollary 6.6, Appendix C (five letters, brackets)	Three-letter variant with the same bounds, proved here (Proposition 7.1)
Rare-type lower bound	Proposition 6.4(iii)–(iv), for a single word	Every realized type and every f (Proposition 4.7)
Fan-out-one pseudovariety boundary	Sufficiency for $\mathbf{LI} \vee \mathbf{Grp}$ only (Remark 7.7)	New (Theorem 7.3)
Typed versus untyped class	Separations at $f = 1$	At $f \geq 2$, a non-context-free member outside Yoshinaka's class (Proposition 3.18)
Necessity of the factor $\|G\|\bar{\theta}_G$	—	New at every f (Proposition 6.1)
Fan-out two: power transfer	—	New (Theorem 8.2)

Scope of the claims. All bounds concern the total encoded length of characteristic samples for the specified set-driven operator, relative to the stated presentation; they bound neither an intrinsic grammar size nor a stage of Gold convergence, and their exponents are discussed at the end of Section 9. Lower bounds on data come only from distinguishing targets (Propositions 4.7 and 6.1); thick refinements (Proposition 4.9) limit the witness construction, not the data (Corollary 4.10). The typing is fixed before the target is seen; refining it enlarges the class but changes the reconstruction operator, so we neither infer nor optimize it. Table 2 lists, for each main claim, the parameters that are fixed, the operators it concerns and its side conditions; constants in upper bounds depend on the fixed parameters only.

Organization. Section 2 fixes the setting and Section 3 recalls the reconstruction operator and its qualitative guarantees, and separates the typed from the untyped class (Section 3.4). Section 4 reduces characteristic data to typed thickness and to distribution representatives. Section 5 proves Theorem A, Section 6 proves Theorem B, and Section 7 proves Theorem C and restates the boundary in terms of identification in polynomial time and thick data. Section 8 shows that the witness family does not carry over to fan-out two, and Section 9 discusses the exponents and the open problem. The appendices contain the technical proofs.

Table 2: Scope of the main claims. "Canonical" means both $\mathcal{B}_{f,s}^{\leq r_{\max}}$ and $\mathcal{B}_{f,s}$ (with one common sample); "class-wide" means every set-driven operator with characteristic samples throughout the class. G is a reduced good presentation of $L \cap \Sigma^+$ of fan-out $\leq f$ and rank $\leq r_{\max}$.

Claim	Fixed	Operators	Bound and conditions
Theorem A (Theorem 5.8)	f, $r_{\max}$, Σ, s with $s(\Sigma^+) \in \mathbf{LG}$	canonical	upper $O(\|G\|^2 \bar{\theta}_G)$, every $L \in \mathcal{C}_{f,s}^{\mathrm{mcf}}$ and every G
Theorem B(i) (Proposition 6.1)	f, any s	canonical ($r \geq 1$), class-wide	lower $\Omega(\|G\| \bar{\theta}_G)$ for finite targets; needs $\|\Sigma| \geq 4f+1$ ($\|\Sigma| \geq 2$ if $f = 1$); presentations of rank two
Theorem B(ii) (Corollary 6.2)	f, $r_{\max} \geq 1$, s_{triv}	$\mathcal{B}_{f,s_{\mathrm{triv}}}$	upper $O(\|G\| \bar{\theta}_G)$ (Yoshinaka [14]); attained if $r_{\max} \geq 2$ and $\|\Sigma| \geq 4f+1$ (≥ 2 if $f = 1$)
Theorem C(i) (Theorem 7.3)	$f = 1$, $r_{\max} \geq 2$, a pseudovariety	$\mathcal{B}_{1,s}^{\leq r_{\max}}$ (Definition 7.2)	uniform polynomial data iff $\mathbf{V} \subseteq \mathbf{LG}$
Theorem C(ii)	$f = 1$, s_U on three letters	canonical, class-wide	lower $3 \cdot 2^n - 1$ for presentations of size and thickness $O(n)$
Corollary 7.6	f, $r_{\max}$, s	set-driven	IPTtD for $\mathbf{LG}$ at every f; at $f = 1$ exactly for $\mathbf{V} \subseteq \mathbf{LG}$
Theorem 8.2	any $f \geq 2$, any s	none (membership)	the family of Theorem C(ii) and its letter-disjoint images leave $\mathcal{C}_{f,s}^{\mathrm{mcf}}$, and their forced word is never the only one of its type there; no data bound at $f \geq 2$
Open			whether $\mathbf{LG}$ is the boundary at $f \geq 2$ (Open Problem 9.1)

2 Preliminaries

Throughout, Σ is a finite alphabet, ε the empty word and $\Sigma^+ := \Sigma^* \setminus \{\varepsilon\}$. If $\Sigma = \emptyset$, the only languages are $\emptyset$ and $\{\varepsilon\}$ and are handled directly. For a finite $K \subseteq \Sigma^*$ put $\|K\|_+ := \sum_{w \in K} (|w| + 1)$, and for a tuple $\vec{u} = (u_1, \ldots, u_d)$ put $\|\vec{u}\|_1 := \sum_i |u_i|$. Every polynomial bound below states whether it concerns cardinality or $\|\cdot\|_+$. The table fixes symbols used across sections; all others (sliding-factor labellings, auxiliary morphisms, padding lengths) are defined where they occur.

Symbol	Meaning
G, G_{in}	a (reduced good) MCFG; an arbitrary standard input MCFG
B, C, X, Y	nonterminals; X, Y also denote typed nonterminals where specified
$\|G\|$, θ_G, t_G	encoding size (Definition 2.5), rule thickness, nonterminal thickness
s, $\widehat{s}$, S	positive typing, its fresh-identity completion, its positive image
$\preceq$	kernel refinement (Definition 2.3), also between monoid morphisms on Σ^*
$\mathbb{G}$, $\mathbb{H}$	finite groups ($\mathbb{G}$ the group component of a typing, $\mathbb{H}$ a Rees-matrix group or an auxiliary group)
g	a monoid morphism $\Sigma^* \to \mathbb{G}$ into a finite group
T, N	a finite semigroup; the finite monoid receiving an auxiliary morphism λ
λ, $\lambda_{k,\nu}$	an auxiliary finite-monoid morphism refining $\widehat{s}$ (Section 4); the sliding-factor morphism (Definition 5.7)
T	a derivation tree
n	the index of a grammar family
t_{typ}, τ, μ_m	shortest typed yield, representative length, shortest word of type m (Section 4); none of them is a thickness of G
S_L	the set $s(L \cap \Sigma^+)$ of realized positive types
$\mathcal{B}_{f,s}$, $\mathcal{B}_{f,s}^{\leq r}$	uncapped and rank-capped reconstruction operators (Definition 3.4)

2.1 Positive-Data Learning

Definition 2.1 (Text, TxtEx identification, conservativeness)**.** Fix a pause symbol $\perp \notin \Sigma$. A *text* for $L \subseteq \Sigma^*$ is an infinite sequence over $L \cup \{\perp\}$ in which every element of L occurs; the *content* of a finite prefix is the set of elements of Σ^* occurring in it. A *learner* is a computable map from finite prefixes to grammars. It *TxtEx-identifies* L [1] if on every text for L its outputs are, from some stage on, one fixed grammar H with $L(H) = L$; it TxtEx-identifies a class if it does so for each member. It is *conservative* if it changes its hypothesis only on a datum $w \in \Sigma^*$ not generated by the current hypothesis. We fix a hypothesis H_0 with $L(H_0) = \emptyset$.

Definition 2.2 (Set-driven operator and characteristic sample)**.** A *set-driven reconstruction operator* $\mathcal{B}$ maps each finite $K \subseteq \Sigma^*$ to a grammar $\mathcal{B}(K)$ depending only on the set K; it is *sample-consistent* if $K \subseteq L(\mathcal{B}(K))$ for all K. A finite $\mathcal{F} \subseteq L$ is a *characteristic sample for* $\mathcal{B}$ *and* L if $L(\mathcal{B}(K)) = L$ for every finite K with $\mathcal{F} \subseteq K \subseteq L$ (cf. [4]).

The definition constrains generated languages only: $\mathcal{B}(K)$ need not stabilize syntactically as K grows.

2.2 Fixed Finite-Semigroup Typing

Definition 2.3 (Typing, fresh completion, refinement)**.** A *typing* is a surjective semigroup morphism $s : \Sigma^+ \twoheadrightarrow \mathsf{S}$ onto a finite semigroup, given by the multiplication table of S and the images of the letters, so that $s(w)$ is computable in time $O(|w|)$; S is its *positive image*. (A morphism into a larger finite semigroup is replaced by its corestriction to the image.) Let $\mathsf{S}^\sharp := \mathsf{S} \uplus \{\mathbf{1}_\sharp\}$ be obtained by adjoining a *fresh* identity, even if S already has one, and let $\widehat{s} : \Sigma^* \twoheadrightarrow \mathsf{S}^\sharp$ be the monoid morphism with $\widehat{s}(\varepsilon) = \mathbf{1}_\sharp$ and $\widehat{s}|_{\Sigma^+} = s$. For $\vec{w} \in (\Sigma^*)^d$ put $\widehat{s}^{(d)}(\vec{w}) := (\widehat{s}(w_1), \ldots, \widehat{s}(w_d))$; on $(\Sigma^+)^d$ this is $s^{(d)}(\vec{w})$. For typings s, s' write $s \preceq s'$ if $\ker s' \subseteq \ker s$ on Σ^+, and let (s, s') be the typing $w \mapsto (s(w), s'(w))$ onto its image in $\mathsf{S} \times \mathsf{T}$, where $\mathsf{T} := s'(\Sigma^+)$; thus $s, s' \preceq (s, s')$. Write s_{triv} for the one-element typing.

The fresh completion evaluates grammar templates (including empty terminal gaps) and types tuple yields in the refinement of Section 3.2; its new identity is not a positive-word type. Algebraic conditions such as **LG** concern S, not $\mathsf{S}^\sharp$; indeed $\mathsf{S}^\sharp$ is never a local group, since $\{\mathbf{1}_\sharp, e\} \cong U_1$ for any idempotent e of the minimal ideal of S. A finite-monoid morphism $q : \Sigma^* \to \mathsf{N}$ determines the positive typing $q|_{\Sigma^+}$: by Lemma D.1(i), it gives the same target class and set-driven reconstruction rules as the fresh completion of that typing. Lemma D.1(ii) additionally identifies their trimmed yield-type refinements for *good* grammars. This last identification is not asserted for arbitrary standard grammars with empty-component yields.

Definition 2.4 (Prefix–suffix typing)**.** For $k, \ell \geq 0$ put $m_0 := \max\{1, k + \ell\}$, and for $|w| \geq m_0$ let $\mathrm{pref}_k(w)$, $\mathrm{suf}_\ell(w)$ be the prefix and suffix of the indicated lengths (length-zero windows are ε). Define

$$\widehat{s}_{k,\ell}(w) = \begin{cases} w, & |w| < m_0, \\ \langle \mathrm{pref}_k(w), \mathrm{suf}_\ell(w) \rangle, & |w| \geq m_0, \end{cases}$$

with disjoint tags for short words and pairs, and put $M_{k,\ell} := \widehat{s}_{k,\ell}(\Sigma^*)$. For short x, y and pairs $\langle u, v \rangle, \langle u', v' \rangle$ define

$$x \cdot y := \widehat{s}_{k,\ell}(xy), \qquad x \cdot \langle u, v \rangle := \langle \mathrm{pref}_k(xu), v \rangle,$$
$$\langle u, v \rangle \cdot y := \langle u, \mathrm{suf}_\ell(vy) \rangle, \quad \langle u, v \rangle \cdot \langle u', v' \rangle := \langle u, v' \rangle.$$

Put $s_{k,\ell} := \widehat{s}_{k,\ell}|_{\Sigma^+}$; a typing s with $s \preceq s_{k,\ell}$ for some $k, \ell \geq 0$ is a *window typing*.

A case check on the lengths of $x, y \in \Sigma^*$ gives $\widehat{s}_{k,\ell}(xy) = \widehat{s}_{k,\ell}(x) \cdot \widehat{s}_{k,\ell}(y)$. Since $\widehat{s}_{k,\ell}$ is onto $M_{k,\ell}$, the product is associative and $\widehat{s}_{k,\ell}(\varepsilon)$ is an identity, so $\widehat{s}_{k,\ell}$ is a finite-monoid morphism. The short-word tag separates ε from $\widehat{s}_{k,\ell}(\Sigma^+)$, so $\widehat{s}_{k,\ell}$ is isomorphic to the completion $\widehat{s_{k,\ell}}$.

2.3 Multiple Context-Free Grammars

Definition 2.5 (Standard MCFG presentation and size)**.** An MCFG [28, 29] is $G = (V, \Sigma, P, S, \mathrm{fan})$ with finite V, $S \in V$, $\mathrm{fan}(S) = 1$ and $\mathrm{fan}(A) \geq 1$. A rule of rank $r \geq 0$ is $A \to \rho(B_1, \ldots, B_r)$ with template $\rho = (\alpha_1, \ldots, \alpha_{d_0})$, $d_0 = \mathrm{fan}(A)$, $d_j = \mathrm{fan}(B_j)$, where each α_i is a word over $\Gamma_\rho := \Sigma \uplus \Xi_1 \uplus \cdots \uplus \Xi_r$, $\Xi_j := \{\xi_j^1, \ldots, \xi_j^{d_j}\}$, and every variable occurs at most once in $\alpha_1 \cdots \alpha_{d_0}$ (*linearity*). For $\vec{u}_j \in (\Sigma^*)^{d_j}$, $\rho(\vec{u}_1, \ldots, \vec{u}_r)$ substitutes $u_{j,\kappa}$ for ξ_j^κ; for $r = 0$ the rule is a *constant rule*. The tuple languages $L_A(G) \subseteq (\Sigma^*)^{\mathrm{fan}(A)}$ form the least family closed under the rules, and $L(G) := \{w : (w) \in L_S(G)\}$. We write $A \Rightarrow_G^* \vec{u}$ for a *terminal derivation* (a finite derivation tree rooted at A, evaluated bottom-up); one from S is *successful*. G is an f-MCFG if $\mathrm{fan}(A) \leq f$ for all A, and has rule-rank cap $r_{\max}$ if all ranks are at most $r_{\max}$. Such a presentation, with no further restriction, is called *standard*. Its *size* is

$$\|G\| := |V| + \sum_{(A \to \rho(B_1,\ldots,B_r)) \in P} \Big(1 + r + \textstyle\sum_i |\alpha_i|\Big),$$

where $|\alpha_i|$ counts terminal and variable occurrences. All bounds use this symbol-level size; a bit-level encoding with binary nonterminal names changes it by at most a factor $O(\log \|G\|)$.

$\mathsf{MCFL}(f, r)$ is the class of languages of MCFGs of fan-out $\leq f$ and rule rank $\leq r$; f-MCFL $= \bigcup_{r<\infty} \mathsf{MCFL}(f, r)$, and the minimum rule rank of a language is taken over all its fan-out-$\leq f$ presentations. For nonterminals of equal fan-out d, $A \to B$ abbreviates $A \to (\xi_1^1, \ldots, \xi_1^d)(B)$, and $A \to \vec{c}$ a constant rule. Rule variables are always ξ_j^κ; terminals in examples are in `typewriter` font. Fan-out and rank are distinct resources. At fixed fan-out at least two the rule-rank hierarchy does not collapse in general, although adjacent levels can coincide; in particular $\mathsf{MCFL}(2, 2) = \mathsf{MCFL}(2, 3)$ [22, 31]. By contrast, fan-out-one CFGs can be binarized. The reconstruction below therefore assumes neither a binary normal form nor a known rank cap.

Definition 2.6 (Grammar properties)**.** A rule is *nondeleting* if every variable occurs exactly once in its template; a nondeleting rule is *nonpermuting* if for each j the variables $\xi_j^1, \ldots, \xi_j^{d_j}$ occur in this order in $\alpha_1 \cdots \alpha_{d_0}$, and *nonmerging* if moreover no α_i contains a factor $\xi_j^\kappa \xi_j^{\kappa+1}$. Following [14, Section 2.3], a rule is *ε-free* ("λ-free" there) if no α_i is empty. A rule, template or grammar is *good* if it is ε-free, nondeleting, nonpermuting and nonmerging. A nonterminal A is *reachable* if there is a chain $S = A_0, \ldots, A_m = A$ in which each A_{i+1} is a child in a rule for A_i, and *productive* if $L_A(G) \neq \emptyset$; G is *reduced* if every nonterminal is both.

In a ε-free grammar every component of every derived tuple is nonempty (induction on derivations). ε-freeness concerns template components and is distinct from $\varepsilon \notin L(G)$.

2.4 Tuple Contexts and Substitutability

A *sentence context of arity d* is a word $E = v_0 \Box_1 v_1 \cdots v_{d-1} \Box_d v_d$ with each hole occurring once, $v_0, v_d \in \Sigma^*$ and $v_1, \ldots, v_{d-1} \in \Sigma^+$; $\mathcal{E}_d$ is the set of these *admissible* contexts [14, Section 3.1]. $E[\vec{w}]$ fills $\Box_i$ by w_i, $|E|_{\mathrm{fix}} := \sum_{i=0}^d |v_i|$, and E *accepts* $\vec{w}$ in L if $E[\vec{w}] \in L$. The *d-tuple distribution* of $\vec{x} \in (\Sigma^*)^d$ is $\mathcal{D}_L^{(d)}(\vec{x}) := \{E \in \mathcal{E}_d : E[\vec{x}] \in L\}$; at fan-out one it is the two-sided distribution $\{(u, v) : uxv \in L\}$.

Definition 2.7 (Tuple occurrence)**.** For $w = a_1 \cdots a_n$ number the cut positions $\textcircled{0}a_1\textcircled{1}\cdots a_n\textcircled{n}$ and write $w[i : i'] := a_{i+1} \cdots a_{i'}$. An *arity-d tuple occurrence* in w is a sequence $0 \le i_1 < i'_1 < i_2 < i'_2 < \cdots < i_d < i'_d \le n$; it determines $\vec{x} = (w[i_1 : i'_1], \ldots, w[i_d : i'_d]) \in (\Sigma^+)^d$ and the context $E \in \mathcal{E}_d$ obtained by replacing these intervals by $\Box_1, \ldots, \Box_d$, so that $E[\vec{x}] = w$. For finite K, $\mathrm{Obs}_d(K) := \{\vec{x} \in (\Sigma^+)^d : E[\vec{x}] \in K \text{ for some } E \in \mathcal{E}_d\}$ is the set of tuples arising from such occurrences, and for a fixed fan-out bound $f \ge 1$, put $\mathrm{Obs}(K) := \bigcup_{1 \le d \le f} \mathrm{Obs}_d(K)$.

Definition 2.8 ((f, s)-tuple substitutability and the target class)**.** Let $f \ge 1$ and let s be a typing. A language $L \subseteq \Sigma^*$ is (f, s)*-tuple-substitutable* if for every $1 \le d \le f$ and all $\vec{x}, \vec{y} \in (\Sigma^+)^d$,

$$s^{(d)}(\vec{x}) = s^{(d)}(\vec{y}) \;\text{ and }\; \mathcal{D}_L^{(d)}(\vec{x}) \cap \mathcal{D}_L^{(d)}(\vec{y}) \neq \emptyset \implies \mathcal{D}_L^{(d)}(\vec{x}) = \mathcal{D}_L^{(d)}(\vec{y}).$$

$\mathcal{C}_{f,s}^{\mathrm{mcf}}$ is the class of (f, s)-tuple-substitutable f-MCFLs. For $\vec{x}, \vec{y} \in (\Sigma^+)^d$ write $\vec{x} \equiv_{L,s}^{d} \vec{y}$ (or $\vec{x} \equiv_{L,s} \vec{y}$) if $s^{(d)}(\vec{x}) = s^{(d)}(\vec{y})$ and $\mathcal{D}_L^{(d)}(\vec{x}) = \mathcal{D}_L^{(d)}(\vec{y})$; this is an equivalence relation.

Empty components are never compared, following [14, Section 3.1 and footnote 3] (see [8] for CFGs), and $\varepsilon \in L$ is handled at the start symbol; otherwise already $\mathtt{a}^+$ would fail, since $\mathtt{a}$ and ε share a context but not their distributions. The bias consists of both s and f:

(a) If $s \preceq s'$, equal s'-types are equal s-types, so $\mathcal{C}_{f,s}^{\mathrm{mcf}} \subseteq \mathcal{C}_{f,s'}^{\mathrm{mcf}}$; in particular $\mathcal{C}_{f,s_{\mathrm{triv}}}^{\mathrm{mcf}} \subseteq \mathcal{C}_{f,s}^{\mathrm{mcf}}$. This compares classes only: the operators $\mathcal{B}_{f,s}$ and $\mathcal{B}_{f,s'}$ of Section 3 differ in general (they coincide when $\ker s = \ker s'$).

(b) $\mathcal{C}_{f+1,s}^{\mathrm{mcf}} \cap f\text{-MCFL} \subseteq \mathcal{C}_{f,s}^{\mathrm{mcf}}$, but the class is not monotone in f: for the trivial typing $\{w\#w^R : w \in \{\mathtt{a}, \mathtt{b}\}^*\}$ is 1- but not 2-dimensionally substitutable [14, Section 3.1, Example 5].

(c) For every $\mathsf{Acc} \subseteq \mathsf{S}^\sharp$, $\widehat{s}^{-1}(\mathsf{Acc}) \in \mathcal{C}_{f,s}^{\mathrm{mcf}}$: it is regular, and componentwise equal types give equal total types in every context.

Proposition 2.9 (Intersection and abelian filtering)**.** *Fix f and s.*

(i) If L_1, L_2 are (f, s)-tuple-substitutable, so is $L_1 \cap L_2$.

(ii) If L is (f, s)-tuple-substitutable, A is an abelian group, $g : \Sigma^* \to \mathsf{A}$ a monoid morphism, $\mathsf{H} \le \mathsf{A}$ and $\gamma \in \mathsf{A}$, then $L \cap g^{-1}(\gamma + \mathsf{H})$ is (f, s)-tuple-substitutable.

(iii) If $L \in \mathcal{C}_{f,s}^{\mathrm{mcf}}$, $\mathsf{Acc} \subseteq \mathsf{S}^\sharp$ and, in (ii), A is finite, then $L \cap \widehat{s}^{-1}(\mathsf{Acc}) \cap g^{-1}(\gamma + \mathsf{H}) \in \mathcal{C}_{f,s}^{\mathrm{mcf}}$.

For infinite A the intersection in (ii) belongs to $\mathcal{C}_{f,s}^{\mathrm{mcf}}$ whenever it is an f-MCFL.

Proof. (i) A context accepting equal-type $\vec{x}, \vec{y}$ in $L_1 \cap L_2$ accepts both in each L_i, so their distributions agree in each L_i, hence in the intersection. (ii) If some context E accepts $\vec{x}, \vec{y}$ in $g^{-1}(\gamma + \mathsf{H})$, then $\delta := \sum_i (g(y_i) - g(x_i)) \in \mathsf{H}$, and by commutativity $g(E'[\vec{y}]) - g(E'[\vec{x}]) = \delta$ for every $E' \in \mathcal{E}_d$, so coset membership transfers in both directions. Thus $g^{-1}(\gamma + \mathsf{H})$ is (f, s_{triv})-, hence (f, s)-tuple-substitutable; apply (i). (iii) The language $\widehat{s}^{-1}(\mathsf{Acc})$ is (f, s)-tuple-substitutable by item (c) above, so the intersection is, by (i) and (ii); it is the intersection of an f-MCFL with a regular language, hence an f-MCFL [28]. □

The factorizations in (ii) and (iii) are a tool for constructing members of the class; they are not supplied to the reconstruction.

3 Reconstruction and Qualitative Learning

This section fixes the reconstruction operator analysed later and records its qualitative guarantees, none of which is claimed as new. For fixed f, rank cap and typing, our operator follows the distributional reconstruction pattern for substitutable languages over finite signatures in [15]. We use the MCFG signature of [15, Example 3] with tuple sorts refined by s-types (cf. [15, Section 8]). The related qualitative results are

Here	Related prior result	Treatment here
Prop. 3.10 (soundness)	[15, Lemma 2]	—
Prop. 3.11 (completeness)	[15, Lemma 3]	arbitrary same-type representatives
Thm. 3.13 (exact reconstruction)	[15, Lemmas 2–3]	as for Prop. 3.11
Cor. 3.16 (identification)	[15, Theorem 1]	unbounded rank over the class

The table records analogues, not literal identities: all statements of this section are proved directly for the concrete grammar in Appendices C and D, without embedding into the signature framework. Identifying all of $\mathcal{C}^{\mathrm{mcf}}_{f,s}$, whose members have unbounded minimum rank, uses the sample-dependent cap $\|K\|_+$, outside the bounded-arity assumption of [15]. For a fixed finite signature, that paper bounds the *number* of words of a characteristic sample, polynomially in the grammar; it does not bound their total *length*, which is the subject of this paper.

3.1 Set-Driven Reconstruction

All interaction between templates and contexts rests on one observation.

Lemma 3.1 (Induced child contexts)**.** *Let $\rho = (\alpha_1, \dots, \alpha_{d_0})$ be a good template of rank r with child arities $d_1, \dots, d_r$, let $E \in \mathcal{E}_{d_0}$, let $1 \le j \le r$, and fix $\vec{u}_i \in (\Sigma^+)^{d_i}$ for $i \ne j$. Let E_j be the word obtained from $E[(\alpha_1, \dots, \alpha_{d_0})]$ by replacing ξ^{κ}_j by $\square_\kappa$ and every other variable $\xi^{\kappa'}_i$ by $u_{i,\kappa'}$. Then $E_j \in \mathcal{E}_{d_j}$ and $E_j[\vec{x}] = E[\rho(\vec{u}_1, \dots, \vec{u}_{j-1}, \vec{x}, \vec{u}_{j+1}, \dots, \vec{u}_r)]$ for every $\vec{x} \in (\Sigma^*)^{d_j}$.*

Lemma 3.2 (Composition preserves equivalence)**.** *Let $L \subseteq \Sigma^*$, let ρ be a good template of rank r, and let $\vec{x}_j, \vec{u}_j \in (\Sigma^+)^{d_j}$ satisfy $\vec{u}_j \equiv_{L,s} \vec{x}_j$ for every j. Then $\rho(\vec{u}_1, \dots, \vec{u}_r) \equiv_{L,s} \rho(\vec{x}_1, \dots, \vec{x}_r)$.*

Definition 3.3 (Composition witness)**.** Let K be finite and $r \ge 1$. A *composition witness of rank r* in K is a triple $(\vec{z}, \rho, (\vec{u}_1, \dots, \vec{u}_r))$ in which ρ is a good template of rank r with output arity $d_0 \le f$ and child arities $d_1, \dots, d_r \le f$, $\vec{z} \in \mathrm{Obs}_{d_0}(K)$, $\vec{u}_j \in (\Sigma^+)^{d_j}$, and $\vec{z} = \rho(\vec{u}_1, \dots, \vec{u}_r)$.

Equivalently, a composition witness is a segmentation of the components of an observed $\vec{z}$ into terminal gaps and nonempty intervals labelled by variables, subject to the good-template conditions. If $E[\vec{z}] \in K$, Lemma 3.1 gives $E_j[\vec{u}_j] = E[\vec{z}] \in K$ with E_j admissible, so every $\vec{u}_j \in \mathrm{Obs}(K)$. The children occupy disjoint, nonempty parts of $\vec{z}$, so

$$r \le \textstyle\sum_j \|\vec{u}_j\|_1 \le \|\vec{z}\|_1 \le \max_{w \in K} |w| \le \|K\|_+. \tag{3.1}$$

Definition 3.4 (Capped and uncapped reconstruction operators)**.** Fix f and a typing s. For $r_{\max} \ge 1$ let $\mathcal{B}^{\le r_{\max}}_{f,s}(K) = \widehat{G}_{\le r_{\max}}(K)$ be the grammar with start symbol $\widehat{S}$, a nonterminal $[\vec{x}]$ of arity d for each $\vec{x} \in \mathrm{Obs}_d(K)$, $1 \le d \le f$, and the rules

(Start) $\widehat{S} \to [(w)]$ for $w \in K \cap \Sigma^+$, and $\widehat{S} \to \varepsilon$ if $\varepsilon \in K$;

(Const) $[\vec{x}] \to \vec{x}$ for every $\vec{x} \in \mathrm{Obs}(K)$;

(Comp) $[\vec{z}] \to \rho([\vec{u}_1], \dots, [\vec{u}_r])$ for every composition witness of rank $1 \le r \le r_{\max}$;

(Link) $[\vec{x}] \to [\vec{y}]$ for $\vec{x}, \vec{y} \in \mathrm{Obs}_d(K)$ with $s^{(d)}(\vec{x}) = s^{(d)}(\vec{y})$ and $E[\vec{x}], E[\vec{y}] \in K$ for some $E \in \mathcal{E}_d$.

For $K \neq \emptyset$ put $\mathcal{B}_{f,s}(K) := \widehat{G}(K) := \widehat{G}_{\leq \|K\|_+}(K)$; $\mathcal{B}_{f,s}(\emptyset)$ has no rules. Fixed encoding orders make both operators deterministic and set-driven.

By (3.1), $\mathcal{B}_{f,s}(K)$ contains a Comp rule for every composition witness in K. Both operators are sample-consistent, via $\widehat{S} \to [(w)] \to (w)$ and $\widehat{S} \to \varepsilon$. All non-start templates are ε-free, so $\widehat{S} \Rightarrow^* \varepsilon$ only by the rule $\widehat{S} \to \varepsilon$. For the trivial typing and $\varepsilon \notin K$, Comp/Const, Link and Start correspond to the rules of Types I–III of [14, Section 4.1], and $\mathcal{B}_{f,s}$ corresponds to the operator of the unbounded-rank learner $\mathcal{A}(p, *)$ of [14, Corollary 2].

3.2 Yield-Typed Refinement and Witnesses

Definition 3.5 (Yield-type refinement)**.** Let $G = (V, \Sigma, P, S, \mathrm{fan})$ be a finite nondeleting MCFG and s a typing. For a template ρ and $\mathbf{t}_j \in (\mathsf{S}^\sharp)^{d_j}$, let $\mathrm{out}_\rho(\mathbf{t}_1, \ldots, \mathbf{t}_r) \in (\mathsf{S}^\sharp)^{d_0}$ have ith component the image of α_i under the monoid morphism $\Gamma_\rho^* \to \mathsf{S}^\sharp$ extending $a \mapsto \widehat{s}(a)$ and $\xi_j^\kappa \mapsto (\mathbf{t}_j)_\kappa$. The *full yield-type refinement* $G^{\widehat{s}}$ has a fresh start symbol $\widetilde{S}$, nonterminals $A_{\mathbf{t}}$ for $A \in V$ and $\mathbf{t} \in (\mathsf{S}^\sharp)^{\mathrm{fan}(A)}$, start rules $\widetilde{S} \to S_{(m)}$ for $m \in \mathsf{S}^\sharp$, and for every rule $A \to \rho(B_1, \ldots, B_r)$ (including $r = 0$) and all $\mathbf{t}_j \in (\mathsf{S}^\sharp)^{\mathrm{fan}(B_j)}$ the rule $A_{\mathrm{out}_\rho(\mathbf{t}_1,\ldots,\mathbf{t}_r)} \to \rho(B_{1,\mathbf{t}_1}, \ldots, B_{r,\mathbf{t}_r})$. The *trimmed refinement* $\widetilde{G}_0$ keeps exactly the nonterminals and rules occurring in some successful derivation from $\widetilde{S}$. Unless $G^{\widehat{s}}$ is named, *typed nonterminals* and *typed rules* are the non-start nonterminals and rules of $\widetilde{G}_0$, and $L_X := \{\vec{u} : X \Rightarrow^*_{\widetilde{G}_0} \vec{u}\}$.

Typed rules have the templates of their sources, so $\widetilde{G}_0$ is good whenever G is; each L_X is nonempty, and its tuples have nonempty components if G is ε-free.

Proposition 3.6 (Yield-type invariants)**.** *(i) If $A_{\mathbf{t}} \Rightarrow^* \vec{u}$ in $G^{\widehat{s}}$, then $\widehat{s}^{(\mathrm{fan}(A))}(\vec{u}) = \mathbf{t}$, and erasing indices gives a G-derivation of $\vec{u}$.*

(ii) Every terminal derivation $A \Rightarrow^*_G \vec{u}$ lifts uniquely to $A_{\widehat{s}^{(\mathrm{fan}(A))}(\vec{u})} \Rightarrow^*_{G^{\widehat{s}}} \vec{u}$; a successful derivation of w lifts uniquely through $\widetilde{S} \to S_{(\widehat{s}(w))}$.

(iii) $L(\widetilde{G}_0) = L(G^{\widehat{s}}) = L(G)$.

(iv) For every typed nonterminal X, every terminal derivation $X \Rightarrow^*_{G^{\widehat{s}}} \vec{u}$ uses only rules of $\widetilde{G}_0$; hence $L_X = \{\vec{u} : X \Rightarrow^*_{G^{\widehat{s}}} \vec{u}\}$.

Lemma 3.7 (Canonical contexts for typed nonterminals)**.** *Let G be good. Every typed nonterminal X of arity d has a context $\chi(X) \in \mathcal{E}_d$ with $\chi(X)[\vec{u}] \in L(G)$ for all $\vec{u} \in L_X$; if $\widetilde{S} \to X$ is in $\widetilde{G}_0$, one may take $\chi(X) = \square_1$.*

Lemma 3.8 (Typed nonterminals are distributionally homogeneous)**.** *Let G be good of fan-out $\leq f$ with $L := L(G) \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$. For every typed nonterminal X of arity d, L_X lies in a single $\equiv^d_{L,s}$-class, denoted $\mathrm{Cl}_{L,s}(X)$.*

The semantic class $\mathrm{Cl}_{L,s}(X)$ is distinct from the syntactic reconstruction nonterminal $[\vec{x}]$ of Definition 3.4.

The proofs of these three statements are in Appendix D. The homogeneity of Lemma 3.8 is weaker than that of congruential MCFGs [17], whose contexts may have empty separators.

Definition 3.9 (Canonical tuples, contexts and the witness set)**.** Let G be good. For every typed nonterminal X fix a *canonical tuple* $\omega(X) \in L_X$ and a *canonical context* $\chi(X)$ as in Lemma 3.7, with $\chi(X) = \square_1$ whenever $\widetilde{S} \to X$ is in $\widetilde{G}_0$. Let $\mathsf{W}(\widetilde{G}_0)$ consist of the words

$$\chi(X)[\omega(X)] \ \ (X \text{ typed}), \qquad \chi(X)[\rho(\omega(Y_1), \ldots, \omega(Y_q))] \ \ (X \to \rho(Y_1, \ldots, Y_q) \text{ typed}, \ q \geq 0).$$

Since $\omega(X)$ and $\rho(\omega(Y_1), \ldots, \omega(Y_q))$ belong to L_X, $\mathsf{W}(\widetilde{G}_0)$ is a finite subset of $L(G)$, nonempty if $L(G) \neq \emptyset$, and $|\mathsf{W}(\widetilde{G}_0)|$ is at most the number of typed nonterminals and rules. Canonical objects are properties of the target, not inputs of the operator: qualitatively any choice works, while Section 4 chooses short ones.

3.3 Qualitative Guarantees

Proposition 3.10 (Soundness of every hypothesis)**.** *Let L be (f,s)-tuple-substitutable and $K \subseteq L$ finite. For every $r_{\max} \geq 1$, if $[\vec{x}] \Rightarrow^* \vec{u}$ in $\widehat{G}_{\leq r_{\max}}(K)$, then $\vec{u} \equiv_{L,s} \vec{x}$. Consequently $L(\mathcal{B}_{f,s}^{\leq r_{\max}}(K)) \subseteq L$ and $L(\mathcal{B}_{f,s}(K)) \subseteq L$.*

Proposition 3.11 (Completeness from a witness assignment)**.** *Let G be a finite good MCFG of fan-out $\leq f$, let r_G be the maximum rank of a typed rule of $\widetilde{G}_0$ ($r_G := 0$ if none), and assign to each typed nonterminal $X = A_{\mathbf{t}}$ a tuple $\iota(X) \in (\Sigma^+)^{\mathrm{fan}(A)}$ with $\widehat{s}^{(\mathrm{fan}(A))}(\iota(X)) = \mathbf{t}$ and a context $\chi(X) \in \mathcal{E}_{\mathrm{fan}(A)}$, with $\chi(X) = \square_1$ if $\widetilde{S} \to X$ is in $\widetilde{G}_0$. Suppose K contains $\chi(X)[\iota(X)]$ for each X and $\chi(X)[\rho(\iota(Y_1), \ldots, \iota(Y_q))]$ for each typed rule $X \to \rho(Y_1, \ldots, Y_q)$, $q \geq 0$. Then*

$$L(G) \subseteq L(\mathcal{B}_{f,s}^{\leq r_{\max}}(K)) \quad (r_{\max} \geq \max\{1, r_G\}), \qquad L(G) \subseteq L(\mathcal{B}_{f,s}(K)).$$

Neither substitutability nor $\iota(X) \in L_X$ is assumed in Proposition 3.11; $\iota = \omega$ is the case $\mathsf{W}(\widetilde{G}_0) \subseteq K$.

Exact reconstruction needs a good presentation of the target; the following normalization controls size, fan-out and rank, but not thickness. The rule thickness of G_{good} may exceed that of G_{in}, so all data bounds below are stated relative to a given reduced good presentation and are not transferred through normalization.

Proposition 3.12 (Good normalization with rank preservation)**.** *Let G_{in} be a finite standard f-MCFG of rule rank at most $r_{\max}$. There is a finite good f-MCFG G_{good} of rule rank at most $r_{\max}$ with $L(G_{\mathrm{good}}) = L(G_{\mathrm{in}}) \setminus \{\varepsilon\}$ and, for fixed $f, r_{\max}$, $\|G_{\mathrm{good}}\| = O_{f,r_{\max}}(\|G_{\mathrm{in}}\|)$. If $L(G_{\mathrm{in}}) \cap \Sigma^+ \neq \emptyset$, G_{good} is reduced.*

Theorem 3.13 (Exact reconstruction for a fixed typing)**.** *Fix $f \geq 1$ and a typing s, and let $L \in \mathcal{C}_{f,s}^{\mathrm{mcf}}$. If $L \cap \Sigma^+ = \emptyset$ put $\mathcal{F}_L := L$ and $r_G := 0$. Otherwise let G be a reduced good fan-out-$\leq f$ presentation of $L \cap \Sigma^+$ (Proposition 3.12), let r_G be as in Proposition 3.11, and put $\mathcal{F}_L := \mathsf{W}(\widetilde{G}_0) \cup (\{\varepsilon\} \cap L)$. Then $\mathcal{F}_L \subseteq L$ is finite, and every finite K with $\mathcal{F}_L \subseteq K \subseteq L$ satisfies*

$$L(\mathcal{B}_{f,s}(K)) = L, \qquad L(\mathcal{B}_{f,s}^{\leq r_{\max}}(K)) = L \quad \text{for every } r_{\max} \geq \max\{1, r_G\}.$$

If $L \in \mathsf{MCFL}(f, r)$, G can be chosen of rank $\leq r$, so $r_{\max} = \max\{1, r\}$ suffices (also when $r = 0$).

Theorem 3.14 (Polynomial construction under a rule-rank cap)**.** *Fix $f, r_{\max} \geq 1$ and s. The encoding of $\widehat{G}_{\leq r_{\max}}(K)$ is computable in time $(1 + \|K\|_+)^{O(f r_{\max})}$; its fan-out is at most f and its rank at most $r_{\max}$.*

Definition 3.15 (Conservative sequential learner)**.** The learner $\mathcal{A}_{f,s}$ (respectively $\mathcal{A}_{f,s}^{\leq r_{\max}}$) starts with H_0, keeps its hypothesis on $\perp$ and on every datum the hypothesis generates, and otherwise replaces it by $\mathcal{B}_{f,s}(K)$ (respectively $\mathcal{B}_{f,s}^{\leq r_{\max}}(K)$), where K is the content read so far. This is the policy of [14, Algorithm 1].

Corollary 3.16 (Gold identification)**.** *Fix $f \geq 1$ and s.*

(i) *$\mathcal{A}_{f,s}$ TxtEx-identifies $\mathcal{C}_{f,s}^{\mathrm{mcf}}$.*

(ii) *For fixed $r_{\max}$, $\mathcal{A}_{f,s}^{\leq r_{\max}}$ TxtEx-identifies $\mathcal{C}_{f,s}^{\mathrm{mcf}} \cap \mathsf{MCFL}(f, r_{\max})$, and processes each datum in time polynomial in $\|K\|_+$ for the content K read so far.*

Proofs are in Appendix C (Proposition 3.12: Appendix E). None of these statements bounds the length of characteristic data or the stage of convergence.

3.4 Typed versus Untyped Classes

Regular path languages give simple members of every class.

Lemma 3.17 (Path languages)**.** *Let R be the set of labels of paths from a vertex p to a vertex p' in a finite directed graph in which every letter labels exactly one edge. If $\vec{x}, \vec{y} \in (\Sigma^+)^d$ and $E \in \mathcal{E}_d$ satisfy $E[\vec{x}], E[\vec{y}] \in R$, then $F[\vec{x}] \in R \iff F[\vec{y}] \in R$ for every $F \in \mathcal{E}_d$.*

Proof. A word of Σ^+ lies in R iff the target of the edge of each letter is the source of the edge of the next one, the first edge starts at p and the last ends at p'. The h-th hole of E is preceded by the last letter of v_{h-1}, or by the vertex p when $h = 1$ and $v_0 = \varepsilon$, and is followed likewise; internal separators are nonempty. So E determines the vertices at both ends of each hole, and x_h and y_h label paths between the same two vertices. Replacing every x_h by y_h in $F[\vec{x}]$ therefore preserves all three conditions, and conversely. □

The class grows strictly with the typing already at fan-out two, and outside the context-free languages; the untyped operator then over-generalizes.

Proposition 3.18 (A typed class beyond the untyped one)**.** *Let $\Sigma = \{\mathtt{a}, \mathtt{b}, \mathtt{c}, \mathtt{d}, \#_1, \#_2, \#_3\}$, let $s : \Sigma^+ \twoheadrightarrow \mathbb{Z}_2$ count the letters $\mathtt{a}$ and $\mathtt{c}$ modulo 2, and let*

$$L_{\mathrm{sep}} := \{\mathtt{a}^m \#_1 \mathtt{b}^n \#_2 \mathtt{c}^{m'} \#_3 \mathtt{d}^n : m, n \geq 0,\ m' \in \{m, m-1\},\ m' \geq 0\}.$$

(i) *L_{sep} is a 2-MCFL of rule rank two and is not context-free.*

(ii) *L_{sep} is (f, s)-tuple-substitutable for every $f \geq 1$; hence $L_{\mathrm{sep}} \in \mathcal{C}_{f,s}^{\mathrm{mcf}}$ for every $f \geq 2$.*

(iii) *L_{sep} is not (f, s_{triv})-tuple-substitutable for any $f \geq 1$: the words $\mathtt{a}$ and $\mathtt{aa}$ share the context $\Box_1 \#_1 \#_2 \mathtt{c} \#_3$ but not the context $\Box_1 \#_1 \#_2 \#_3$. Accordingly, on the sample $K = \{\mathtt{a}\#_1\#_2\mathtt{c}\#_3,\ \mathtt{aa}\#_1\#_2\mathtt{c}\#_3,\ \mathtt{a}\#_1\#_2\#_3\} \subseteq L_{\mathrm{sep}}$ the untyped operator $\mathcal{B}_{f,s_{\mathrm{triv}}}$ derives $\mathtt{aa}\#_1\#_2\#_3 \notin L_{\mathrm{sep}}$, whereas $\mathcal{B}_{f,s}$ does not link $\mathtt{a}$ with $\mathtt{aa}$.*

Proof. Let $\psi : \Sigma^* \to \mathbb{Z}^2$ be the morphism with $\psi(\mathtt{a}) = (1, 0)$, $\psi(\mathtt{c}) = (-1, 0)$, $\psi(\mathtt{b}) = (0, 1)$, $\psi(\mathtt{d}) = (0, -1)$ and $\psi(\#_i) = (0, 0)$, and let $R := \mathtt{a}^*\#_1\mathtt{b}^*\#_2\mathtt{c}^*\#_3\mathtt{d}^*$. Then $L_{\mathrm{sep}} = R \cap \psi^{-1}(P)$ with $P := \{(0, 0), (1, 0)\}$, and R is the path language from 0 to 3 of the graph on $\{0, 1, 2, 3\}$ with loops $\mathtt{a}, \mathtt{b}, \mathtt{c}, \mathtt{d}$ at $0, 1, 2, 3$ and edges $\#_h : h - 1 \to h$; every letter labels exactly one edge.

(i) The rules $E \to (\varepsilon,\varepsilon) \mid (\mathtt{a}\xi_1^1, \mathtt{c}\xi_1^2)(E)$, $D \to (\varepsilon,\varepsilon) \mid (\mathtt{b}\xi_1^1, \mathtt{d}\xi_1^2)(D)$ and

$$S \to (\xi_1^1 \#_1 \xi_2^1 \#_2 \xi_1^2 \#_3 \xi_2^2)(E, D) \mid (\mathtt{a}\xi_1^1 \#_1 \xi_2^1 \#_2 \xi_1^2 \#_3 \xi_2^2)(E, D)$$

form a standard MCFG of fan-out two and rule rank two generating L_{sep}, since E derives the pairs $(\mathtt{a}^k, \mathtt{c}^k)$ and D the pairs $(\mathtt{b}^n, \mathtt{d}^n)$. The words of L_{sep} in which $\mathtt{a}$ and $\mathtt{c}$ occur an even number of times altogether are exactly those with $m' = m$; this is an intersection with a regular language, and erasing the markers maps the result onto $\{\mathtt{a}^m\mathtt{b}^n\mathtt{c}^m\mathtt{d}^n : m, n \geq 0\}$, which is not context-free. Context-free languages are closed under both operations.

(ii) Let $1 \leq d \leq f$, let $\vec{x}, \vec{y} \in (\Sigma^+)^d$ have equal componentwise s-types, and let $E \in \mathcal{E}_d$ accept both. Put $(\delta_1, \delta_2) := \sum_i (\psi(x_i) - \psi(y_i)) = \psi(E[\vec{x}]) - \psi(E[\vec{y}]) \in P - P$, so $\delta_2 = 0$ and $\delta_1 \in \{-1, 0, 1\}$. The first coordinate of ψ is congruent to s modulo 2 on every word, so δ_1 is even; hence $\psi(F[\vec{x}]) = \psi(F[\vec{y}])$ for every $F \in \mathcal{E}_d$. By Lemma 3.17, $F[\vec{x}] \in R$ iff $F[\vec{y}] \in R$. So F accepts $\vec{x}$ iff it accepts $\vec{y}$. With (i), $L_{\mathrm{sep}} \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$ for $f \geq 2$.

(iii) The two contexts give $\mathtt{a}\#_1\#_2\mathtt{c}\#_3$, $\mathtt{aa}\#_1\#_2\mathtt{c}\#_3 \in L_{\mathrm{sep}}$ ($m' = m$ and $m' = m - 1$), $\mathtt{a}\#_1\#_2\#_3 \in L_{\mathrm{sep}}$ and $\mathtt{aa}\#_1\#_2\#_3 \notin L_{\mathrm{sep}}$. On K, the first context gives the Link rule $[(\mathtt{a})] \to [(\mathtt{aa})]$ for the trivial typing, and $((\mathtt{a}\#_1\#_2\#_3), (\xi_1^1\#_1\#_2\#_3), ((\mathtt{a})))$ is a composition witness of rank one, so $\widehat{S} \to [(\mathtt{a}\#_1\#_2\#_3)] \to [(\mathtt{a})]\#_1\#_2\#_3 \to [(\mathtt{aa})]\#_1\#_2\#_3 \to \mathtt{aa}\#_1\#_2\#_3$. Under s the types of $\mathtt{a}$ and $\mathtt{aa}$ differ, so no such Link rule exists. □

The positive image of s is the group $\mathbb{Z}_2$, a local group, so Theorem A gives L_{sep} characteristic data of length $O(\|G\|^2\bar{\theta}_G)$ for each of its reduced good presentations, at every fan-out $f \geq 2$, while L_{sep} lies outside the class of Yoshinaka's untyped learner [14] at every fan-out.

4 Characteristic-Data Bounds

Throughout this section f, $r_{\max} \geq 1$, Σ and the typing $s : \Sigma^+ \twoheadrightarrow \mathsf{S}$ are fixed; $G = (V, \Sigma, P, S, \mathrm{fan})$ is a reduced good MCFG of fan-out $\leq f$ and rule rank $\leq r_{\max}$, and $\widetilde{G}_0$ is its trimmed $\widehat{s}$-refinement. Under a rank cap, Theorem 3.13 already supplies $O(\|G\|)$ characteristic words; the task is to bound their total length *in terms of* G. All bounds follow one scheme.

(1) *Witness length* (Lemma 4.2). If, for an auxiliary finite-monoid morphism λ refining s, every tuple of every nonterminal can be exchanged for a tuple of the same nonterminal and the same λ-vector of total length at most β, then $\mathsf{W}(\widetilde{G}_0)$ is characteristic and has total length $O(\|G\|(\bar{\theta}_G + \beta))$.

(2) *Minimal counterexamples* (Section 5.1). In a shortest derivation with a prescribed root λ-vector, a large antichain of heavy subtrees would admit a simultaneous replacement preserving that vector; so antichains are small, and Lemma 5.1 turns this into $\beta = O(\|G\|\bar{\theta}_G)$ or $O(\|G\|^2\bar{\theta}_G)$.

(3) *Regimes.* Only λ and the replacement step change: window typings keep marked boundary letters; window–group typings in addition preserve ordered group products (Proposition 5.2); local-group typings reduce to these through sliding-factor coordinates (Section 5.3).

The full yield-type refinement is not used as the primary thickness measure, since it may be exponentially thicker than G (Proposition 4.9). The proofs of this section are in Appendix D.

4.1 Thickness Scales and Witness Length

For a rule R of G let

$$\theta_G(R) := \min\{\, |w| : w \in L(G) \text{ has a successful derivation using } R \,\},$$

the length of the derived *sentence*, not of the derivation. Put $\theta_G := \max_{R\in P} \theta_G(R)$, $\bar{\theta}_G := \max\{1, \theta_G\}$ and $t_G := \max_{A\in V} \min\{\|\vec{w}\|_1 : \vec{w} \in L_A(G)\}$ (rule and nonterminal thickness; θ_G is the thickness of [14, Section 4.3]).

If a target L contains ε, every statement below is applied to $L \cap \Sigma^+$ and the datum ε is added, as in Theorem 3.13; this adds one to the encoded length.

Lemma 4.1 (Thickness lemma)**.** *Let G be reduced and good.*

(i) *Every rule writes at most θ_G terminal symbols explicitly.*

(ii) *$t_G \le \theta_G$: every $A \in V$ has some $\vec{u} \in L_A(G)$ with $\|\vec{u}\|_1 \le \theta_G$.*

(iii) *$\theta_G \le 2\|G\| \max\{1, t_G\}$.*

(iv) *Every $A \in V$ of fan-out d has a context $E_A \in \mathcal{E}_d$ with $|E_A|_{\mathrm{fix}} \le \theta_G$ that results from a successful derivation by deleting one occurrence of an A-subtree; $E_S = \square_1$, and $E_A[\vec{u}] \in L(G)$ for all $\vec{u} \in L_A(G)$. For every typed nonterminal $X = A_{\mathbf{t}}$, the choice $\chi(X) := E_A$ meets Definition 3.9 and is* derivational *for X: it results from a successful derivation of $\widetilde{G}_0$ by deleting an occurrence of an X-subtree.*

Thus θ_G and t_G are polynomially equivalent on reduced good presentations; Yoshinaka's efficiency criterion measures data in target size and rule thickness [14, Definition 1]; see also [10].

For a finite-monoid morphism $\lambda : \Sigma^* \to \mathsf{N}$ we write $s \preceq \lambda$ if $\ker(\lambda|_{\Sigma^+}) \subseteq \ker s$; since $\widehat{s}$ separates ε, this is the relation of Definition 2.3 whenever λ also separates ε from Σ^+. For $\beta \ge 0$ consider

($\mathrm{R}^{\lambda}_{\beta}$) for every $B \in V$ and $\vec{u} \in L_B(G)$ there is $\vec{u}' \in L_B(G)$ with $\lambda^{(\mathrm{fan}(B))}(\vec{u}') = \lambda^{(\mathrm{fan}(B))}(\vec{u})$ and $\|\vec{u}'\|_1 \le \beta$.

Put

$$C_s(G) := |V||\mathsf{S}^\sharp|^f + |P||\mathsf{S}^\sharp|^{f r_{\max}} \;\le\; |\mathsf{S}^\sharp|^{f r_{\max}}\, \|G\|,$$

using $|V| + |P| \le \|G\|$; it bounds the numbers of typed nonterminals and typed rules together, since a typed rule is determined by its source rule and the type vectors of its at most $r_{\max}$ children.

Lemma 4.2 (Witness length)**.** *Let $\lambda : \Sigma^* \to \mathsf{N}$ be a finite-monoid morphism with $s \preceq \lambda$.*

(i) (Type-preserving exchange.) *If $X = B_{\mathbf{t}}$ is a typed nonterminal, $\vec{u} \in L_X$, $\vec{u}' \in L_B(G)$ and $\lambda^{(\mathrm{fan}(B))}(\vec{u}') = \lambda^{(\mathrm{fan}(B))}(\vec{u})$, then $\vec{u}' \in L_X$. Hence under ($\mathrm{R}^{\lambda}_{\beta}$) there are canonical tuples with $\|\omega(X)\|_1 \le \beta$ for every X.*

(ii) (Length.) *Let $\chi(X) := E_B$ for $X = B_{\mathbf{t}}$, as in Lemma 4.1(iv), and let ι assign to every typed X a tuple $\iota(X)$ of nonempty words of its arity with $\|\iota(X)\|_1 \le b$. Let W_ι consist of the words $\chi(X)[\iota(X)]$ and $\chi(X)[\rho(\iota(Y_1), \dots, \iota(Y_q))]$ for typed rules $X \to \rho(Y_1, \dots, Y_q)$, $q \ge 0$; thus $\mathsf{W}_\omega = \mathsf{W}(\widetilde{G}_0)$. Every word of W_ι has length at most $2\bar{\theta}_G + r_{\max} b$, and*

$$|\mathsf{W}_\iota| \le C_s(G), \qquad \|\mathsf{W}_\iota\|_+ \le C_s(G)\,\big(1 + 2\bar{\theta}_G + r_{\max} b\big).$$

(iii) (Refined thickness.) *In (ii), each rule word* $\chi(X)[\rho(\omega(Y_1),\dots,\omega(Y_q))]$ *is the yield of a successful derivation of* $\widetilde{G}_0$ *using the typed rule* $X \to \rho(Y_1,\dots,Y_q)$*; hence* $\theta_{\widetilde{G}_0} \le 2\bar{\theta}_G + r_{\max} b$.

(iv) (Characteristic sample.) *Let* $L(G) = L \cap \Sigma^+$ *for some* $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$*, and in* (ii) *let* $\iota(X) \in \mathrm{Cl}_{L,s}(X)$ *for every* X *(Lemma 3.8), for instance* $\iota = \omega$*. Then* $\mathsf{W}_\iota \subseteq L$*, and* $\mathsf{W}_\iota \cup (\{\varepsilon\} \cap L)$ *is a characteristic sample for* $\mathcal{B}^{\le r_{\max}}_{f,s}$ *and for* $\mathcal{B}_{f,s}$.

Items (i)–(iii) of Lemma 4.2 do not use substitutability.

4.2 Typed Thickness and Characteristic Data

Recall that $\widetilde{G}_0$ is the trimmed full yield-type refinement of G (Definition 3.5). Define

$$t_{\mathrm{typ}}(G,s) := \max_X \min_{\vec{u} \in L_X} \|\vec{u}\|_1,$$

where X ranges over the non-start typed nonterminals, and let $N_{\mathrm{typ}}(G,s)$ denote their number plus the number of non-start typed rules. In particular $N_{\mathrm{typ}} \le C_s(G) \le |\mathsf{S}^\sharp|^{f r_{\max}} \|G\|$.

Theorem 4.3 (Characteristic data from typed thickness)**.** *Let* $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$ *and* G *be a reduced good fan-out-*$\le f$*, rank-*$\le r_{\max}$ *grammar with* $L(G) = L \cap \Sigma^+$*. Take shortest canonical tuples and the derivational contexts* E_B *of Lemma 4.1(iv). Then* $K = \mathsf{W}(\widetilde{G}_0) \cup (\{\varepsilon\} \cap L)$ *is a characteristic sample for both* $\mathcal{B}^{\le r_{\max}}_{f,s}$ *and* $\mathcal{B}_{f,s}$*, with*

$$|\mathsf{W}(\widetilde{G}_0)| \le N_{\mathrm{typ}}, \qquad \|\mathsf{W}(\widetilde{G}_0)\|_+ \le N_{\mathrm{typ}}(1 + 2\bar{\theta}_G + r_{\max} t_{\mathrm{typ}}).$$

For the full sample K*, the length bound has the additional term* $\mathbf{1}_{\{\varepsilon \in L\}}$.

The numerical length estimate holds for any typing; substitutability is used only for the characteristic-sample claim.

For the trivial typing, every nonterminal of the ε-free G has one realized type vector, so $t_{\mathrm{typ}} = t_G$ and $N_{\mathrm{typ}} = |V| + |P| \le \|G\|$. The theorem recovers the $O(\|G\|\bar{\theta}_G)$ untyped scale, although Yoshinaka's original sample has the sharper bounds $|K_G| \le |P|$ and $\|K_G\| \le |P|\theta_G$ [14, Lemma 7 and Theorem 1]. A nontrivial typing may instead make t_{typ} exponential.

Remark 4.4 (Typed rule thickness). Let $\theta_{\mathrm{typ}}(G,s) := \max\{1, \theta_{\widetilde{G}_0}\}$ be the rule thickness of the trimmed refinement itself. Every rule of G has a retained typed copy, and erasing typed labels from a successful derivation recovers a G-derivation, so $\theta_G \le \theta_{\mathrm{typ}}$. Moreover $t_{\mathrm{typ}} \le \theta_{\mathrm{typ}} \le 2\bar{\theta}_G + r_{\max} t_{\mathrm{typ}}$: the first inequality is Lemma 4.1(ii) for the reduced good G_0, and the second is Lemma 4.2(iii) with shortest canonical tuples, which covers the start links as well. So the bound of Theorem 4.3 may equally be read in terms of the thickness of the refined grammar.

Presentation-independent lower bounds from rare types. For $L \subseteq \Sigma^*$ and $m \in \mathsf{S}$ put

$$\mu_m(L) := \min\{|w| : w \in L \cap \Sigma^+,\ s(w) = m\}, \qquad (\min \emptyset := \infty).$$

For a nonempty set of realized types $\mathsf{S}_L := s(L \cap \Sigma^+)$, put $\mu^\star(L,s) := \max_{m \in \mathsf{S}_L} \mu_m(L)$; set $\mu^\star(L,s) = 0$ when $\mathsf{S}_L = \emptyset$. A typed start nonterminal $S_{(m)}$ exists for each $m \in \mathsf{S}_L$, and $L_{S_{(m)}} = \{(w) : w \in L \cap \Sigma^+,\ s(w) = m\}$ (Proposition 3.6). Thus, whenever $L \cap \Sigma^+ \ne \emptyset$,

$$\mu^\star(L,s) \le t_{\mathrm{typ}}(G,s) \qquad \text{for every reduced good presentation } G. \tag{4.1}$$

Lemma 4.5 (Deleting an entire yield type)**.** *If $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$ and $m \in \mathsf{S}$, then $L^{\setminus m} := L \setminus s^{-1}(m)$ also belongs to $\mathcal{C}^{\mathrm{mcf}}_{f,s}$. Moreover, $L \setminus \{\varepsilon\} \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$.*

Proof. Apply Proposition 2.9(iii) with the trivial group and $\mathsf{Acc} = \mathsf{S}^\sharp \setminus \{m\}$, respectively $\mathsf{Acc} = \mathsf{S}$. □

All lower bounds of this paper rest on one principle.

Lemma 4.6 (Distinguishing sub-targets)**.** *Let $L, L' \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$ with $L' \subsetneq L$. Every characteristic sample of L for $\mathcal{B}^{\leq r}_{f,s}$ ($r \geq 1$) or $\mathcal{B}_{f,s}$, and every characteristic sample of L for a set-driven operator having one for L' as well, contains a word of $L \setminus L'$.*

Proof. Let $F \subseteq L'$ be finite. For the canonical operators, Proposition 3.10 applied to the target L' gives $L(\mathcal{B}(F)) \subseteq L' \neq L$, so F is not characteristic for L. For the other operator, let F' be characteristic for L'; then $F \cup F' \subseteq L' \subseteq L$, so if F were characteristic for L, the output on $F \cup F'$ would be both L and L'. □

Proposition 4.7 (Rare types force positive data)**.** *Let $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$, and let $m \in \mathsf{S}_L$. Every characteristic sample of L for $\mathcal{B}^{\leq r}_{f,s}$ ($r \geq 1$) or for $\mathcal{B}_{f,s}$ contains a word of type m. The same is true for any set-driven operator having characteristic samples for both L and $L^{\setminus m}$, in particular for an operator having characteristic samples for the entire class $\mathcal{C}^{\mathrm{mcf}}_{f,s}$. If $\varepsilon \in L$, its sample for either canonical operator must also contain ε; the same holds for a set-driven operator admitting a characteristic sample for $L \setminus \{\varepsilon\}$ as well. Consequently, for the canonical operators and for class-wide set-driven operators, every characteristic sample F satisfies*

$$\|F\|_+ \geq \sum_{m \in \mathsf{S}_L} \big(\mu_m(L) + 1\big) + \mathbf{1}_{\{\varepsilon \in L\}}.$$

Corollary 4.8 (Characteristic-data sandwich)**.** *Suppose $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$, $L \cap \Sigma^+ \neq \emptyset$, and G is a reduced good fan-out-$\leq f$, rank-$\leq r_{\max}$ presentation of $L \cap \Sigma^+$. Write $\mathrm{cs}^{\leq r_{\max}}_{f,s}(L)$ for the minimum total encoded length of a characteristic sample for $\mathcal{B}^{\leq r_{\max}}_{f,s}$. Then*

$$\sum_{m \in \mathsf{S}_L} \big(\mu_m(L) + 1\big) + \mathbf{1}_{\{\varepsilon \in L\}} \;\leq\; \mathrm{cs}^{\leq r_{\max}}_{f,s}(L) \;\leq\; N_{\mathrm{typ}}\big(1 + 2\bar{\theta}_G + r_{\max} t_{\mathrm{typ}}(G,s)\big) + \mathbf{1}_{\{\varepsilon \in L\}}.$$

The lower bound is independent of the presentation and applies to every set-driven operator with characteristic samples throughout $\mathcal{C}^{\mathrm{mcf}}_{f,s}$.

Proposition 4.9 (Exponential blow-up of full yield-type refinement)**.** *Let $\Sigma = \{\mathtt{a}, \mathtt{c}\}$ and let $s_{\mathtt{c}}$ be the typing onto $U_1 = \{1, \zeta\}$ ($\zeta^2 = \zeta$, Section 7) with $s_{\mathtt{c}}(\mathtt{a}) = 1$ and $s_{\mathtt{c}}(\mathtt{c}) = \zeta$. For $n \geq 1$ the fan-out-one, rank-two grammar*

$$S \to SS \mid \mathtt{a} \mid \mathtt{c} \mid A_0 \mid \cdots \mid A_n, \qquad A_0 \to \mathtt{a}, \qquad A_i \to A_{i-1}A_{i-1} \mid \mathtt{c} \quad (1 \leq i \leq n)$$

is reduced and good, generates Σ^+, and has $\|G_n\| = O(n)$ and $\theta_{G_n} \leq 2$, but $t_{\mathrm{typ}}(G_n, s_{\mathtt{c}}) \geq 2^n$ and $\theta_{\widetilde{G}_{n,0}} \geq 2^n$.

This blow-up concerns the witness construction, not the data, because the tuples used for a typed nonterminal need not be its yields. For a typed nonterminal X, Lemma 3.8 places L_X in one class $\mathrm{Cl}_{L,s}(X)$; the *representative length* is

$$\tau_X := \min\{\|\vec{v}\|_1 : \vec{v} \in \mathrm{Cl}_{L,s}(X)\}, \qquad \tau(G,s) := \max_X \tau_X \quad (\max \emptyset := 0).$$

Corollary 4.10 (Characteristic data from equivalent representatives)**.** *Under the hypotheses of Theorem 4.3,*

$$\mu^\star(L,s) \le \tau(G,s) \le t_{\mathrm{typ}}(G,s),$$

and L has a characteristic sample K for both $\mathcal{B}_{f,s}^{\le r_{\max}}$ and $\mathcal{B}_{f,s}$ with

$$\|K\|_+ \le N_{\mathrm{typ}}\bigl(1 + 2\bar{\theta}_G + r_{\max}\tau(G,s)\bigr) + \mathbf{1}_{\{\varepsilon\in L\}}.$$

In Proposition 4.9, $L = \Sigma^+$ and every admissible context accepts every tuple, so two tuples are equivalent exactly when their types agree; single letters represent both types, and $\tau(G_n, s_{\mathsf{c}}) = 1$ although $t_{\mathrm{typ}}(G_n, s_{\mathsf{c}}) \ge 2^n$. Corollary 4.10 then gives characteristic data of length $O(\|G_n\|)$. The corollary does not bound τ itself; the proofs below bound t_{typ} and therefore work on G.

5 Upper Bound: Local Groups at Every Fan-Out

This section proves Theorem A. The difficulty specific to multiple context-free grammars is that a subtree contributes several blocks to different root components, interleaved with blocks of other subtrees; for noncommutative groups the effect of replacing several subtrees at once depends on this order.

5.1 Short Representatives by Minimal Counterexamples

Let $\lambda : \Sigma^* \to \mathsf{N}$ be a finite-monoid morphism. A vector $\mathbf{t} \in \mathsf{N}^d$ is *realizable at B* (of fan-out d) if $\mathbf{t} = \lambda^{(d)}(\vec{u})$ for some $\vec{u} \in L_B(G)$. A *λ-minimal* derivation for $(B, \mathbf{t})$ is a terminal derivation from B with root λ-vector $\mathbf{t}$ that minimizes first the total yield length and then the number of nodes. For a threshold b, a node is *heavy* if its yield has total length greater than b.

Let $n_\lambda(G)$ be the number of pairs $(C, \lambda^{(\mathrm{fan}(C))}(\vec{u}))$ with $C \in V$ and $\vec{u} \in L_C(G)$, the *realized labels*; thus $n_\lambda(G) \le |V||\mathsf{N}|^f$.

Lemma 5.1 (Heavy-node count and shortening principle)**.** (i) *Let $b \ge \bar{\theta}_G$ and $c \ge 0$, and let T be λ-minimal for $(B, \mathbf{t})$. If every antichain of heavy nodes of T has at most c elements, then*

$$\|\textit{root tuple of } T\|_1 \le (r_{\max}+1)\, b\, \max\{1, c\, n_\lambda(G)\}.$$

(ii) (Shortening principle.) *Let $c, c' \ge 0$ and $b = \bar{\theta}_G$. Suppose that for every λ-minimal T and every antichain Q of heavy nodes of T there is $Q_0 \subseteq Q$ with $|Q \setminus Q_0| \le c'$ such that, if $|Q_0| > c$, the subtrees at the nodes of some nonempty $J \subseteq Q_0$ can be replaced simultaneously by derivations, from the same nonterminals, of tuples of total length at most $\bar{\theta}_G$ without changing the root λ-vector. Then every heavy antichain of a λ-minimal derivation has at most $c + c'$ elements, and $(\mathrm{R}_\beta^\lambda)$ holds with $\beta = (r_{\max}+1)\,\bar{\theta}_G \max\{1, (c+c')\, n_\lambda(G)\}$.*

Proof. (i) Label each node by its nonterminal and the λ-vector of its yield; only realized labels occur. If an ancestor and a proper descendant had the same label, replacing the ancestor's subtree by the descendant's would keep the root λ-vector (templates evaluate λ homomorphically), would not lengthen the root yield (by linearity the descendant's yield is part of the ancestor's), and would remove nodes, contradicting minimality. So every root-to-leaf path has at most $n_\lambda(G)$ nodes. If the root is light its yield has length at most b. Otherwise the heavy nodes form an ancestor-closed subtree whose leaves form an antichain, so there are at most $c\, n_\lambda(G)$ heavy nodes. Every terminal of the root yield is written by a heavy node (at most $\bar{\theta}_G \le b$ each, Lemma 4.1(i)) or lies in a light child of a heavy node (at most $r_{\max}$ children of length at most b).

(ii) If $|Q_0| > c$, the replacement gives a derivation from B (the nodes of J are incomparable) with root λ-vector $\mathbf{t}$ and a strictly shorter root yield, since every replaced yield had total length greater than $\bar{\theta}_G$ and, by linearity and nondeletion, occurs exactly once in the root yield. This contradicts minimality, so $|Q| \le c + c'$. For $\vec{u} \in L_B(G)$ apply (i) to a λ-minimal derivation for $(B, \lambda^{(\mathrm{fan}(B))}(\vec{u}))$. □

Group constants. For a finite abelian group $\mathbb{H}$, $\mathsf{d}(\mathbb{H})$ is the maximal length of a zero-sum-free sequence over $\mathbb{H}$ (one with no nonempty subsequence summing to the identity; thus $\mathsf{d}(\mathbb{H}) + 1$ is the Davenport constant; $\mathsf{d}(\{1\}) = 0$). For a finite group $\mathbb{G}$ and $f \ge 1$ put

$$N_{\mathbb{G},f} = \max\{3, |\mathbb{G}|^f\}, \quad \mathcal{P}_f = f^{f+1} f!, \quad M_{\mathbb{G},f} = \mathcal{P}_f \, N_{\mathbb{G},f}^{2^{f^2}},$$

$$c_{\mathbb{G},f} := \begin{cases} \mathsf{d}(\mathbb{G}^f), & \mathbb{G} \text{ abelian}, \\ |\mathbb{G}| - 1, & \mathbb{G} \text{ nonabelian}, \ f = 1, \\ M_{\mathbb{G},f} - 1, & \text{otherwise}. \end{cases}$$

$\mathcal{P}_f$ bounds the number of block patterns of one owner, and $N_{\mathbb{G},f} \ge 3$ is needed by the order classification in Appendix A.1. The double exponent 2^{f^2} comes from applying the Erdős–Szekeres-type extraction of Lemma A.8 once for each of up to f^2 ordered pairs of block positions; we do not know whether it is necessary. For abelian groups no order information is needed, and $c_{\mathbb{G},f} = \mathsf{d}(\mathbb{G}^f)$ is linear in f (Remark A.3). At fan-out one every owner has a single block on a single row, no homogenization is needed, and the prefix-product pigeonhole alone gives $|\mathbb{G}| - 1$ (Lemma A.2(c)); since $\mathsf{d}(\mathbb{G}) \le |\mathbb{G}| - 1$, the abelian value is never larger.

Simultaneous block replacement. A *block-replacement family* of width f over $\mathbb{G}$ (Definition A.1) consists of $d \le f$ rows, each an ordered product in $\mathbb{G}$ of fixed constants and *blocks*; each block has an *owner* in $[m] := \{1, \dots, m\}$, an original value and a replacement value, and each owner owns at most f blocks, which may interleave with other owners' blocks arbitrarily. $\mathrm{val}_\varrho(J)$ is the value of row ϱ when exactly the blocks of owners in $J \subseteq [m]$ take their replacement values.

Proposition 5.2 (Simultaneous block replacement)**.** *In a block-replacement family of width f over a finite group $\mathbb{G}$ with $m > c_{\mathbb{G},f}$ owners there is a nonempty $J \subseteq [m]$ with $\mathrm{val}_\varrho(J) = \mathrm{val}_\varrho(\varnothing)$ for every row ϱ.*

Proof. This is Lemma A.2. For abelian $\mathbb{G}$ one selects a nonempty zero-sum subsequence of the owners' difference vectors in $\mathbb{G}^d \le \mathbb{G}^f$. At $f = 1$ a pigeonhole on prefix-product states of the single row selects an interval of owners. For arbitrary $\mathbb{G}$ and f no factors are commuted: among owners with a common block pattern (at most $\mathcal{P}_f$ patterns) one homogenizes all pairwise order comparisons (Lemma A.8), which decomposes every row into consecutive rank chunks (Lemma A.4); a pigeonhole on prefix-product boundary states then yields an interval of owners preserving all ordered row products (Lemma A.5). □

Individual block or owner values need not be preserved by the replacement, and owner contributions cannot be commuted: if the two rows are $\mathfrak{b}_{1,1}\mathfrak{b}_{2,1}$ and $\mathfrak{b}_{2,2}\mathfrak{b}_{1,2}$, with value (12) on the blocks of owner 1 and (23) on those of owner 2 in S_3, the row values are $(12)(23) \ne (23)(12)$. Example A.6 shows a replacement preserving rows of opposite owner orientation.

Lemma 5.3 (Short representatives for window–group invariants)**.** *Fix $k, \ell \geq 0$, a finite group $\mathbb{G}$ (possibly trivial) and a morphism $g : \Sigma^* \to \mathbb{G}$. Put $\lambda := (\widehat{s}_{k,\ell}, g) : \Sigma^* \to \mathsf{N} := M_{k,\ell} \times \mathbb{G}$. Then $(\mathrm{R}^{\lambda}_{\beta})$ holds with*

$$\beta = b_{k,\ell,\mathbb{G}} := (r_{\max}+1)\,\bar{\theta}_G\,(c_{\mathbb{G},f}+f(k+\ell+1))\,|V||\mathsf{N}|^f \;\leq\; (r_{\max}+1)(c_{\mathbb{G},f}+f(k+\ell+1))|\mathsf{N}|^f\,\|G\|\,\bar{\theta}_G.$$

Proof. We verify the hypothesis of Lemma 5.1(ii) with $c = c_{\mathbb{G},f}$ and $c' = f(k+\ell+1)$; since $c + c' \geq 1$ and $n_\lambda(G) \leq |V||\mathsf{N}|^f$, it gives the bound, and the second inequality uses $|V| \leq \|G\|$. Let T be λ-minimal with heavy threshold $b = \bar{\theta}_G$. In each root component of length at least $m_0 = \max\{1, k+\ell\}$ mark the first k and the last ℓ positions (one arbitrary position if $k = \ell = 0$), and in each shorter component mark all positions: at most c' marks. Let Q be an antichain of heavy nodes. Incomparable subtrees have disjoint yields, so at most c' nodes of Q contain a mark; let Q_0 be the others.

Let $|Q_0| > c_{\mathbb{G},f}$. For $v \in Q_0$ with nonterminal C_v fix a tuple of $L_{C_v}(G)$ of total length at most $\bar{\theta}_G$ (Lemma 4.1(ii)). By linearity and nondeletion each $v \in Q_0$ contributes at most f disjoint contiguous blocks to the root components, and the g-value of each root component is the ordered product of the g-values of fixed material and of these blocks. This is a block-replacement family over $\mathbb{G}$ with owners Q_0, so Proposition 5.2 gives a nonempty $J \subseteq Q_0$ whose simultaneous replacement preserves every root g-value. The replaced blocks contain no mark: short root components are untouched, and in a long component they lie strictly between the marked prefix and suffix, which keep their letters and positions relative to the ends, while the component keeps its $k+\ell$ marked letters (at least one if $k = \ell = 0$) and stays long. So the root $\widehat{s}_{k,\ell}$-vector is preserved as well, and with it the root λ-vector. □

Remark 5.4 (Window representatives). For trivial $\mathbb{G}$ one has $c_{\mathbb{G},f} = 0$ and $\mathsf{N} \cong M_{k,\ell}$, so $(\mathrm{R}^{\widehat{s}_{k,\ell}}_{\beta})$ holds with $\beta = (r_{\max}+1)\bar{\theta}_G\, f(k+\ell+1)\,|V||M_{k,\ell}|^f$ for window typings.

5.2 Window and Window–Group Typings: Explicit Constants

Window–group typings are local-group typings (Proposition B.6(i)), so Theorem A covers them; this subsection is the base case of the replacement argument, with explicit constants. A typing s is a *window–group typing* if $\widehat{s} \preceq (\widehat{s}_{k,\ell}, g)$ for some $k, \ell \geq 0$, a finite group $\mathbb{G}$ and a morphism $g : \Sigma^* \to \mathbb{G}$; for trivial $\mathbb{G}$ this means that s is a window typing.

Proposition 5.5 (Polynomial characteristic data for window–group typings)**.** *Let $\mathbb{G}$ be a finite group (possibly trivial), $g : \Sigma^* \to \mathbb{G}$ a morphism, and $\widehat{s} \preceq (\widehat{s}_{k,\ell}, g)$; let $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$ and $L(G) = L \cap \Sigma^+$. Then $\mathsf{W}(\widetilde{G}_0) \cup (\{\varepsilon\} \cap L)$ is a characteristic sample for both $\mathcal{B}^{\leq r_{\max}}_{f,s}$ and $\mathcal{B}_{f,s}$ with $|\mathsf{W}(\widetilde{G}_0)| \leq C_s(G) = O(\|G\|)$ and*

$$\|\mathsf{W}(\widetilde{G}_0)\|_+ \leq K_{k,\ell,\mathbb{G}}\,\|G\|^2\bar{\theta}_G,$$

$$K_{k,\ell,\mathbb{G}} := |\mathsf{S}^\sharp|^{f r_{\max}}\big(3 + r_{\max}(r_{\max}+1)(c_{\mathbb{G},f} + f(k+\ell+1))(|M_{k,\ell}||\mathbb{G}|)^f\big).$$

Proof. Lemma 5.3 and Lemma 4.2(i),(ii),(iv) with $\lambda = (\widehat{s}_{k,\ell}, g)$ and $b = b_{k,\ell,\mathbb{G}}$ give the sample and $\|\mathsf{W}\|_+ \leq C_s(G)(1 + 2\bar{\theta}_G + r_{\max} b_{k,\ell,\mathbb{G}})$; insert $C_s(G) \leq |\mathsf{S}^\sharp|^{f r_{\max}}\|G\|$, $1 + 2\bar{\theta}_G \leq 3\|G\|\bar{\theta}_G$ and $|V| \leq \|G\|$. □

The group may be noncommutative and its blocks may interleave across components. For trivial $\mathbb{G}$, i.e. $s \preceq s_{k,\ell}$, one has $c_{\mathbb{G},f} = 0$.

Remark 5.6 (Abelian groups). If $\mathbb{G}$ is abelian, Lemma A.2(a) gives $c_{\mathbb{G},f} = \mathsf{d}(\mathbb{G}^f)$, and $\mathsf{d}(\mathbb{G}^f) = O(\exp(\mathbb{G})\, f \log|\mathbb{G}|)$ (Remark A.3; $\exp(\mathbb{G})$ is the least common multiple of the element orders), so

$K_{k,\ell,\mathbb{G}}$ is polynomial in f apart from the factors $|\mathsf{S}^\sharp|^{f r_{\max}}$ and $(|M_{k,\ell}||\mathbb{G}|)^f$. For nonabelian $\mathbb{G}$ and $f \geq 2$, by contrast, $c_{\mathbb{G},f} = M_{\mathbb{G},f} - 1$ grows doubly exponentially in f.

At fan-out one, polynomial time and thick-data learning for fixed windows is due to Yoshinaka [12, Theorem 1]; the bounds above are relative to a reduced good presentation and are not claimed to improve on that result after normalization. What is new is that window and ordered group values are preserved *simultaneously* across up to f interleaved components.

5.3 Local-Group Typings

LI is the pseudovariety of finite semigroups T satisfying $e\mathsf{T}e = \{e\}$ for every idempotent $e \in \mathsf{T}$, **Grp** that of finite groups, and $\mathbf{LI} \vee \mathbf{Grp}$ consists of the divisors of products $\mathsf{T} \times \mathbb{G}$ with $\mathsf{T} \in \mathbf{LI}$ and $\mathbb{G}$ a finite group [26, 27]. By Lemma B.5 (Appendix B), every typing with $\mathsf{S} \in \mathbf{LI} \vee \mathbf{Grp}$ is a window–group typing, with k, $\mathbb{G}$ and g depending only on s, so Proposition 5.5 applies to it.

A finite semigroup is a *local group*, $\mathsf{T} \in \mathbf{LG}$, if $e\mathsf{T}e$ is a group for every idempotent $e \in \mathsf{T}$. A typing is a *local-group typing* if its positive image S is a local group; the condition is never tested on $\mathsf{S}^\sharp$. By Proposition B.6, $\mathbf{LI} \vee \mathbf{Grp} \subsetneq \mathbf{LG}$.

Definition 5.7 (Sliding-factor morphism)**.** Fix $k \geq 1$, a finite group $\mathbb{G}$ and a map $\nu : \Sigma^{k+1} \to \mathbb{G}$ (the *sliding-factor labelling*). For $|w| \geq 2k$ let $\mathrm{fac}_k(w) = (F_1, \ldots, F_{|w|-k})$ be the factors of length $k+1$ of w in order of their starting positions, and put $\gamma_{k,\nu}(w) := \nu(F_1)\nu(F_2)\cdots\nu(F_{|w|-k})$ (ordered product in $\mathbb{G}$); put $\gamma_{k,\nu}(w) := 1_{\mathbb{G}}$ for $|w| < 2k$. The *sliding-factor morphism* is $\lambda_{k,\nu}(w) := (\widehat{s}_{k,k}(w), \gamma_{k,\nu}(w))$; by Lemma B.1 it is a morphism onto a finite monoid $M_{k,\nu}$. The typing s is *local-group-determined* if $\widehat{s} \preceq \lambda_{k,\nu}$ for some k, $\mathbb{G}$ and ν.

A sliding-factor morphism can encode a group contribution determined by a sliding factor, rather than by each letter alone. For example, $\lambda_{1,\nu}$ with $\nu(\mathtt{ab}) = 1 \in \mathbb{Z}_2$ and zero on other pairs records the parity of occurrences of $\mathtt{ab}$. Even on $\{\mathtt{a}, \mathtt{b}\}$ this is not determined by any fixed window combined with a finite-group morphism (Proposition B.6(ii)). The local-group characterization below expresses this additional power without requiring a new learner or a target-specific typing.

Theorem 5.8 (Polynomial characteristic data for local-group typings)**.** *Let $\mathsf{S} \in \mathbf{LG}$. Every reduced good fan-out-$\leq f$, rank-$\leq r_{\max}$ grammar G satisfies $t_{\mathrm{typ}}(G, s) = O_{f,r_{\max},s,\Sigma}(\|G\|\bar{\theta}_G)$. If moreover $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$ and $L(G) = L \cap \Sigma^+$, then $\mathsf{W}(\widetilde{G}_0) \cup (\{\varepsilon\} \cap L)$ is a characteristic sample for both $\mathcal{B}^{\leq r_{\max}}_{\bar{f},s}$ and $\mathcal{B}_{f,s}$ with*

$$|\mathsf{W}(\widetilde{G}_0)| = O(\|G\|), \qquad \|\mathsf{W}(\widetilde{G}_0)\|_+ = O_{f,r_{\max},s,\Sigma}(\|G\|^2\bar{\theta}_G).$$

Proof. By Theorem B.4, $\widehat{s} \preceq \lambda := \lambda_{k,\nu}$ for $k = |\mathsf{S}|$ and some $\mathbb{G}, \nu$ depending only on s. Lemma B.8 gives $(\mathrm{R}^{\lambda}_{\beta})$ with $\beta = \beta_\lambda = O(\|G\|\bar{\theta}_G)$: heavy subtrees are replaced by arbitrary short yields of their nonterminals, after the owners have been thinned so that their blocks lie at least $2k$ letters apart and each block has been enlarged by k unchanged letters on both sides (Lemma B.7); the enlarged blocks keep their window types, so the ordered sliding-factor products are controlled by Lemma B.2 and Proposition 5.2. This step uses no substitutability. Lemma 4.2(i) gives $t_{\mathrm{typ}} \leq \beta_\lambda$, and Theorem 4.3 gives $\|\mathsf{W}\|_+ \leq N_{\mathrm{typ}}(1 + 2\bar{\theta}_G + r_{\max}\beta_\lambda) = O(\|G\|^2\bar{\theta}_G)$. □

The Rees and sliding-factor coordinates are classical; their combination with owner-wise block shortening is the quantitative step. Local-group and window–group typings thus have the same quadratic dependence on $\|G\|$; Proposition 5.5 gives explicit constants for the latter.

6 Lower Bound: The Linear Factor at Every Fan-Out

This section proves Theorem B.

The upper side of Corollary 4.8 grows with $\|G\|$ and the lower side does not. Nevertheless the product $\|G\|\bar{\theta}_G$ is necessary at every fan-out, under every typing over an alphabet of at least $4f+1$ letters (two at fan-out one) and for presentations within a constant factor of the smallest.

Proposition 6.1 (The linear factor is necessary at every fan-out)**.** *Let $f \geq 1$, let Σ contain $4f+1$ distinct letters $\mathtt{a}_1, \ldots, \mathtt{a}_{2f}, \mathtt{d}_1, \ldots, \mathtt{d}_{2f}, \mathtt{z}$, and let s be any typing. For integers $q \geq n \geq 1$ put $W := 2fq + 2^n + 2f$ and*

$$w_i := \mathtt{a}_1^i \mathtt{d}_1 \mathtt{a}_2^i \mathtt{d}_2 \cdots \mathtt{a}_{2f}^i \mathtt{d}_{2f} \mathtt{z}^{2f(q-i)+2^n} \quad (0 \leq i \leq q), \qquad L_{q,n}^{(f)} := \{w_0, \ldots, w_q\}.$$

(i) *Every subset of $L_{q,n}^{(f)}$ belongs to $\mathcal{C}_{f,s_{\mathrm{triv}}}^{\mathrm{mcf}} \subseteq \mathcal{C}_{f,s}^{\mathrm{mcf}}$.*

(ii) *For $\mathcal{B}_{f,s}$, for $\mathcal{B}_{f,s}^{\leq r}$ ($r \geq 1$), and for every set-driven operator with characteristic samples throughout $\mathcal{C}_{f,s}^{\mathrm{mcf}}$, the only characteristic sample of $L_{q,n}^{(f)}$ is $L_{q,n}^{(f)}$ itself.*

(iii) *$L_{q,n}^{(f)}$ has a reduced good presentation $G_{q,n}^{(f)}$ of fan-out f and rule rank two with $\|G_{q,n}^{(f)}\| = (7f+11)q + 6n + 3f + 10 \leq (7f+17)(q+1)$ and $\bar{\theta}_{G_{q,n}^{(f)}} = W$. Every reduced good fan-out-$\leq f$ presentation G of $L_{q,n}^{(f)}$ has $\bar{\theta}_G = W$ and $\|G\| \geq q+1$.*

(iv) *For $f = 1$,* (i)–(iii) *hold over any alphabet containing two letters $\mathtt{a}, \mathtt{d}$, for the words $w_i = \mathtt{a}^i\mathtt{d}\mathtt{a}^i\mathtt{d}\mathtt{a}^{2(q-i)+2^n}$ obtained by identifying $\mathtt{a}_1, \mathtt{a}_2, \mathtt{z}$ with $\mathtt{a}$ and $\mathtt{d}_1, \mathtt{d}_2$ with $\mathtt{d}$, and the grammar identified in the same way.*

Consequently every characteristic sample F in (ii) *satisfies*

$$\|F\|_+ = (q+1)(W+1) \;\geq\; \tfrac{1}{7f+17}\, \|G_{q,n}^{(f)}\|\, \bar{\theta}_{G_{q,n}^{(f)}}.$$

In w_i the position of one marker $\mathtt{d}_h$, and the distance of two, determine i. Let a d-context be shared by two different words. Its fixed part carries at most $d-1$ of the $2f$ markers: a marker in the leftmost fixed word would have the same position in both words, a marker in the rightmost one the same distance from the end (all words have one length), and two markers in one fixed word the same distance; each determines the word. So the holes carry at least $2f - d + 1 \geq d + 1$ markers, some component of each tuple carries two, and it identifies the word. With $q = n$ one has $\|G_{n,n}^{(f)}\| = \Theta(n)$ and $\bar{\theta}_{G_{n,n}^{(f)}} = 2^n + \Theta(n)$; with n fixed, $\bar{\theta}_{G_{q,n}^{(f)}} = \Theta(\|G_{q,n}^{(f)}\|)$. By (iii) the presentations are within a constant factor of the smallest reduced good ones, so the bound is not obtained by padding G; with the trivial typing the proposition shows that Yoshinaka's bound $\|K_G\| \leq |P|\theta_G$ is attained up to a constant factor. All yields of a nonterminal of $G_{q,n}^{(f)}$ have one total length, so $t_{\mathrm{typ}} = \bar{\theta}_G = W$ for this family, and it says nothing about the further factor $\|G\|$ in Proposition 5.5 and Theorem 5.8. The proof is in Appendix G.

Corollary 6.2 (The trivial typing)**.** *Fix f and $r_{\max} \geq 1$. For the trivial typing, every $L \in \mathcal{C}_{f,s_{\mathrm{triv}}}^{\mathrm{mcf}}$ and every reduced good rank-$\leq r_{\max}$ presentation G of $L \cap \Sigma^+$, the operator $\mathcal{B}_{f,s_{\mathrm{triv}}}$ has a characteristic sample of total length $O(\|G\|\bar{\theta}_G)$, as for Yoshinaka's sample [14, Lemma 7 and Theorem 1]. Suppose moreover that $r_{\max} \geq 2$ and that Σ has at least $4f+1$ letters, or that $r_{\max} \geq 2$, $f = 1$ and Σ has at least two letters. Then this bound is attained up to a constant factor: for the targets $L_{q,n}^{(f)}$ and presentations $G_{q,n}^{(f)}$ of Proposition 6.1, every characteristic sample has length at least $\|G_{q,n}^{(f)}\|\bar{\theta}_{G_{q,n}^{(f)}}/(7f+17)$.*

Proof. Every typed nonterminal is an untyped one, so $t_{\mathrm{typ}}(G, s_{\mathrm{triv}}) \le \bar{\theta}_G$ by Lemma 4.1, and Theorem 4.3 gives $O(\|G\|\bar{\theta}_G)$. Proposition 6.1 gives the second assertion; the presentations $G^{(f)}_{q,n}$ have rule rank two, which is why $r_{\max} \ge 2$ is needed there. □

A characteristic sample stays characteristic when finitely many words of the target are added, so only the length of a *shortest* characteristic sample is meaningful; the corollary bounds it from above for every target and shows that the bound is attained by one family.

7 The Boundary at Fan-Out One

This section proves Theorem C.

A three-letter doubling family. Let $\mathtt{a}, \mathtt{b}, \mathtt{c}$ be distinct letters, let $U_1 = \{1, \zeta\}$ with identity 1 and $\zeta^2 = \zeta$, and let $s_U : \{\mathtt{a}, \mathtt{b}, \mathtt{c}\}^+ \twoheadrightarrow U_1$ send $\mathtt{a}, \mathtt{b}$ to 1 and $\mathtt{c}$ to ζ; thus $s_U(w) = 1$ exactly for the nonempty words without $\mathtt{c}$. For $n \ge 1$ let G_n be the fan-out-one grammar

$$S \to E_n, \qquad E_0 \to \mathtt{a}, \qquad E_i \to \mathtt{b}E_{i-1}E_{i-1}\mathtt{b} \mid \mathtt{c}E_{i-1}\mathtt{c} \quad (1 \le i \le n),$$

put $L^{(i)} := L_{E_i}(G_n)$, so $L^{(0)} = \{\mathtt{a}\}$ and $L^{(i)} = \mathtt{b}L^{(i-1)}L^{(i-1)}\mathtt{b} \cup \mathtt{c}L^{(i-1)}\mathtt{c}$, put $L_n := L^{(n)} = L(G_n)$, $W_0 := \mathtt{a}$ and $W_i := \mathtt{b}W_{i-1}W_{i-1}\mathtt{b}$. This is a three-letter variant of the bracketed doubling family of the companion CFG manuscript [23, Proposition 6.4], in which the opening and closing brackets are different letters; all its properties are proved here.

Proposition 7.1 (Three-letter doubling family)**.** *For every $n \ge 1$:*

(i) *G_n is reduced and good, of fan-out one and rule rank two, with $\|G_n\| = 13n + 7$ and $\bar{\theta}_{G_n} = 4n$.*

(ii) *L_n is Clark–Eyraud substitutable: for all $x, y \in \Sigma^+$, $\mathcal{D}^{(1)}_{L_n}(x) \cap \mathcal{D}^{(1)}_{L_n}(y) \ne \emptyset$ implies $\mathcal{D}^{(1)}_{L_n}(x) = \mathcal{D}^{(1)}_{L_n}(y)$. Hence $L_n \in \mathcal{C}^{\mathrm{mcf}}_{1,s}$ for every typing s on every alphabet $\Sigma \supseteq \{\mathtt{a}, \mathtt{b}, \mathtt{c}\}$; in particular $L_n \in \mathcal{C}^{\mathrm{mcf}}_{1,s_U}$.*

(iii) *W_n is the only word of L_n without $\mathtt{c}$, hence the only one of s_U-type 1, and $|W_n| = 3 \cdot 2^n - 2$.*

(iv) *Every characteristic sample F of L_n for $\mathcal{B}^{\le r}_{1,s_U}$ ($r \ge 1$) or $\mathcal{B}_{1,s_U}$, or for any set-driven operator having characteristic samples throughout $\mathcal{C}^{\mathrm{mcf}}_{1,s_U}$, contains W_n; hence $\|F\|_+ \ge 3 \cdot 2^n - 1$.*

The proof is in Appendix F: (iii) holds because a derivation without $\mathtt{c}$ is the complete binary tree, and (iv) follows from (iii) and Proposition 4.7.

Definition 7.2 (Uniform polynomial characteristic-data length)**.** Let $f, r_{\max} \ge 1$. A pseudovariety $\mathbf{V}$ of finite semigroups has *uniform polynomial characteristic-data length at fan-out f and rank $r_{\max}$* if for every nonempty finite alphabet Σ and every surjective typing $s : \Sigma^+ \twoheadrightarrow \mathsf{S}$ with $\mathsf{S} \in \mathbf{V}$ there is a polynomial $p = p_{f,r_{\max},s,\Sigma}$ such that for every $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$ and every reduced good fan-out-$\le f$, rank-$\le r_{\max}$ presentation G of $L \cap \Sigma^+$, the operator $\mathcal{B}^{\le r_{\max}}_{f,s}$ has a characteristic sample F for L with $\|F\|_+ \le p(\|G\|, \bar{\theta}_G)$.

The quantifiers are ordered as written: p may depend on f, $r_{\max}$, Σ and the typing s, but not on the target L or the presentation G.

Theorem 7.3 (Fan-out one: a pseudovariety boundary for data length)**.** *Let $r_{\max} \ge 2$ and let $\mathbf{V}$ be a pseudovariety of finite semigroups.*

(i) $\mathbf{V}$ *has uniform polynomial characteristic-data length at fan-out one and rank* $r_{\max}$ *if and only if* $\mathbf{V} \subseteq \mathbf{LG}$.

(ii) *If* $\mathbf{V} \subseteq \mathbf{LG}$, *one may take* $p(\|G\|, \bar{\theta}_G) = O_{r_{\max},s,\Sigma}(\|G\|^2 \bar{\theta}_G)$ *in Definition 7.2.*

(iii) *If* $\mathbf{V} \not\subseteq \mathbf{LG}$, *the typing* s_U *of Proposition 7.1 has positive image* $U_1 \in \mathbf{V}$, *and the targets* $L_n \in \mathcal{C}^{\mathrm{mcf}}_{1,s_U}$ *have reduced good rank-two presentations* G_n *with* $\|G_n\| = O(n)$ *and* $\bar{\theta}_{G_n} = O(n)$, *while every characteristic sample of* L_n *has total encoded length at least* $3 \cdot 2^n - 1$. *This lower bound holds for* $\mathcal{B}^{\leq r_{\max}}_{1,s_U}$, *for* $\mathcal{B}_{1,s_U}$, *and for every set-driven operator having characteristic samples throughout* $\mathcal{C}^{\mathrm{mcf}}_{1,s_U}$.

Proof. (ii), and hence the *if* direction of (i), is Theorem 5.8 with $f = 1$, since every $\mathsf{S} \in \mathbf{V}$ then lies in $\mathbf{LG}$. For (iii), pick $\mathsf{S} \in \mathbf{V} \setminus \mathbf{LG}$. By Theorem B.4(ii), S has a subsemigroup isomorphic to U_1, and $U_1 \in \mathbf{V}$ because pseudovarieties are closed under subsemigroups. The typing s_U maps onto U_1; the remaining claims are Proposition 7.1(i) and (iv), using $\|G_n\| = O(n)$ and $\theta_{G_n} \leq 4n$. Since $3 \cdot 2^n - 1$ is not bounded by any polynomial in $\|G_n\|$ and $\bar{\theta}_{G_n}$, the *only if* direction of (i) follows. □

Remark 7.4 (Scope of Theorem 7.3). (a) The lower bound uses rank two: the doubling rules $E_i \to \mathtt{b}E_{i-1}E_{i-1}\mathtt{b}$ are binary. For targets represented by linear CFGs the companion manuscript gives, for every fixed typing, characteristic data polynomial in the grammar size alone [23, Theorem 8.3]. (b) The theorem concerns pseudovarieties; which individual typings have polynomial data is not characterized here. (c) The upper bound is for the fixed operator $\mathcal{B}^{\leq r_{\max}}_{1,s}$, whereas the lower bound holds for every set-driven operator with characteristic samples throughout the class. (d) At fan-out $f \geq 2$ the *if* direction persists (Theorem 5.8), but the *only if* direction is open (Open Problem 9.1); the family L_n cannot serve there (Theorem 8.2).

Polynomial time and thick data. Eyraud, Heinz and Yoshinaka [10, Definition 2.12], following [12], call a class identifiable in polynomial time and thick data (IPTtD) if some algorithm returns, in time polynomial in the size of a given sample, a hypothesis consistent with it, and every representation R of a target has a characteristic sample of size polynomial in the size of R and its thickness. By Lemma 4.1(ii),(iii), their thickness (the longest shortest yield of a nonterminal, t_G) and our rule thickness are polynomially equivalent given $\|G\|$. The sample sizes $\sum_{w \in K} |w|$ and $\|K\|_+$ differ by at most the number of words. Their definition asks the algorithm to return a hypothesis in a fixed class of representations; the raw output of $\mathcal{B}^{\leq r_{\max}}_{f,s}$ may contain the rule $\widehat{S} \to \varepsilon$, so we first remove it.

Lemma 7.5 (Reduced good hypotheses)**.** *Fix* f, $r_{\max} \geq 1$ *and* s, *let* $K \subseteq \Sigma^*$ *be finite with* $K \cap \Sigma^+ \neq \emptyset$, *and let* G^+_K *be* $\mathcal{B}^{\leq r_{\max}}_{f,s}(K)$ *without the rule* $\widehat{S} \to \varepsilon$. *Then* G^+_K *is a reduced good MCFG of fan-out* $\leq f$ *and rank* $\leq r_{\max}$, $L(G^+_K) = L(\mathcal{B}^{\leq r_{\max}}_{f,s}(K)) \setminus \{\varepsilon\}$, *and* G^+_K *is computable from* K *in time* $(1 + \|K\|_+)^{O(f r_{\max})}$.

Proof. Start and Link rules have identity templates, Const rules write tuples of nonempty words, and Comp rules have good templates (Definition 3.3); so every rule of G^+_K is good, and no nonterminal derives a tuple with an empty component. Hence ε is derived only by the removed rule, which gives the language. Every $[\vec{x}]$ is productive by its Const rule, and $\widehat{S}$ is productive since K has a nonempty word. For reachability let $\vec{x} \in \mathrm{Obs}_d(K)$, say $w = E[\vec{x}] \in K$ with $E = v_0 \Box_1 v_1 \cdots \Box_d v_d \in \mathcal{E}_d$. Then $w \neq \varepsilon$, $(w) \in \mathrm{Obs}_1(K)$ with the context $\Box_1$, and the template $\rho = (v_0 \xi^1_1 v_1 \cdots \xi^d_1 v_d)$ is good: it is ε-free, uses each variable once and in order, and is nonmerging since $v_1, \ldots, v_{d-1}$ are nonempty. So $((w), \rho, (\vec{x}))$ is a composition witness of rank one, and $\widehat{S} \to [(w)] \to \rho([\vec{x}])$ reaches $[\vec{x}]$. Fan-out, rank and running time are those of Theorem 3.14. □

Corollary 7.6 (Identification in polynomial time and thick data)**.** *Fix $f \geq 1$, $r_{\max} \geq 1$ and a typing s. Let $\mathcal{R}$ consist of the pairs (G, z) of a bit $z \in \{0,1\}$ and either a reduced good fan-out-$\leq f$, rank-$\leq r_{\max}$ MCFG G or a fixed grammar $G_\emptyset$ without rules; the pair represents $L(G) \cup \{\varepsilon : z = 1\}$. The members of $\mathcal{C}^{\mathrm{mcf}}_{f,s} \cap \mathsf{MCFL}(f, r_{\max})$ are represented by $(G, [\varepsilon \in L])$ with G a reduced good fan-out-$\leq f$, rank-$\leq r_{\max}$ presentation of $L \cap \Sigma^+$, or $G = G_\emptyset$ if $L \cap \Sigma^+ = \emptyset$.*

(i) *If $s(\Sigma^+) \in \mathbf{LG}$, the algorithm returning $(G^+_K, [\varepsilon \in K])$ of Lemma 7.5, or $(G_\emptyset, [\varepsilon \in K])$ if $K \cap \Sigma^+ = \emptyset$, identifies $\mathcal{C}^{\mathrm{mcf}}_{f,s} \cap \mathsf{MCFL}(f, r_{\max})$ in polynomial time and thick data for the class $\mathcal{R}$.*

(ii) *If $f = 1$ and $r_{\max} \geq 2$, no algorithm, whatever its running time, satisfies the data condition of IPTtD for $\mathcal{C}^{\mathrm{mcf}}_{1,s_U} \cap \mathsf{MCFL}(1, r_{\max})$.*

(iii) *Consequently, for $r_{\max} \geq 2$ and a pseudovariety $\mathbf{V}$, the following are equivalent: $\mathbf{V} \subseteq \mathbf{LG}$; for every typing s with $s(\Sigma^+) \in \mathbf{V}$, the class $\mathcal{C}^{\mathrm{mcf}}_{1,s} \cap \mathsf{MCFL}(1, r_{\max})$ is IPTtD.*

Proof. (i) By Lemma 7.5 the hypothesis lies in $\mathcal{R}$, is computed in time polynomial in $\|K\|_+$, and represents $L(\mathcal{B}^{\leq r_{\max}}_{f,s}(K))$, which contains K (each $w \in K \cap \Sigma^+$ is derived by its Start and Const rules). Theorem 5.8 gives a sample F of length $O(\|G\|^2 \bar{\theta}_G)$ with $L(\mathcal{B}^{\leq r_{\max}}_{f,s}(K)) = L$ whenever $F \subseteq K \subseteq L$, so the hypothesis then represents L; and $\bar{\theta}_G \leq 2\|G\| \max\{1, t_G\}$. (ii) In [10, Definition 2.12] an algorithm reads a sample, which is a finite set of strings, so it is a set-driven operator. Such an algorithm has a characteristic sample for every member of the class, since every member L with $L \cap \Sigma^+ \neq \emptyset$ has a reduced good presentation of rank $\leq r_{\max}$ (Proposition 3.12). The targets L_n of Proposition 7.1 belong to the class, and so does $L_n \setminus \{W_n\}$: it lies in $\mathcal{C}^{\mathrm{mcf}}_{1,s_U}$ by Lemma 4.5 and is finite. By Proposition 4.7, every characteristic sample of L_n contains W_n, of length $3 \cdot 2^n - 2$, while $\|G_n\| = O(n)$ and $t_{G_n} \leq \theta_{G_n} \leq 4n$. (iii) follows from (i) with $f = 1$ and from (ii), since a pseudovariety not contained in $\mathbf{LG}$ contains U_1 (Theorem 7.3). □

The hypotheses lie in $\mathcal{R}$ on every sample; on samples that contain no characteristic sample their languages need not belong to the class, as for the learner of [14]. Item (i) holds at every fan-out, item (ii) only at fan-out one (Theorem 8.2).

8 Fan-Out Two: The Lower Bound Does Not Lift

This section shows that the exponential family of Theorem C(ii), and its images under letter-disjoint morphisms, do not carry over to fan-out two (Theorem 8.2).

Lemma 8.1 (Power transfer across two tuple components)**.** *Let $f \geq 2$, $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$, $u \in \Sigma^+$, and let $i_0, p \geq 1$ be an eventual index and period of $s(u)$, i.e. $s(u)^{t+p} = s(u)^t$ for $t \geq i_0$. Suppose $i, j \geq i_0$, $\delta \in p\mathbb{Z}_{>0}$, and some word of L contains the factor $u^{i+\delta+j+1}$. Then for all $w_{\mathrm{L}}, w_{\mathrm{R}} \in \Sigma^*$ and $w_{\mathrm{M}} \in \Sigma^+$,*

$$w_{\mathrm{L}} u^i w_{\mathrm{M}} u^{\delta+j} w_{\mathrm{R}} \in L \quad \Longleftrightarrow \quad w_{\mathrm{L}} u^{i+\delta} w_{\mathrm{M}} u^j w_{\mathrm{R}} \in L.$$

Proof. Write the given word as $y u^{i+\delta+j+1} z$. The tuples $\vec{x} = (u^i, u^{\delta+j})$ and $\vec{x}' = (u^{i+\delta}, u^j)$ have the same componentwise s-types by eventual periodicity. They also have the common admissible accepting context $E = y\Box_1 u \Box_2 z$, since $E[\vec{x}] = E[\vec{x}'] = y u^{i+\delta+j+1} z \in L$. The (f,s)-condition thus equates their two-component distributions. Test them in the admissible context $w_{\mathrm{L}}\Box_1 w_{\mathrm{M}}\Box_2 w_{\mathrm{R}}$; the condition $w_{\mathrm{M}} \neq \varepsilon$ is essential. □

Theorem 8.2 (The doubling families do not lift)**.** *Let $s : \Sigma^+ \twoheadrightarrow \mathsf{S}$ be any fixed finite-semigroup typing, let $u_{\mathtt{a}}, u_{\mathtt{b}}, u_{\mathtt{c}} \in \Sigma^+$ be letter-disjoint, let φ be the morphism $\mathtt{a} \mapsto u_{\mathtt{a}}$, $\mathtt{b} \mapsto u_{\mathtt{b}}$, $\mathtt{c} \mapsto u_{\mathtt{c}}$, and let $H_n := \varphi(L_n)$ for the family L_n of Proposition 7.1 (so $H_n = L_n$ when φ is the identity). Let $n \geq 2|\mathsf{S}| + 2$ and $f \geq 2$. Every $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$ containing $\varphi(W_n)$ contains another word with the same type, the same Parikh vector and no letter of $u_{\mathtt{c}}$. In particular $H_n \notin \mathcal{C}^{\mathrm{mcf}}_{f,s}$, and in no member of $\mathcal{C}^{\mathrm{mcf}}_{f,s}$ is $\varphi(W_n)$ the only word of its type.*

Proof. The morphism φ is injective, and a word of H_n without the letters of $u_{\mathtt{c}}$ is the image of a word of L_n without $\mathtt{c}$, so $\varphi(W_n)$ is the only such word. Choose an eventual index $i_0 \geq 1$ and period $p \geq 1$ for $s(u_{\mathtt{b}})$ with $i_0 + p \leq |\mathsf{S}| + 1$; then $n \geq 2i_0 + p + 1$. Put $i = i_0$, $\delta = p$ and $j = n - i - \delta - 1 \geq i_0$. The word $\varphi(W_n) = u_{\mathtt{b}}\,\varphi(W_{n-1})\,\varphi(W_{n-1})\,u_{\mathtt{b}}$ starts with $u_{\mathtt{b}}^n u_{\mathtt{a}}$, so it contains $u_{\mathtt{b}}^{i+\delta+j+1}$. Let $u_{\mathtt{b}}^i$ be the i copies of $u_{\mathtt{b}}$ after the first one, and let $u_{\mathtt{b}}^{\delta+j}$ be the $\delta + j \leq n - 1$ copies immediately before the first $u_{\mathtt{a}}$ of the second copy of $\varphi(W_{n-1})$, which is preceded by at least $n - 1$ copies of $u_{\mathtt{b}}$. Then

$$\varphi(W_n) = w_{\mathrm{L}} u_{\mathtt{b}}^i w_{\mathrm{M}} u_{\mathtt{b}}^{\delta+j} w_{\mathrm{R}}$$

with $w_{\mathrm{L}} = u_{\mathtt{b}}$ and $w_{\mathrm{M}} \neq \varepsilon$ containing the first $u_{\mathtt{a}}$. Let $L \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$ contain $\varphi(W_n)$. By Lemma 8.1, $w' := w_{\mathrm{L}} u_{\mathtt{b}}^{i+\delta} w_{\mathrm{M}} u_{\mathtt{b}}^j w_{\mathrm{R}} \in L$. The two filled tuples have equal componentwise types, so $s(w') = s(\varphi(W_n))$; w' has the Parikh vector of $\varphi(W_n)$ and no letter of $u_{\mathtt{c}}$, but begins with $u_{\mathtt{b}}^{n+\delta} u_{\mathtt{a}}$, not $u_{\mathtt{b}}^n u_{\mathtt{a}}$, so $w' \neq \varphi(W_n)$ since $u_{\mathtt{a}}$ and $u_{\mathtt{b}}$ have no common letter. For $L = H_n$ this contradicts the uniqueness of $\varphi(W_n)$ shown above. □

Remark 8.3. Power transfer preserves the total length and Parikh vector of the word, so it does not show that rare-type representatives are short; it only excludes the family of Proposition 7.1 and its letter-disjoint images, also after words are added.

9 Conclusion and an Open Problem

For a reduced good presentation G and fixed alphabet, fan-out, rank and typing, Theorem 4.3 reduces the total encoded length of characteristic samples to the shortest typed yield $t_{\mathrm{typ}}(G, s)$, and Corollary 4.10 further to the representative length $\tau(G, s)$. Owner-wise simultaneous replacement gives $O(\|G\|^2 \bar{\theta}_G)$ for window–group and local-group typings at every fan-out (Theorem A), and at every fan-out the product $\|G\|\bar{\theta}_G$ is necessary under every typing over a sufficiently large alphabet (Theorem B). At fan-out one, **LG** is the exact boundary for pseudovarieties, also for identification in polynomial time and thick data, and outside it the failure is exponential (Theorem C, Corollary 7.6). At fan-out two the exponential fan-out-one family does not lift (Theorem 8.2).

Open Problem 9.1 (The boundary at fan-out two). Fix $f \geq 2$ and $r_{\max} \geq 2$. Is **LG** the pseudovariety boundary at fan-out f, that is, does every pseudovariety of finite semigroups not contained in **LG** fail to have uniform polynomial characteristic-data length at fan-out f and rank $r_{\max}$ (Definition 7.2)?

Pseudovarieties contained in **LG** have uniform polynomial characteristic-data length at every fan-out (Theorem 5.8). Every pseudovariety not contained in **LG** contains U_1 (Theorem B.4), so the answer is positive as soon as one typing onto U_1 needs superpolynomial data at fan-out f; by Proposition 4.7 a family of targets whose shortest word of some type is superpolynomially long in $\|G\|$ and $\bar{\theta}_G$ would suffice, and Theorem 8.2 excludes the doubling family. A negative answer would show that the boundary depends on the fan-out.

For a different quantity the arity does matter: for the minimum size of a finite-monoid observation that preserves guarded tuple substitution through arity f, the least arity at which this minimum stabilizes uniformly over a pseudovariety of finite monoids is 1, 2 or ∞ [24]. Whether the characteristic-data boundary depends on the fan-out is the corresponding question for data length.

Tightness of the exponents. In $O(\|G\|^2\bar{\theta}_G)$, for window–group and local-group typings alike, one factor $\|G\|$ counts the witness words and the other bounds root-to-leaf paths in a λ-minimal derivation by the number of node labels (Lemma 5.1). The path factor is genuine for the witness construction: the grammars $S \to SS \mid \mathtt{a} \mid \mathtt{b} \mid A_0 \mid \cdots \mid A_n$, $A_0 \to \mathtt{b}$, $A_i \to \mathtt{a} \mid A_{i-1}\mathtt{b}$ $(1 \le i \le n)$ generate $\{\mathtt{a},\mathtt{b}\}^+$ with $\|G_n\| = 10n + 16$ and $\bar{\theta}_{G_n} = 2$, but $\mathtt{b}^{n+1}$ is the only yield of A_n of $s_{1,1}$-type $\langle \mathtt{b}, \mathtt{b} \rangle$, so $t_{\mathrm{typ}}(G_n, s_{1,1}) \ge n + 1$. For this target, however, $\tau(G_n, s_{1,1}) \le 2$ (Corollary 4.10), so the example bounds the witness method, not the data. For the trivial typing the upper bound $O(\|G\|\bar{\theta}_G)$ is attained by the marker families (Corollary 6.2); for every typing with $\mathsf{S} \in \mathbf{LG}$ the remaining gap is a single factor $\|G\|$, a gap between the known examples and the bounds rather than a proved ratio. Neither lower-bound argument closes it: for a fixed typing the rare-type argument yields at most $|\mathsf{S}|(t_{\mathrm{typ}} + 1)$, and the marker families consist of $O(\|G\|)$ words of length $\bar{\theta}_G = t_{\mathrm{typ}}$. For noncommutative groups the constant $M_{\mathbb{G},f}$ is doubly exponential in f because of the Ramsey-type homogenization (Lemma A.8); $|\mathbb{G}|$ owners suffice at $f = 1$, and whether $\mathrm{poly}(|\mathbb{G}|^f)$ owners suffice for $f \ge 2$, as for abelian groups, is open.

Acknowledgments

The author thanks Makoto Kanazawa and Ryo Yoshinaka for valuable discussions and encouragement.

Declaration of competing interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Data availability

No empirical dataset is associated with this theoretical paper. All proofs are given in full.

A Simultaneous Block Replacement in Finite Groups

This appendix proves Proposition 5.2, used in Lemmas 5.3 and B.8. The abelian case is a zero-sum argument. The general case never commutes factors; it has three parts: an order classification for homogeneous point configurations (Lemma A.4), an interval selection by boundary states (Lemma A.5), and a Ramsey-type extraction of homogeneous indices (Lemmas A.7 and A.8).

Roadmap of the general case. The proof of Lemma A.2(b), at the end of this appendix, applies the steps below in this order; each step keeps the owners it does not discard unchanged and absorbs the blocks of discarded owners into the constants (they keep their original values in every row).

Step	Input → output	Invariant gained
1. Common pattern	$m \geq M_{\mathbb{G},f}$ owners → $m/\mathcal{P}_f$ owners	same number of blocks, lines and within-owner order
2. Lemmas A.7, A.8	→ $N_{\mathbb{G},f} = \max\{3, \lvert\mathbb{G}\rvert^f\}$ owners	each comparison "block t of i left of block u of j" is constant over $i < j$
3. Lemma A.4	all lines → at most f class regions in total, each a run of $N_{\mathbb{G},f}$ chunks	one chunk per owner, increasing or decreasing; no foreign points inside a region
4. Lemma A.5	regions → an interval $J = (a, b]$ of owners	the states $(\eta_p^{\mathcal{Q}})_{\mathcal{Q}}$, $\eta_p^{\mathcal{Q}} = P_p Q_p^{-1}$, agree at a and b, so every ordered row product is preserved

For $f = 2$ and $\mathbb{G} = S_3$, Example A.6 shows the output of Step 3 (one increasing region on row 1, one decreasing region on row 2) and the use of η in Step 4: the replaced owners do not preserve their own contributions, only the two ordered row products are preserved.

Definition A.1 (Block-replacement family)**.** Let $\mathbb{G}$ be a finite group and $f \geq 1$. A *block-replacement family* of width f over $\mathbb{G}$ with index set $[m]$ consists of $d \leq f$ *lines*; on line ϱ, a left-to-right sequence of blocks $\mathfrak{b}_{\varrho,1}, \ldots, \mathfrak{b}_{\varrho,q_\varrho}$ and constants $a_{\varrho,0}, \ldots, a_{\varrho,q_\varrho} \in \mathbb{G}$; for every block $\mathfrak{b}$ an *owner* $o(\mathfrak{b}) \in [m]$ and two values $x_\mathfrak{b}, x'_\mathfrak{b} \in \mathbb{G}$; every index owns at most f blocks. For $J \subseteq [m]$ put

$$\mathrm{val}_\varrho(J) := a_{\varrho,0}\, y_{\varrho,1}\, a_{\varrho,1} \cdots y_{\varrho,q_\varrho}\, a_{\varrho,q_\varrho}, \qquad y_{\varrho,j} := \begin{cases} x'_{\mathfrak{b}_{\varrho,j}}, & o(\mathfrak{b}_{\varrho,j}) \in J, \\ x_{\mathfrak{b}_{\varrho,j}}, & o(\mathfrak{b}_{\varrho,j}) \notin J. \end{cases}$$

Lemma A.2 (Block replacement)**.** *In a block-replacement family of width f over $\mathbb{G}$ there is a nonempty $J \subseteq [m]$ with $\mathrm{val}_\varrho(J) = \mathrm{val}_\varrho(\emptyset)$ for every line ϱ, provided that either* (a) *$\mathbb{G}$ is abelian and $m > \mathsf{d}(\mathbb{G}^f)$, or* (b) *$\mathbb{G}$ is arbitrary and $m \geq M_{\mathbb{G},f}$, or* (c) *$\mathbb{G}$ is arbitrary, $f = 1$ and $m \geq \lvert\mathbb{G}\rvert$, with d and $M_{\mathbb{G},f}$ as in Section 5.1.*

Proof of (a). Write $\mathbb{G}$ additively. By commutativity $\mathrm{val}_\varrho(J) - \mathrm{val}_\varrho(\emptyset) = \sum_{i \in J} \delta_{i,\varrho}$, where $\delta_{i,\varrho}$ is the sum of $x'_\mathfrak{b} - x_\mathfrak{b}$ over the blocks $\mathfrak{b}$ of line ϱ owned by i. The sequence $(\delta_i)_{i \in [m]}$ with $\delta_i := (\delta_{i,\varrho})_\varrho \in \mathbb{G}^d$ has length $m > \mathsf{d}(\mathbb{G}^f) \geq \mathsf{d}(\mathbb{G}^d)$ ($\mathbb{G}^d$ embeds in $\mathbb{G}^f$), so it has a nonempty zero-sum subsequence $(\delta_i)_{i \in J}$. □

Remark A.3 (Size of the zero-sum constant). The prefix-sum pigeonhole gives $\mathsf{d}(\mathbb{G}^f) \leq \lvert\mathbb{G}\rvert^f - 1$, but much more is true. Write $\mathbb{G} \cong C_{q_1} \oplus \cdots \oplus C_{q_\ell}$ and $q = \exp(\mathbb{G}) \geq 2$. Then

$$f \sum_{i=1}^{\ell} (q_i - 1) \;\leq\; \mathsf{d}(\mathbb{G}^f) \;\leq\; (q-1) + q \log \frac{\lvert\mathbb{G}\rvert^f}{q},$$

where the lower bound is attained by a basis sequence and the upper bound is due to van Emde Boas and Kruyswijk [32, Proposition 5.1.8 and Theorem 5.5.5]; for a p-group the lower bound is an equality [32, Theorem 5.5.9]. Hence $\mathsf{d}(\mathbb{G}^f) = O(\exp(\mathbb{G})\, f \log \lvert\mathbb{G}\rvert)$; for $\mathbb{G} = C_2$ the antichain bound drops from $2^f - 1$ to f. Since zero-sum-free sequences of length $\mathsf{d}(\mathbb{G}^d)$ exist, the argument cannot do better; we do not claim that a grammar realizes them.

A.1 Arbitrary Finite Groups: Order Homogenization

Lemma A.4 (A homogeneous block-order lemma)**.** *Let $\mathcal{T}$ be a finite set of types and $n \geq 3$, and let (t, i), $t \in \mathcal{T}$, $i \in [n]$, be distinct points on a line. Suppose that* (i) *the order of (t, i) and (u, i) does not depend on i, and* (ii) *for each ordered pair (t, u), including $t = u$, the order of (t, i) and*

(u, j) is the same for all $i < j$. Then $\mathcal{T}$ is partitioned into classes such that: the points of each class occupy a consecutive region containing no other points; each class has a direction (increasing or decreasing in i); and its region consists of n successive chunks*, one per index in that direction, each containing exactly one point of every type of the class, in an order independent of the index.*

Proof. By (ii) with $t = u$, the points of each type are monotone in i. For distinct t, u let X mean $(t, i) < (u, j)$ and Y mean $(u, i) < (t, j)$ for $i < j$; both are constant by (ii).

Both increasing. $X \wedge Y$ puts all points of index i of both types before those of index $j > i$ (*interleaved by rank*). Under $X \wedge \neg Y$ all comparisons of different indices put t left of u, and so do equal indices, since $(u, i) < (t, i)$ would give $(t, 1) < (t, 2) < (u, 1) < (t, 1)$; so t and u are *separated*. $\neg X \wedge Y$ is symmetric, and $\neg X \wedge \neg Y$ gives $(t, 1) < (t, 2) < (u, 1) < (u, 2) < (t, 1)$.

t increasing, u decreasing. $X \wedge Y$ gives $(t, 2) < (u, 3) < (u, 1) < (t, 2)$ and $\neg X \wedge \neg Y$ gives $(t, 3) < (u, 2) < (t, 1) < (t, 3)$. Under $X \wedge \neg Y$ the comparisons of different indices put t left of u, and $(u, i) < (t, i)$ would give $(t, 1) < (u, 2) < (u, 1) < (t, 1)$; so they are separated, and $\neg X \wedge Y$ is symmetric. Reversing indices covers two decreasing types.

Thus two types are separated, or have the same direction and are interleaved. Interleaving is transitive: if t, u and u, v are interleaved (all increasing, say) but t, v are separated, then $(v, 1) < (u, 2) < (t, 3) < (v, 1)$ if t is left of v, and $(t, 1) < (u, 2) < (v, 3) < (t, 1)$ otherwise. So interleaving is an equivalence relation. A type outside a class is separated from every type of the class and lies on the same side of all of them, since otherwise two interleaved types would be separated; hence class regions contain no other points. Inside a class, interleaving gives the chunks, and (i) gives the within-chunk order. □

The case analysis for two distinct types t, u, with X and Y as in the proof, is summarized below; two decreasing types are the increasing case with indices reversed.

Directions	X, Y	Outcome	Cycle excluding the other cases
both increasing	X, Y	interleaved by index	—
	$X, \neg Y$	separated, t left of u	—
	$\neg X, Y$	separated, u left of t	—
	$\neg X, \neg Y$	impossible	$(t, 1) < (t, 2) < (u, 1) < (u, 2) < (t, 1)$
t increasing, u decreasing	X, Y	impossible	$(t, 2) < (u, 3) < (u, 1) < (t, 2)$
	$X, \neg Y$	separated, t left of u	—
	$\neg X, Y$	separated, u left of t	—
	$\neg X, \neg Y$	impossible	$(t, 3) < (u, 2) < (t, 1) < (t, 3)$

Lemma A.5 (Simultaneous finite-group interval shortening)**.** *Let $\mathbb{G}$ be a finite group and consider a block-replacement family of width f over $\mathbb{G}$ with $n \geq \max\{3, |\mathbb{G}|^f\}$ indices, in which all indices have the same number of blocks, on the same lines, in the same within-index order, and in which, for every two block positions t, u on a common line, the truth value of "block t of index i lies left of block u of index j" is the same for all $i < j$. Then a nonempty interval J of indices satisfies $\mathrm{val}_\varrho(J) = \mathrm{val}_\varrho(\emptyset)$ for every line ϱ.*

Proof. Apply Lemma A.4 on each line; there are $c \leq f$ class regions in total. Fix a region $\mathcal{Q}$ and number its chunks $1, \ldots, n$ from left to right. Assign constants between chunks to the chunk on their left, and absorb fixed constants at the ends of the region into its first or last chunk. Each chunk has original and replacement products with the *same* fixed constants in the same order. Let P_p be the original value of chunks $1, \ldots, p$ and Q_p their value when all of them are replaced ($P_0 = Q_0 = 1_\mathbb{G}$), and put $\eta_p^\mathcal{Q} := P_p Q_p^{-1}$. Replacing exactly chunks $p + 1, \ldots, p'$ gives the ordered value $P_p\,(Q_p^{-1} Q_{p'})\,(P_{p'}^{-1} P_n)$, which equals P_n iff $P_p Q_p^{-1} \cdot Q_{p'} P_{p'}^{-1} = 1_\mathbb{G}$, that is, iff $\eta_p^\mathcal{Q} = \eta_{p'}^\mathcal{Q}$; no

factors are commuted. The order in $\eta_p^{\mathcal{Q}} = P_p Q_p^{-1}$ is forced by the position of the replaced chunks between an original prefix and an original suffix; the element $Q_p^{-1} P_p$ need not detect this in a noncommutative group. If $\mathcal{Q}$ is increasing, the owners in $(a, b]$ occupy exactly chunks $(a, b]$; if it is decreasing, chunks $(n-b, n-a]$. For $j = 0, \dots, n$ let the state be the vector in $\mathbb{G}^c$ whose $\mathcal{Q}$-entry is $\eta_j^{\mathcal{Q}}$ for increasing and $\eta_{n-j}^{\mathcal{Q}}$ for decreasing regions. Since $n+1 > |\mathbb{G}|^c$, two states coincide, at $a < b$. Replacing the owners $a+1, \dots, b$ then preserves every region value, and nothing outside the regions changes. □

Example A.6 (Opposite orientations and noncommuting values). In S_3 let $\sigma_1 = (12)$, $\sigma_2 = (23)$. Three owners have blocks U_i, V_i on row 1 and W_i on row 2:

$$\text{row 1: } (U_1\sigma_2 V_1)\,\sigma_1\,(U_2\sigma_2 V_2)\,\sigma_1\,(U_3\sigma_2 V_3), \qquad \text{row 2: } W_3\,\sigma_2\,W_2\,\sigma_2\,W_1,$$

so row 1 is one increasing class and row 2 one decreasing class. Originally $U_i = \sigma_2$, $V_i = W_i = 1$, and both rows have value 1. Let owner 1 have the trivial replacement $U_1' = U_1$, $W_1' = W_1$, and replace exactly owners $J = \{2, 3\}$ by $U_2' = 1$, $W_2' = \sigma_2\sigma_1\sigma_2$, $U_3' = \sigma_1\sigma_2\sigma_1\sigma_2$, $W_3' = \sigma_1$, $V_i' = 1$. Then row 1 becomes $\sigma_1 \cdot \sigma_2\sigma_1 \cdot \sigma_1\sigma_2\sigma_1 = 1$ and row 2 becomes $\sigma_1\sigma_2 \cdot \sigma_2\sigma_1\sigma_2 \cdot \sigma_2 = 1$, computed in order. In the notation of the proof, $(a, b] = (1, 3]$ and $\eta_1 = \eta_3 = 1$ on row 1, while $\eta_{3-b} = \eta_0 = 1 = \eta_2 = \eta_{3-a}$ on row 2: one owner interval matches two different boundary-index orientations, although neither replaced owner preserves its own contribution.

Lemma A.7 (Two monotone sequences). *Let $m \geq 3$, $m' \geq (m-1)^2 + 1$ and $\mathrm{lt} : \{(i, j) : 1 \leq i < j \leq m'\} \to \{0, 1\}$ be monotone in each argument with fixed directions: for each fixed i it is monotone in j, for each fixed j monotone in i, and each of these two directions is the same for all values of the fixed argument. Then* lt *is constant on the pairs $i < j$ of some $I \subseteq [m']$ with $|I| = m$.*

Proof. Replacing lt by $1 - \mathrm{lt}$ if necessary, rows are nondecreasing in j. It suffices to find an index set of size at least m, since constancy persists in subsets.

If columns are nonincreasing in i, let $\mathrm{nx}(i) := \min\{j > i : \mathrm{lt}(i, j) = 1\}$ (∞ if none); nx is nondecreasing. The chain $i_1 = 1$, $i_{r+1} = \mathrm{nx}(i_r)$ satisfies $\mathrm{lt}(i_a, i_b) = 1$ for $a < b$; if it has m terms we are done. Otherwise its at most $m-1$ terms cut $[m']$ into at most $m-1$ intervals $[i_r, i_{r+1})$, $[i_L, m']$, one of which has at least $m'/(m-1) > m-1$, hence at least m, indices, and inside it $j < \mathrm{nx}(i_r) \leq \mathrm{nx}(i)$ gives $\mathrm{lt}(i, j) = 0$.

If columns are nondecreasing, let i^* be the least index with $\mathrm{lt}(i^*, i^* + 1) = 1$ ($i^* = m'$ if none). Then $\mathrm{lt}(i, j) = 1$ for $i^* \leq i < j$ and $\mathrm{lt}(i, j) \leq \mathrm{lt}(j-1, j) = 0$ for $i < j \leq i^*$, and one of $[1, i^*]$, $[i^*, m']$ has at least $(m'+1)/2 \geq m$ indices because $(m-1)^2 + 1 \geq 2m - 1$ for $m \geq 3$. □

Lemma A.8 (Polynomial order homogenization). *Let $n_0 \geq 3$, $m_0 := n_0$ and $m_{q+1} := (m_q - 1)^2 + 1$, so $m_q \leq n_0^{2^q}$. Let $\mathcal{T}$ be a finite set of types, each assigned to a line, and for $i \in [n]$ and $t \in \mathcal{T}$ let $p_t(i)$ be a point on the line of t, the points on each line being distinct. Let $\mathcal{O}$ be the set of ordered pairs (t, u), $t = u$ allowed, of types on a common line, If $n \geq m_{|\mathcal{O}|}$, there is $I \subseteq [n]$ with $|I| = n_0$ such that for every $(t, u) \in \mathcal{O}$ the truth value of $p_t(i) < p_u(j)$ is the same for all $i < j$ in I.*

Proof. Treat the pairs one at a time, the pairs (t, t) first; each step passes from an index set of size at least $(m-1)^2 + 1$ to a subset of size at least m, and earlier homogeneity persists in subsets. For (t, t), Erdős–Szekeres gives a monotone subsequence of length m. For $t \neq u$ (both now monotone), relabel the current indices $1, \dots, m'$ and put $\mathrm{lt}(i, j) := [p_t(i) < p_u(j)]$ for $i < j$. For fixed i, lt is monotone in j in the direction of p_u; for fixed j, monotone in i in the direction opposite to p_t. Since $m \geq m_0 = n_0 \geq 3$, Lemma A.7 applies. □

Proof of Lemma A.2(c). An index owning no block can be chosen alone, so let every index own one block; at $f = 1$ all blocks lie on the single line. Renumber the indices by the positions of their blocks, so that they form one increasing region with one block per chunk. As in the proof of Lemma A.5, replacing exactly the indices $a+1, \dots, b$ preserves the line value iff $\eta_a = \eta_b$, and the $m+1 > |\mathbb{G}|$ values $\eta_0, \dots, \eta_m \in \mathbb{G}$ cannot all be distinct. □

Proof of Lemma A.2(b). An index owning no block can be chosen alone, so assume every index owns a block. The *pattern* of an index records the number of its blocks, the line of each, and their order on shared lines; there are at most $f \cdot f^f \cdot f! = \mathcal{P}_f$ patterns, so at least $N_{\mathbb{G},f}^{2^{f^2}}$ indices share one. Absorb the blocks of all other indices into the constants: these indices are never replaced, so their blocks keep their original values in every row. The blocks of the common pattern give at most f types and at most f^2 ordered pairs on common lines, so Lemma A.8 with $n_0 = N_{\mathbb{G},f}$ yields $N_{\mathbb{G},f}$ indices with homogeneous order comparisons. The blocks of the indices discarded in this extraction are absorbed into the constants in the same way. The common pattern fixes the order of the blocks of one index, which is hypothesis (i) of Lemma A.4, and Lemma A.8 gives hypothesis (ii); a comparison with $i > j$ is the negation of the comparison for the reversed pair, since the points are distinct. The remaining $N_{\mathbb{G},f}$ indices then satisfy the hypotheses of Lemma A.5, whose interval is the required J. □

For fixed f, $M_{\mathbb{G},f} = \mathcal{P}_f \max\{3, |\mathbb{G}|^f\}^{2^{f^2}}$ is polynomial in $|\mathbb{G}|$; its degree $f \cdot 2^{f^2}$ is exponential in f^2, so $M_{\mathbb{G},f}$ itself is doubly exponential in f.

B Local-Group Coordinates and MCFG Block Shortening

LG is always tested on the positive image $\mathsf{S} = \widehat{s}(\Sigma^+)$. The classical identity $\mathbf{LG} = \mathbf{Grp} * \mathbf{D}$ places it within the sliding-window and delay theory [34, 35], and the exclusion of $U_1 = \{1, \zeta\}$ ($\zeta^2 = \zeta$) is [33, Proposition 2.3]. Write x^ω for the idempotent power of an element x of a finite semigroup. Equivalently, **LG** is defined by the single pseudo-identity $(x^\omega y x^\omega)^\omega = x^\omega$: for an idempotent $e = x^\omega$ the finite monoid $e\mathsf{S}e = \{eye : y \in \mathsf{S}\}$ is a group exactly when e is its only idempotent, and every idempotent has the form x^ω. The coordinates below are a self-contained implementation of this algebra, not a new characterization; what is specific to MCFG learning is the factorization of group values into fixed separators and replaceable blocks (Lemma B.2). Throughout, $k \geq 1$, $\mathbb{G}$ is a finite group, $\lambda_{k,\nu}$ is as in Definition 5.7, a word is *short* if $|w| < 2k$ (recorded exactly by $\widehat{s}_{k,k}$) and *long* otherwise, and for $x, y \in \Sigma^k$ we put $\Delta_\nu(x,y) := \prod_{i=1}^{k} \nu((xy)[i..i+k])$, the ordered product over the k factors of length $k+1$ of xy that start in x.

Pumping. For a finite semigroup T, every product of at least $|\mathsf{T}|$ elements lies in $\mathsf{T}^1 E(\mathsf{T}) \mathsf{T}^1$ [33, Lemma 2.2], [25]: either two of its first $|\mathsf{T}|$ prefix products coincide, $t_i = t_j$ with $i < j$, and then $t_i = t_i x^\omega$ for the product x of the factors $i+1, \dots, j$; or these prefix products exhaust T and one of them is idempotent. All three uses below (Lemmas B.3 and B.5, Theorem B.4) rest on this fact.

Lemma B.1 (Sliding-factor maps are morphisms)**.** *$\lambda_{k,\nu}(uv)$ is a function of $\lambda_{k,\nu}(u)$ and $\lambda_{k,\nu}(v)$; hence $\lambda_{k,\nu}$ is a morphism onto a finite monoid $M_{k,\nu}$ with at most $|\Sigma|^{<2k} + |\Sigma|^{2k}|\mathbb{G}|$ elements. For long u, v,*

$$\lambda_{k,\nu}(uv) = \big(\langle \mathrm{pref}_k\, u, \mathrm{suf}_k\, v\rangle,\ \gamma_{k,\nu}(u)\, \Delta_\nu(\mathrm{suf}_k\, u, \mathrm{pref}_k\, v)\, \gamma_{k,\nu}(v)\big).$$

Proof. The window component is a morphism. For long u, v the factors of length $k+1$ of uv are, in order, those of u, the k factors starting in the last k positions of u (inside $\mathrm{suf}_k(u)\,\mathrm{pref}_k(v)$), and those of v. If u is short and v long, the factors starting in u lie inside $u\,\mathrm{pref}_k(v)$ and the others are

those of v; the case of long u and short v is symmetric; and if both are short, uv is determined exactly. □

Lemma B.2 (Block decomposition)**.** *Let $w = x_0\mathfrak{b}_1x_1\cdots\mathfrak{b}_qx_q$ and $w' = x_0\mathfrak{b}'_1x_1\cdots\mathfrak{b}'_qx_q$ with $\widehat{s}_{k,k}(\mathfrak{b}'_j) = \widehat{s}_{k,k}(\mathfrak{b}_j)$ for all j. Then $\widehat{s}_{k,k}(w') = \widehat{s}_{k,k}(w)$, and if w is long there are $z_0,\dots,z_q \in \mathbb{G}$, depending only on the x_j and the window types of the $\mathfrak{b}_j$, with*

$$\gamma_{k,\nu}(w) = z_0\,\gamma_{k,\nu}(\mathfrak{b}_1)\,z_1\cdots\gamma_{k,\nu}(\mathfrak{b}_q)\,z_q, \qquad \gamma_{k,\nu}(w') = z_0\,\gamma_{k,\nu}(\mathfrak{b}'_1)\,z_1\cdots\gamma_{k,\nu}(\mathfrak{b}'_q)\,z_q.$$

Proof. The window claim holds because $\widehat{s}_{k,k}$ is a morphism. A short block equals its replacement and is absorbed into the fixed words, so let all blocks be long. A factor of length $k+1$ either lies inside some block or meets every block it touches in a prefix or suffix of length at most k; factors of the second kind are spelled by the fixed words and the first and last k letters of the blocks, so they are the same in w and w' and occur in the same order. Ordered by starting position, the factors inside $\mathfrak{b}_j$ form one run with product $\gamma_{k,\nu}(\mathfrak{b}_j)$; let z_j be the ordered product of the factors between consecutive runs. □

Lemma B.3 (Sliding-factor coordinates)**.** *Let s be a typing whose positive image S has all its idempotents in its minimal ideal $\mathsf{K} = \mathcal{M}(\mathbb{H}; I, \Lambda; \mathsf{P})$, a Rees matrix semigroup over a group $\mathbb{H}$ with sandwich matrix $\mathsf{P} = (p_{ji})_{j\in\Lambda, i\in I}$, whose elements $(i,\gamma,j) \in I\times\mathbb{H}\times\Lambda$ multiply by $(i,\gamma,j)(i',\gamma',j') = (i,\gamma p_{ji'}\gamma',j')$. Let $k \geq 1$ satisfy $\mathsf{S}^k \subseteq \mathsf{K}$ (by pumping, $\mathsf{S}^{|\mathsf{S}|} \subseteq \mathsf{S}^1E(\mathsf{S})\mathsf{S}^1 \subseteq \mathsf{K}$, so $k = |\mathsf{S}|$ will do; the least such k gives the smallest windows). There is $\nu : \Sigma^{k+1} \to \mathbb{H}$ such that every w with $|w| \geq 2k$ satisfies*

$$s(w) = \big(\mathrm{row}(s(\mathrm{pref}_k\,w)),\ \mathrm{grp}(s(\mathrm{pref}_k\,w))\,\gamma_{k,\nu}(w),\ \mathrm{col}(s(\mathrm{suf}_k\,w))\big), \tag{$\star$}$$

where row, grp, col *are the three coordinates of an element of K. In particular $s(w)$ is determined by its first and last k letters and an ordered product in $\mathbb{H}$.*

Proof. (1) Coordinates. Fix $i_0 \in I$; for $j \in \Lambda$ the element $r_j := (i_0, p_{ji_0}^{-1}, j)$ is a right identity for the elements of K with Λ-coordinate j. For $y \in \mathsf{S}$ write $r_jy = (i',\gamma',j')$ and $\mathrm{act}(j,y) := p_{ji'}\gamma'$; then $(i,\gamma,j)y = (i,\gamma\,\mathrm{act}(j,y),j')$, so right multiplication fixes the I-coordinate. Dually, left multiplication fixes the Λ-coordinate of an element of K: for any $j'' \in \Lambda$ the idempotent $e' := (i, p_{j''i}^{-1}, j'') \in \mathsf{K}$ satisfies $e'(i,\gamma,j) = (i,\gamma,j)$, so $y(i,\gamma,j) = (ye')(i,\gamma,j)$ with $ye' \in \mathsf{K}$, and the Rees product of two elements of K has the Λ-coordinate of its right factor.

(2) Group coordinate. Let $w = a_1\cdots a_m$ with letters a_t, $m \geq 2k$, and $s(a_1\cdots a_k) = (i,\gamma,j_k)$. For $t \geq k$ the Λ-coordinate j_t of $s(a_1\cdots a_t)$ equals that of $s(a_{t-k+1}\cdots a_t)$ by $\mathsf{S}^k \subseteq \mathsf{K}$ and (1). Put $\nu(F) := \mathrm{act}\big(\mathrm{col}(s(F[1..k])), s(F[k+1])\big)$. Applying the right-action identity for $t = k,\dots,m-1$ in order gives $s(w) = (i,\ \gamma\prod_{t=k}^{m-1}\nu(a_{t-k+1}\cdots a_{t+1}),\ j_m)$, and the product is $\gamma_{k,\nu}(w)$. □

Theorem B.4 (Local-group criteria in sliding-factor coordinates)**.** *For a typing s with positive image S the following are equivalent:* (i) $\mathsf{S} \in \mathbf{LG}$*;* (ii) S *has no subsemigroup isomorphic to U_1;* (iii) *every idempotent of S lies in the minimal ideal of S;* (iv) *s is local-group-determined, and one may take $k = |\mathsf{S}|$, or any k with $\mathsf{S}^k = \mathsf{K}$.*

Proof. By [33, Proposition 2.3, items (1), (3) and (4)], (i) and (ii) are each equivalent to $\mathsf{S}^m = \mathsf{K}$ for some m. If $\mathsf{S}^m = \mathsf{K}$ then $e = e^m \in \mathsf{K}$ for each idempotent e; conversely, if $E(\mathsf{S}) \subseteq \mathsf{K}$ then $\mathsf{S}^k \subseteq \mathsf{S}^1E(\mathsf{S})\mathsf{S}^1 \subseteq \mathsf{K}$ for $k = |\mathsf{S}|$ by pumping, and $\mathsf{S}^k \supseteq \mathsf{K}$ since S^k is an ideal. So (i)–(iii) are equivalent.

(iii)⇒(iv): Lemma B.3 with $k = |\mathsf{S}|$ (or any k with $\mathsf{S}^k = \mathsf{K}$) determines $s(w)$ for $|w| \geq 2k$ from $\lambda_{k,\nu}(w)$, and $\widehat{s}_{k,k}$ records shorter words and ε exactly; so $\widehat{s} \preceq \lambda_{k,\nu}$.

(iv)⇒(i): S is a homomorphic image of $\mathsf{T} := \lambda_{k,\nu}(\Sigma^+)$ and **LG** is closed under homomorphic images, so it suffices that $\mathsf{T} \in \mathbf{LG}$. No short element is idempotent (squaring changes the exactly recorded word or makes it long). Let $e = (\langle p, q\rangle, \gamma_e)$ be idempotent. By Lemma B.1, every element of $e\mathsf{T}e$ has window $\langle p, q\rangle$, and long elements with this window multiply by $(\langle p, q\rangle, \gamma)(\langle p, q\rangle, \gamma') = (\langle p, q\rangle, \gamma\Delta\gamma')$, $\Delta := \Delta_\nu(q, p)$. This operation makes $\{\langle p, q\rangle\} \times \mathbb{G}$ a group isomorphic to $\mathbb{G}$ via $\gamma \mapsto \gamma\Delta$, and $e\mathsf{T}e$ is a finite nonempty subsemigroup of it, hence a subgroup. □

Lemma B.5 (Locally trivial–group typings admit window–group refinements)**.** *Let $\Sigma \neq \emptyset$. If $\mathsf{S} \in \mathbf{LI} \vee \mathbf{Grp}$, there are $k \geq 1$, a finite group $\mathbb{G}$ and a morphism $g : \Sigma^* \to \mathbb{G}$ with $\widehat{s} \preceq (\widehat{s}_{k,k}, g)$; if $\mathsf{S} \in \mathbf{LI}$, $\mathbb{G}$ can be trivial.*

Proof. Choose a finite $\mathsf{S}' \in \mathbf{LI}$, a finite group $\mathbb{G}$, a subsemigroup $\mathsf{T} \leq \mathsf{S}' \times \mathbb{G}$ and a surjective morphism $\vartheta : \mathsf{T} \to \mathsf{S}$ (for $\mathsf{S} \in \mathbf{LI}$: $\mathsf{S}' = \mathsf{S}$, $\mathbb{G} = \{1\}$). Choosing for every letter a preimage in T of its s-image defines morphisms $s' : \Sigma^+ \to \mathsf{S}'$ and g with $\vartheta(s'(w), g(w)) = s(w)$; put $g(\varepsilon) = 1_{\mathbb{G}}$. The subsemigroup $\mathsf{S}_0 := s'(\Sigma^+) \leq \mathsf{S}'$ is locally trivial; let $k := |\mathsf{S}_0|$. For idempotents e, e' of S_0 and $y \in \mathsf{S}_0^1$ one has $eye' = ey(e'ee') = (e(ye')e)e' = ee'$. For $|w| \geq 2k$ write $w = pxz$ with $|p| = |z| = k$. By pumping, $s'(p) = y_1 e y_2$ and $s'(z) = y_3 e' y_4$ with $y_1, \ldots, y_4 \in \mathsf{S}_0^1$ and e, e' idempotent, so $s'(w) = y_1 e\,(y_2\, s'(x)\, y_3)\, e' y_4 = y_1 e e' y_4$ (with $s'(\varepsilon) := 1$), which depends only on $p = \mathrm{pref}_k(w)$ and $z = \mathrm{suf}_k(w)$. Shorter words are recorded exactly by $\widehat{s}_{k,k}$, which also separates ε. Hence equal $(\widehat{s}_{k,k}, g)$-values imply equal (s', g)-values and equal s-values. □

Proposition B.6 (Window–group typings are local-group typings, strictly)**.** (i) *If $\widehat{s} \preceq (\widehat{s}_{k,\ell}, g)$ for a morphism g into a finite group, then $\widehat{s} \preceq \lambda_{\hat{k},\nu}$ with $\hat{k} = \max\{k, \ell, 1\}$ and $\nu(F) := g(\text{last letter of } F)$. Moreover $\mathbf{LI} \vee \mathbf{Grp} \subseteq \mathbf{LG}$.*

(ii) *Already for $\Sigma = \{\mathtt{a}, \mathtt{b}\}$ and the typing $s_{\mathtt{ab}} := \lambda_{1,\nu}|_{\Sigma^+}$ with $\nu(\mathtt{ab}) = 1 \in \mathbb{Z}_2$ and $\nu = 0$ otherwise, $s_{\mathtt{ab}}(\Sigma^+) \in \mathbf{LG} \setminus (\mathbf{LI} \vee \mathbf{Grp})$.*

Proof. (i) For $|w| \geq 2\hat{k}$, $g(w) = g(\mathrm{pref}_{\hat{k}}\, w)\, \gamma_{\hat{k},\nu}(w)$, and $\mathrm{pref}_{\hat{k}}, \mathrm{suf}_{\hat{k}}$ determine $\widehat{s}_{k,\ell}(w)$; shorter words are recorded exactly. The inclusion holds because **LG** is a pseudovariety containing **LI** ($e\mathsf{T}e = \{e\}$) and **Grp**. (ii) Membership in **LG** follows from Theorem B.4. Let $k, \ell \geq 0$ and let $g' : \Sigma^* \to \mathbb{G}'$ map into an arbitrary finite group. Choose $m \geq k + \ell + 1$ and $n \geq 1$ divisible by the order of $g'(\mathtt{a})$, and put $u = \mathtt{b}^m \mathtt{a}^n \mathtt{b}^m$ and $v = \mathtt{b}^{2m}$. They have equal length-k prefix and length-ℓ suffix, are long for $s_{k,\ell}$, and satisfy $g'(u) = g'(\mathtt{b})^{2m} = g'(v)$, but $\#_{\mathtt{ab}}(u) = 1$ and $\#_{\mathtt{ab}}(v) = 0$. Thus $s_{\mathtt{ab}}$ is not a window–group typing, and Lemma B.5 excludes $\mathbf{LI} \vee \mathbf{Grp}$. □

Lemma B.7 (Thinned and padded owners)**.** *Fix $k \geq 1$. Consider pairwise disjoint nonempty intervals in at most f words (*rows*), each interval belonging to one of m owners and each owner holding at most f intervals, and suppose that no interval meets the first or last k letters of its row. Then there is a set Q_1 of at least $m/(4kf + 1)$ owners with the following property. For arbitrary nonempty replacement words of the intervals of the owners in Q_1, there are pairwise disjoint* enlarged blocks*, at most f per owner, such that* (a) *each enlarged block contains intervals of exactly one owner in Q_1, and every interval of an owner in Q_1 lies in exactly one enlarged block;* (b) *replacing, inside an enlarged block, the intervals of its owner yields a word of length at least $2k + 1$ with the same first and last k letters as the block, which itself has length at least $2k + 1$; and* (c) *for every $J \subseteq Q_1$, replacing the intervals of the owners in J equals replacing their enlarged blocks as in* (b)*.*

Proof. Join two distinct owners if some interval of one and some interval of the other lie in the same row with fewer than $2k$ letters between them. A neighbouring interval to the right of an interval $[l, r)$ (in cut positions) starts at one of the $2k$ cuts $r, \ldots, r + 2k - 1$, and disjointness and

nonemptiness allow at most one interval per cut; the same holds on the left. So each owner has at most $4kf$ neighbours, and greedily choosing an owner and deleting its closed neighbourhood gives an independent set Q_1 of at least $m/(4kf+1)$ owners.

Enlarge every interval of an owner in Q_1 by k letters on each side; the boundary hypothesis keeps the enlargement inside its row. Intervals of distinct owners in Q_1 are at least $2k$ letters apart, so their enlargements have disjoint interiors; merge overlapping enlargements of the same owner. This gives (a) and does not increase the number of blocks per owner. Let $[l, r)$ be the union of the merged enlargements and l_0 the smallest left end of an interval in it, so $l = l_0 - k$. No interval of an owner in Q_1 meets $[l, l_0)$: one of the same owner would have been merged with a smaller left end, and one of another owner in Q_1 would end at least $2k$ letters before l_0. Symmetrically for the last k letters. Replacing only the owner's intervals therefore keeps the first and last k letters, and the replacement contains at least one nonempty word besides these $2k$ letters, which gives (b). Letters of owners outside Q_1 inside a block are never replaced, so (c) holds for every $J \subseteq Q_1$. □

Lemma B.8 (Short tuples with prescribed local-group type)**.** *Let G be reduced and good of fan-out $\le f$ and rank $\le r_{\max}$, and $\lambda = \lambda_{k,\nu}$ with group $\mathbb{G}$. Put $C^{\mathrm{buf}}_{k,\mathbb{G},f} := 2kf + (4kf+1)c_{\mathbb{G},f}$ and $\bar{n}_\lambda := |V||M_{k,\nu}|^f$. For every $B \in V$ and every realizable $\mathbf{t} \in M_{k,\nu}^{\mathrm{fan}(B)}$ there is $\vec{u} \in L_B(G)$ with $\lambda^{(\mathrm{fan}(B))}(\vec{u}) = \mathbf{t}$ and*

$$\|\vec{u}\|_1 \le \beta_\lambda := (r_{\max}+1)\,\bar{\theta}_G\, C^{\mathrm{buf}}_{k,\mathbb{G},f}\, \bar{n}_\lambda.$$

Proof. We verify the hypothesis of Lemma 5.1(ii) with $c = (4kf+1)c_{\mathbb{G},f}$ and $c' = 2kf$, so $c + c' = C^{\mathrm{buf}}_{k,\mathbb{G},f} \ge 1$; with $n_\lambda(G) \le \bar{n}_\lambda$ it gives the bound. Let T be λ-minimal with heavy threshold $b = \bar{\theta}_G$. In each root component of length at least $2k$ mark the first and the last k letters, and in each shorter component mark every letter: at most $2kf$ marks. Let Q be an antichain of heavy nodes. By linearity and nondeletion their components occupy disjoint nonempty intervals of the root components, at most f per node, and at most $2kf$ nodes contain a mark; let Q_0 be the others, so their intervals avoid the first and last k letters of every root component and lie in components of length at least $2k$.

Let $|Q_0| > (4kf+1)c_{\mathbb{G},f}$. Lemma B.7 gives a set $Q_1 \subseteq Q_0$ of more than $c_{\mathbb{G},f}$ owners. For $v \in Q_1$ with nonterminal C_v fix a tuple of $L_{C_v}(G)$ of total length at most $\bar{\theta}_G$ (Lemma 4.1(ii)); its components are nonempty, and no window or group condition is imposed on it. By Lemma B.7(b) every enlarged block and its replacement are long and have the same first and last k letters, hence the same $\widehat{s}_{k,k}$-type. So Lemma B.2, applied with the enlarged blocks, writes $\gamma_{k,\nu}$ of every long root component as an ordered product of the block values and constants that do not depend on which owners are replaced. This is a block-replacement family of width f over $\mathbb{G}$ with more than $c_{\mathbb{G},f}$ owners, and Proposition 5.2 gives a nonempty $J \subseteq Q_1$ preserving every root $\gamma_{k,\nu}$-value.

Replace the subtrees at the nodes of J by derivations of the chosen tuples. This is a derivation from B because J is an antichain, and by Lemma B.7(c) its root components are those of the enlarged-block replacement. Short root components contain no replaced letter and are unchanged; long ones keep their marked first and last k letters and stay long, so their $\widehat{s}_{k,k}$-types are unchanged, and their $\gamma_{k,\nu}$-values are preserved by the choice of J. Hence the root λ-vector is unchanged. □

Substitutability is not used in Lemma B.8.

C Auxiliary Verification of Qualitative Reconstruction

Proof of Lemma 3.1. The identity holds because substitutions commute. By linearity and nondeletion each $\square_\kappa$ occurs exactly once in E_j, and by nonpermutation in the order $\kappa = 1, \dots, d_j$, since E

places $\alpha_1, \ldots, \alpha_{d_0}$ from left to right. Consider ξ_j^{κ} and $\xi_j^{\kappa+1}$. If both lie in one α_i, nonmerging puts a terminal or a variable $\xi_i^{\kappa'}$ with $i \neq j$ between them, which becomes a nonempty word. Otherwise they lie in $\alpha_i, \alpha_{i'}$ with $i < i'$, and the material between them contains the separator $v_i \in \Sigma^+$ of E. So every internal separator of E_j is nonempty. □

Proof of Lemma 3.2. Put $\vec{z}_j := \rho(\vec{u}_1, \ldots, \vec{u}_j, \vec{x}_{j+1}, \ldots, \vec{x}_r)$. For $E \in \mathcal{E}_{d_0}$, Lemma 3.1 gives an admissible E_j with $E[\vec{z}_{j-1}] = E_j[\vec{x}_j]$ and $E[\vec{z}_j] = E_j[\vec{u}_j]$; since $\vec{u}_j \equiv_{L,s} \vec{x}_j$, E accepts $\vec{z}_{j-1}$ iff it accepts $\vec{z}_j$. So $\mathcal{D}_L(\vec{z}_0) = \mathcal{D}_L(\vec{z}_r)$. The types agree because $\widehat{s}^{(d_0)}(\rho(\cdots)) = \mathrm{out}_\rho$ of the children's types. □

Proof of Proposition 3.10. Induction on the height of a derivation from $[\vec{x}]$. Const derives $\vec{x}$. Link $[\vec{x}] \to [\vec{y}]$ has same-type $\vec{x}, \vec{y}$ with a common context in $K \subseteq L$, so $\vec{x} \equiv_{L,s} \vec{y}$ by substitutability; combine with the hypothesis for $[\vec{y}]$. For Comp $[\vec{z}] \to \rho([\vec{u}_1], \ldots, [\vec{u}_r])$ with $\vec{z} = \rho(\vec{u}_1, \ldots, \vec{u}_r)$, the children derive tuples equivalent to the $\vec{u}_j$, and Lemma 3.2 transfers this to $\vec{z}$. For $\widehat{S} \to [(w)]$ with $w \in K \subseteq L$, a derived (v) satisfies $(v) \equiv_{L,s} (w)$, and $\square_1$ accepts (w), so $v \in L$; $\widehat{S} \to \varepsilon$ occurs only if $\varepsilon \in K \subseteq L$. The uncapped grammar is the case of cap $\|K\|_+$. □

Proof of Proposition 3.11. This is the argument of [15, Lemma 3] for the concrete grammar. Each $\iota(X)$ and each $\vec{z}_R := \rho(\iota(Y_1), \ldots, \iota(Y_q))$ for a typed rule $R : X \to \rho(Y_1, \ldots, Y_q)$ is observed, via $\chi(X)$. They have the same type (out_ρ of the children's types is the type of X) and the common context $\chi(X)$, so Link gives $[\iota(X)] \to [\vec{z}_R]$. If $q = 0$, Const gives $[\vec{z}_R] \to \vec{z}_R$; otherwise $(\vec{z}_R, \rho, (\iota(Y_j))_j)$ is a composition witness of rank $q \leq r_G$ (and $q \leq \|K\|_+$ by (3.1)), so Comp gives $[\vec{z}_R] \to \rho([\iota(Y_1)], \ldots, [\iota(Y_q)])$. By induction, $X \Rightarrow^*_{\widetilde{G}_0} \vec{u}$ implies $[\iota(X)] \Rightarrow^* \vec{u}$. For $w \in L(G) = L(\widetilde{G}_0)$ (Proposition 3.6(iii)) a successful derivation starts with a start link $\widetilde{S} \to X$; then $\chi(X) = \square_1$, so $\iota(X) = (w')$ with $w' \in K$, and Start gives $\widehat{S} \to [\iota(X)] \Rightarrow^* (w)$. □

Proof of Theorem 3.13. If $L \cap \Sigma^+ = \emptyset$, then $K = L \subseteq \{\varepsilon\}$ and Definition 3.4 gives the claim. Otherwise $\mathcal{F}_L \subseteq L$ is finite, Proposition 3.11 with $\iota = \omega$ (canonical tuples have the right type by Proposition 3.6(i)) gives $L \cap \Sigma^+ \subseteq L(\mathcal{B}(K))$, Proposition 3.10 gives $L(\mathcal{B}(K)) \subseteq L$, and $\varepsilon \in L(\mathcal{B}(K))$ iff $\varepsilon \in K$ iff $\varepsilon \in L$. Rank-preserving normalization gives the last sentence. □

Proof of Theorem 3.14 and Corollary 3.16. As in [14, Lemma 6]: an occurrence is a sample word with at most $2f$ cut positions, and a composition witness of rank $\leq r_{\max}$ adds at most $2fr_{\max}$ more, so there are $(1 + \|K\|_+)^{O(fr_{\max})}$ candidate rules, each checked in polynomial time. Identification is the conservative-learner argument of [14, Corollary 1 and Theorem 1]: once the characteristic sample of Theorem 3.13 has appeared, every new hypothesis is correct, and an earlier hypothesis that is never changed is sound by Proposition 3.10 and complete because every later datum is generated by it. Membership is decidable in polynomial time for fixed f and $r_{\max}$ [14, Proposition 4]. None of this bounds the stage of convergence. □

D Proofs for Typed Witnesses and Characteristic Data

Proof of Proposition 3.6. At every rule application the morphism property gives

$$\widehat{s}^{(\mathrm{fan}(A))}(\rho(\vec{u}_1, \ldots, \vec{u}_r)) = \mathrm{out}_\rho(\widehat{s}^{(d_1)}(\vec{u}_1), \ldots, \widehat{s}^{(d_r)}(\vec{u}_r)).$$

Induction on height gives (i) and the existence and uniqueness of the lift in (ii), since each typed label is forced by its yield. Erasing labels gives $L(G^{\widehat{s}}) \subseteq L(G)$, (ii) gives the converse, and trimming keeps every rule of every successful derivation. For (iv), insert the terminal derivation at an occurrence

of X in a successful derivation of $\widetilde{G}_0$: typed rules depend only on typed symbols, so the result is a successful derivation of $G^{\widehat{s}}$, and trimming retains all its rules. (A rule of $G^{\widehat{s}}$ whose child has no terminal derivation, such as $A_{(0)} \to \mathtt{a}B_{(0)}$ when B derives only words of another type, is not retained; it occurs in no terminal derivation.) □

Definitions 2.8, 3.4 and 3.5 apply verbatim with any finite-monoid morphism $q : \Sigma^* \to \mathsf{N}$ in place of $\widehat{s}$ (tuple types being $q^{(d)}$); write $\mathcal{C}^{\mathrm{mcf}}_{f,q}$ for the class. The class and the operator use types of nonempty components only, so they depend only on $q|_{\Sigma^+}$.

Lemma D.1 (Monoid implementations)**.** *Let $q : \Sigma^* \to \mathsf{N}$ be a finite-monoid morphism, $\mathsf{S}_0 := q(\Sigma^+)$ and $s_0 := q|_{\Sigma^+}$.*

(i) *On $(\Sigma^+)^d$, equality of $q^{(d)}$-, $s_0^{(d)}$- and $\widehat{s_0}^{(d)}$-values coincide. Hence $\mathcal{C}^{\mathrm{mcf}}_{f,q} = \mathcal{C}^{\mathrm{mcf}}_{f,s_0}$, and the operators built with q and with $\widehat{s_0}$ return the same grammar on every sample.*

(ii) *For every good MCFG G, the trimmed refinements of G under q and under $\widehat{s_0}$ coincide (the untrimmed ones need not); hence the same canonical tuples, contexts and witness words may be chosen.*

Proof. (i) The three maps agree on Σ^+. (ii) By ε-freeness and Proposition 3.6(i), every typed nonterminal of a successful derivation under either morphism is $A_{\mathbf{t}}$ with $\mathbf{t} \in \mathsf{S}_0^{\mathrm{fan}(A)}$, and every start link used is $\widetilde{S} \to S_{(m)}$ with $m \in \mathsf{S}_0$. For children indexed in S_0, each nonempty template component is evaluated in S_0 identically by both morphisms, so both refinements have the same rules among S_0-indexed symbols, which carry all successful derivations. □

Proof of Lemma 3.7. Take a successful typed derivation containing X and follow the path from the root to an occurrence of X, starting with $\Box_1$; at each rule apply Lemma 3.1 with the (nonempty) sibling yields fixed. This gives $\chi(X) \in \mathcal{E}_d$, and replacing the X-subtree by any X-derivation gives a successful derivation of $\chi(X)[\vec{u}]$. If $\widetilde{S} \to X$ is a rule, take the path of length zero. □

Proof of Lemma 3.8. For $\vec{u}, \vec{v} \in L_X$, Proposition 3.6(i) gives equal types, $\chi(X)$ accepts both, and the arity is at most f; substitutability gives $\mathcal{D}_L(\vec{u}) = \mathcal{D}_L(\vec{v})$. □

Proof of Lemma 4.1. For a rule R with left-hand side A fix a successful derivation D_R of length $\theta_G(R)$ that uses R; it exists because G is reduced. By nondeletion every terminal written by R, and every component of the tuple produced at that occurrence of A, survives exactly once in the sentence. This gives (i), and (ii) because every productive A has a rule.

(iv) For $A = S$ take $\Box_1$. Otherwise delete the A-subtree of D_R and put ordered holes in place of its components; iterating Lemma 3.1 along the path to the root (siblings have nonempty yields) shows $E_A \in \mathcal{E}_d$, and its fixed letters belong to the sentence of D_R. Inserting any A-derivation gives a successful derivation, so E_A accepts $L_A(G) \supseteq L_X$ (Proposition 3.6(i)). A typed nonterminal with a start link has the form $S_{\mathbf{t}}$, and $E_S = \Box_1$. For derivationality, insert the erasure of any $\widetilde{G}_0$-derivation D_X from X into the hole; the unique lift of the resulting successful derivation (Proposition 3.6(ii)) labels every node by its nonterminal and the $\widehat{s}$-vector of its yield, as D_X does, so it contains D_X at the hole, and deleting that X-subtree leaves E_A.

(iii) Follow a shortest dependency path from S to A; it repeats no nonterminal, so its rules are distinct and differ from R. Complete every sibling on the path and every child of R by a shortest terminal tuple (length $\leq t_G$) and apply R. The distinct rules used write at most $\|G\|$ terminals and have at most $\|G\|$ child occurrences, so the sentence has length at most $\|G\| + \|G\| t_G \leq 2\|G\| \max\{1, t_G\}$. □

Proof of Lemma 4.2. (i) All components are nonempty because G is ε-free, so $s \preceq \lambda$ and Proposition 3.6(i) give $\widehat{s}^{(\mathrm{fan}(B))}(\vec{u}') = \widehat{s}^{(\mathrm{fan}(B))}(\vec{u}) = \mathbf{t}$. By Proposition 3.6(ii) a G-derivation of $\vec{u}'$ lifts to a $\widehat{G^{s}}$-derivation from X, so $\vec{u}' \in L_X$ by Proposition 3.6(iv). For the second claim apply $(\mathrm{R}^{\lambda}_{\beta})$ to some $\vec{u} \in L_X \neq \emptyset$.

(ii) A word of W_ι is $\chi(X)[\iota(X)]$ or a rule word. The context has at most $\bar{\theta}_G$ fixed letters, a typed rule has the template of its source and so writes at most $\bar{\theta}_G$ letters (Lemma 4.1(i)), and there are at most $r_{\max}$ child tuples. There is one word per typed nonterminal and per typed rule.

(iii) Apply the typed rule to $\widetilde{G}_0$-derivations of the $\omega(Y_j) \in L_{Y_j}$, erase the labels, and insert the result into the successful G-derivation from which E_B was obtained. As in the proof of Lemma 4.1(iv), the unique lift contains the typed derivation at the hole, and every rule of a successful derivation of $\widehat{G^{s}}$ is retained by the definition of $\widetilde{G}_0$. A start link $\widetilde{S} \to S_{\mathbf{t}}$ is used by the successful derivation of $\omega(S_{\mathbf{t}})$, of length at most b.

(iv) Since $\chi(X)$ accepts every tuple of L_X and $\iota(X)$ is equivalent to them, $\chi(X)[\iota(X)] \in L$. For a typed rule $X \to \rho(Y_1, \ldots, Y_q)$ and $\vec{u}_j \in L_{Y_j}$, Lemma 3.2 gives $\rho(\iota(Y_1), \ldots, \iota(Y_q)) \equiv_{L,s} \rho(\vec{u}_1, \ldots, \vec{u}_q) \in L_X$, so the rule word lies in L as well. Equivalent tuples have equal types, so $\widehat{s}^{(\mathrm{fan}(B))}(\iota(X)) = \mathbf{t}$ for $X = B_{\mathbf{t}}$. Hence Proposition 3.11 ($r_G \leq r_{\max}$, and $\chi(S_{\mathbf{t}}) = E_S = \Box_1$) and Proposition 3.10 give $L(\mathcal{B}(K)) \cap \Sigma^+ = L \cap \Sigma^+$ for every finite K with $\mathsf{W}_\iota \subseteq K \subseteq L$, for both operators; the Start rule handles ε. □

Proof of Theorem 4.3. Apply Lemma 4.2(ii),(iv) with $\iota = \omega$ and $b = t_{\mathrm{typ}}$, counting the non-start typed nonterminals and rules instead of using the upper bound $C_s(G)$. □

Proof of Proposition 4.7. Apply Lemma 4.6 to $L' = L^{\setminus m}$, which lies in $\mathcal{C}^{\mathrm{mcf}}_{f,s}$ by Lemma 4.5, and, if $\varepsilon \in L$, to $L' = L \setminus \{\varepsilon\}$; the canonical operators have characteristic samples for both by Theorem 3.13. Distinct positive types demand distinct sample words, and a word of type m costs at least $\mu_m(L) + 1$. □

Proof of Corollary 4.8. Proposition 4.7 supplies the lower bound; Theorem 4.3 supplies the upper bound. □

Proof of Proposition 4.9. All templates are nonempty and use each variable once in order, so G_n is good; it is reduced and generates Σ^+ via $S \to SS \mid \mathtt{a} \mid \mathtt{c}$. Each rule has a successful use of length at most 2: complete every A_i by $\mathtt{c}$ ($i \geq 1$) or $\mathtt{a}$ ($i = 0$). A yield of A_n has type 1 iff it contains no $\mathtt{c}$, so the unique yield of $(A_n)_{(1)}$ is $\mathtt{a}^{2^n}$, and $t_{\mathrm{typ}}(G_n, s_{\mathtt{c}}) \geq 2^n$; the typed rule $S_{(1)} \to (A_n)_{(1)}$ is retained (via $S \to A_n$), and every successful derivation using it has length at least 2^n. □

Proof of Corollary 4.10. $L_X \subseteq \mathrm{Cl}_{L,s}(X)$ gives $\tau \leq t_{\mathrm{typ}}$. Apply Lemma 4.2(ii),(iv) with $\iota(X)$ a shortest tuple of $\mathrm{Cl}_{L,s}(X)$ and $b = \tau$, counting as in Theorem 4.3; $K := \mathsf{W}_\iota \cup (\{\varepsilon\} \cap L)$. Finally, for $m \in \mathsf{S}_L$ the typed start nonterminal $X = S_{(m)}$ has $\chi(X) = \Box_1$, so $\Box_1 \in \mathcal{D}_L(\iota(X))$; thus $\iota(X) = (v)$ with $v \in L$ and $s(v) = m$, and $\tau_X \geq \mu_m(L)$. □

E Details of the Good Normalization

The four conversions below give Proposition 3.12; each satisfies a tuple-language identity from which the size bound is read off. Correctness of the nondeleting and nonmerging conversions is standard [14, 28, 30]; the size bounds are not stated there and are derived here. For $\vec{w} = (w_1, \ldots, w_d)$ and nonempty $I \subseteq [d]$, $\pi_I(\vec{w})$ keeps the components indexed by I in order, $\sigma\vec{w}$ permutes components and $\psi(\vec{w})$ concatenates consecutive blocks of components; so $\|\pi_I(\vec{w})\|_1 \leq \|\vec{w}\|_1 = \|\sigma\vec{w}\|_1 = \|\psi(\vec{w})\|_1$.

Lemma E.1 (Stagewise tuple languages)**.** *Let G_0 be a reduced f-MCFG of rank at most $r_{\max}$. Each conversion below preserves linearity, does not increase fan-out, rank or any template, and multiplies the numbers of nonterminals and rules by a factor depending only on f and $r_{\max}$.*

(a) Nondeletion *(construction (f3) in the proof of [28, Lemma 2.2]). Introduce A^I for nonempty $I \subseteq [\mathrm{fan}(A)]$. For $R : A \to \rho(B_1, \ldots, B_q)$ let J_j be the components of B_j used by the I-indexed outputs of ρ; keep these outputs, use $B_j^{J_j}$ if $J_j \neq \emptyset$ and omit B_j otherwise. The rules are nondeleting and $L_{A^I} = \pi_I(L_A(G_0))$.*

(b) Nonpermutation. *Decorate a fan-out-d nonterminal by $\sigma \in \mathfrak{S}_d$ (the symmetric group on $\{1, \ldots, d\}$), the child's permutation being determined by the order in which the parent template uses its variables. The decorated nonterminal generates σL_A.*

(c) Empty components. *Let $\mathsf{Pat}(A)$ be the set of support patterns of L_A, computed by saturation. For realizable child patterns, replace variables of empty components by ε, delete empty parent components, omit all-empty children, and introduce A^I for each nonempty $I \in \mathsf{Pat}(A)$. Then $L_{A^I} = \pi_I\{\vec{u} \in L_A : \mathrm{supp}(\vec{u}) = I\}$, the result is ε-free, and $S^{\{1\}}$ generates the nonempty part of the language.*

(d) Nonmerging *(construction in the proof of [14, Section 2.3, Lemma 1]). Decorate a nonterminal by a composition ψ of its fan-out, the child's composition being given by the maximal runs of its consecutive variables in the parent template. The decorated nonterminal generates the image $\psi(L_A)$.*

Stages (b)–(d) preserve the properties established by earlier stages.

The copies made at each stage, from the proof below, are as follows; templates are never lengthened, so the size grows by at most the product of the factors in the last column. Correctness of (a), (b) and (d) is taken from the cited constructions; the counts, and stage (c) with its identity, are ours.

Stage	Decoration of A	Copies of A	Copies of a rule
(a) nondeletion	nonempty $I \subseteq [\mathrm{fan}(A)]$	$\leq 2^f$	$\leq 2^f$
(b) nonpermutation	$\sigma \in \mathfrak{S}_{\mathrm{fan}(A)}$	$\leq f!$	$\leq f!$
(c) empty components	support pattern $I \in \mathsf{Pat}(A)$	$\leq 2^f$	$\leq 2^{f r_{\max}}$
(d) nonmerging	composition ψ of $\mathrm{fan}(A)$	$\leq 2^{f-1}$	$\leq 2^{f-1}$

In (a), (b) and (d) the decoration of the left-hand side determines those of the children; in (c) the children's patterns determine that of the left-hand side.

Proof. (a) For any child tuples, $\pi_I(\rho(\vec{v}_1, \ldots, \vec{v}_q)) = \rho^I((\pi_{J_j}\vec{v}_j)_{J_j \neq \emptyset})$; induction gives $\subseteq$, and for $\supseteq$ the omitted children are completed by terminal derivations, which exist because G_0 is reduced. There are at most 2^f copies of each nonterminal and rule. (b) Linearity and nondeletion make the child permutations well defined; induction gives σL_A. There are at most $f!$ copies. (c) Retained variables occur once and in order; induction gives the identity, and omitted all-empty children are completed by derivations of their empty pattern. A retained child B^J has only the components indexed by J, and each of them is used by the parent template exactly once and in the original order, so nondeletion and nonpermutation persist. There are at most 2^f patterns per nonterminal and $2^{f r_{\max}}$ choices per rule. (d) Nonpermutation makes the runs compatible with each child's order and nondeletion makes them cover its components; then $\rho^\psi(\psi_1(\vec{v}_1), \ldots, \psi_q(\vec{v}_q)) = \psi(\rho(\vec{v}_1, \ldots, \vec{v}_q))$, and induction gives the image identity (merging need not be injective, so no inverse-image claim is

made). Concatenating nonempty components yields nonempty components, and each merged block of a child is used once and in order, so ε-freeness, nondeletion and nonpermutation persist. The size is linear for fixed $f, r_{\max}$ [14, Section 2.3, after Example 2]. □

Proof of Proposition 3.12. If $L(G_{\text{in}}) \cap \Sigma^+ = \emptyset$, take G_{good} to consist of the start symbol alone, with no rules; it is good and generates $\emptyset = L(G_{\text{in}}) \setminus \{\varepsilon\}$. Otherwise apply (a)–(d) to the trimmed input and trim again. Each stage multiplies the size by a factor depending only on $f, r_{\max}$, so $\|G_{\text{good}}\| = O_{f,r_{\max}}(\|G_{\text{in}}\|)$, and $L(G_{\text{good}}) = L(G_{\text{in}}) \setminus \{\varepsilon\}$ by (c). If this language is nonempty, the final trimming leaves a reduced grammar, and trimming preserves goodness. □

F The Three-Letter Doubling Family

This appendix proves Proposition 7.1; the proof of (ii) adapts the deterministic-parser argument of [23, Appendix C]. Fix $n \geq 1$, and for $x, y \in \Sigma^*$ write $x \equiv y$ if $\{(u, v) : uxv \in L_n\} = \{(u, v) : uyv \in L_n\}$; this is a monoid congruence.

The parser. A *stack* is a word over $E_0, \ldots, E_n$ and the closing markers $\hat{\mathtt{b}}_i, \hat{\mathtt{c}}_i$ $(1 \leq i \leq n)$, with its top on the left; ε is the empty stack. For every stack $\mathcal{Z}$ and $1 \leq i \leq n$ the moves are

$$E_0\mathcal{Z} \xrightarrow{\mathtt{a}} \mathcal{Z}, \quad E_i\mathcal{Z} \xrightarrow{\mathtt{b}} E_{i-1}E_{i-1}\hat{\mathtt{b}}_i\mathcal{Z}, \quad E_i\mathcal{Z} \xrightarrow{\mathtt{c}} E_{i-1}\hat{\mathtt{c}}_i\mathcal{Z},$$
$$\hat{\mathtt{b}}_i\mathcal{Z} \xrightarrow{\mathtt{b}} \mathcal{Z}, \quad \hat{\mathtt{c}}_i\mathcal{Z} \xrightarrow{\mathtt{c}} \mathcal{Z},$$

and there are no others; in particular no move leaves ε. Write $\mathcal{X} \xrightarrow{w} \mathcal{Y}$ if reading w from $\mathcal{X}$ ends in $\mathcal{Y}$. Three properties are used.

(D1) *Determinism.* The top symbol and the letter determine the move, so every word acts as a partial map on stacks.

(D2) *Weight.* Give E_i weight 4^i and each marker weight 1, and let $\mathrm{wt}(\mathcal{X})$ be the total weight. Every move lowers wt, since $4^i > 2 \cdot 4^{i-1} + 1$; hence $\mathcal{X} \xrightarrow{w} \mathcal{X}$ only for $w = \varepsilon$.

(D3) *Locality.* Moves read and rewrite only the top symbol, and markers are only popped. Hence $\mathcal{X} \xrightarrow{w} \mathcal{Y}$ implies $\mathcal{X}\mathcal{Z} \xrightarrow{w} \mathcal{Y}\mathcal{Z}$. Conversely, if $\mathcal{X}_1 \neq \varepsilon$ and $\mathcal{X}_1\mathcal{Z} \xrightarrow{w} \mathcal{Y}$, then either (a) the run reaches $\mathcal{Z}$: $w = w_1w_2$ with $\mathcal{X}_1 \xrightarrow{w_1} \varepsilon$ and $\mathcal{Z} \xrightarrow{w_2} \mathcal{Y}$, so $\mathrm{wt}(\mathcal{Y}) \leq \mathrm{wt}(\mathcal{Z})$; or (b) it does not: $\mathcal{Y} = \mathcal{Y}_1\mathcal{Z}$ with $\mathcal{X}_1 \xrightarrow{w} \mathcal{Y}_1 \neq \varepsilon$, so $\mathrm{wt}(\mathcal{Y}) > \mathrm{wt}(\mathcal{Z})$, and $\mathcal{Y}_1$ ends with the last symbol of $\mathcal{X}_1$ whenever that symbol is a marker. For fixed $\mathcal{Z}$, comparing $\mathrm{wt}(\mathcal{Y})$ with $\mathrm{wt}(\mathcal{Z})$ decides the case.

A letter $\mathtt{b}$ or $\mathtt{c}$ both opens and closes, so a word need not act injectively on all stacks; injectivity holds on the *reachable* stacks, those $\mathcal{X}$ with $E_n \xrightarrow{u} \mathcal{X}$ for some u.

Lemma F.1. (i) *For $0 \leq i \leq n$ and every stack $\mathcal{Z}$, $E_i\mathcal{Z} \xrightarrow{w} \mathcal{Z}$ if and only if $w \in L^{(i)}$.*

(ii) *Put $\mathcal{F}_i := \{E_{i-1}\hat{\mathtt{b}}_i, \hat{\mathtt{b}}_i, \hat{\mathtt{c}}_i\}$. Every reachable stack is ε or $\mathcal{Y}F_{j+1}\cdots F_n$ with $F_i \in \mathcal{F}_i$ and $\mathcal{Y} \in \{E_j, \hat{\mathtt{b}}_j, \hat{\mathtt{c}}_j\}$ ($\mathcal{Y} = E_j$ with $0 \leq j \leq n$, or a marker with $1 \leq j \leq n$). Every word acts as a partial injection on reachable stacks.*

Proof. (i) A run $E_i\mathcal{Z} \xrightarrow{w} \mathcal{Z}$ is in case (a) of (D3), since case (b) ends in $\mathcal{Y}_1\mathcal{Z} \neq \mathcal{Z}$; so $w = w_1w_2$ with $E_i \xrightarrow{w_1} \varepsilon$ and $\mathcal{Z} \xrightarrow{w_2} \mathcal{Z}$, and $w_2 = \varepsilon$ by (D2). With the lifting in (D3), it suffices to show $E_i \xrightarrow{w} \varepsilon$ iff $w \in L^{(i)}$, by induction on i. From E_0 only the $\mathtt{a}$-move applies, and ε admits none. For $i \geq 1$ the first letter is $\mathtt{b}$ or $\mathtt{c}$. After $\mathtt{c}$, the stack $E_{i-1}\hat{\mathtt{c}}_i$ must be emptied; case (a) of (D3) with $\mathcal{X}_1 = E_{i-1}$,

$\mathcal{Z} = \hat{\mathtt{c}}_i$ splits the rest as zz' with $E_{i-1} \xrightarrow{z} \varepsilon$ and $z' = \mathtt{c}$. After $\mathtt{b}$, two such splittings of $E_{i-1}E_{i-1}\hat{\mathtt{b}}_i$ give $z_1 z_2 \mathtt{b}$ with $E_{i-1} \xrightarrow{z_j} \varepsilon$.

(ii) The set of such stacks contains E_n and is closed under the moves: from $E_j F_{j+1} \cdots F_n$, $\mathtt{b}$ gives $E_{j-1}(E_{j-1}\hat{\mathtt{b}}_j)F_{j+1} \cdots F_n$ and $\mathtt{c}$ gives $E_{j-1}\hat{\mathtt{c}}_j F_{j+1} \cdots F_n$; popping the top leaves $F_{j+1} \cdots F_n$, which is ε or again of the stated form, a first frame $E_j\hat{\mathtt{b}}_{j+1}$ being read as $\mathcal{Y} = E_j$ followed by the frame $\hat{\mathtt{b}}_{j+1}$.

For injectivity it suffices to consider letters, since moves map reachable stacks to reachable stacks. A *frame product* is a stack $F_{i+1} \cdots F_n$ with $F_k \in \mathcal{F}_k$ (the empty stack if $i = n$). Each frame F_k contains exactly one marker, of index k, so a frame product determines i: it is n for ε and otherwise one less than the least marker index. In a frame product a symbol E is always followed by a marker $\hat{\mathtt{b}}$. The moves of a letter on reachable stacks, and the shape of their results, are:

Letter	Move (reachable source)	Result begins with	Source recovered from
$\mathtt{a}$	$E_0\mathcal{Z} \to \mathcal{Z}$	a frame product	$\mathcal{Z}$
$\mathtt{b}$, opening	$E_i\mathcal{Z} \to E_{i-1}E_{i-1}\hat{\mathtt{b}}_i\mathcal{Z}$	two symbols E	i and $\mathcal{Z}$ read off
$\mathtt{b}$, closing	$\hat{\mathtt{b}}_i\mathcal{Z} \to \mathcal{Z}$, $\mathcal{Z}$ a frame product	a marker, $E\hat{\mathtt{b}}$, or ε	i from $\mathcal{Z}$
$\mathtt{c}$, opening	$E_i\mathcal{Z} \to E_{i-1}\hat{\mathtt{c}}_i\mathcal{Z}$	E followed by $\hat{\mathtt{c}}$	i and $\mathcal{Z}$ read off
$\mathtt{c}$, closing	$\hat{\mathtt{c}}_i\mathcal{Z} \to \mathcal{Z}$, $\mathcal{Z}$ a frame product	a marker, $E\hat{\mathtt{b}}$, or ε	i from $\mathcal{Z}$

In the closing rows $\mathcal{Z}$ is a frame product because a reachable stack with a marker $\hat{\mathtt{b}}_i$ or $\hat{\mathtt{c}}_i$ on top is $\mathcal{Y}F_{i+1} \cdots F_n$ with $\mathcal{Y}$ that marker. For $\mathtt{b}$, an opening result begins with two symbols E and a closing result never does, since in a frame product a symbol E is followed by a marker; for $\mathtt{c}$, an opening result begins with $E\hat{\mathtt{c}}$, which never occurs in a frame product. So the openings and the closings of one letter have disjoint images, each row is injective by its last column, and every letter, hence every word, acts as a partial injection on reachable stacks. □

Lemma F.2. *If $\mathcal{X}$ is reachable, $w \in L^{(i)}$ and $\mathcal{X} \xrightarrow{w} \mathcal{Y}$, then $\mathcal{X} = E_i\mathcal{Y}$. Consequently $w \equiv w'$ for all $w, w' \in L^{(i)}$.*

Proof. Induct on i; for $i = 0$, $\mathtt{a}$ only pops E_0. Let $i \geq 1$; all stacks met are reachable. If $w = \mathtt{c}v\mathtt{c}$ with $v \in L^{(i-1)}$ and the first $\mathtt{c}$ opens E_j, the induction hypothesis for v at $E_{j-1}\hat{\mathtt{c}}_j\mathcal{Z}$ gives $j = i$, and the last $\mathtt{c}$ pops $\hat{\mathtt{c}}_i$. If the first $\mathtt{c}$ pops a marker, the rest is ε or a product of frames; to read v it must begin with E_{i-1} by the induction hypothesis, so its first frame is $E_{i-1}\hat{\mathtt{b}}_i$, and after v the top is $\hat{\mathtt{b}}_i$, on which $\mathtt{c}$ has no move. If $w = \mathtt{b}v_1v_2\mathtt{b}$ and the first $\mathtt{b}$ opens E_j, the induction hypothesis gives $j = i$, the words v_1, v_2 pop the two copies of E_{i-1}, and the last $\mathtt{b}$ pops $\hat{\mathtt{b}}_i$. If the first $\mathtt{b}$ pops a marker, then as before v_1 leaves the top $\hat{\mathtt{b}}_i$, from which v_2 cannot be read by the induction hypothesis. For the consequence, if $uwv \in L_n$, the run $E_n \xrightarrow{u} \mathcal{X} \xrightarrow{w} \mathcal{Y} \xrightarrow{v} \varepsilon$ has $\mathcal{X} = E_i\mathcal{Y}$, and $E_i\mathcal{Y} \xrightarrow{w'} \mathcal{Y}$ by Lemma F.1(i); so $uw'v \in L_n$, and conversely. □

Lemma F.3 (Branch synchronization)**.** *Let $\mathcal{X} \xrightarrow{x} \mathcal{Y}$ and $\mathcal{X} \xrightarrow{y} \mathcal{Y}$, where x, y are nonempty and begin with different letters. Then $\mathcal{X} = E_i\mathcal{Z}$ for some $1 \leq i \leq n$ and stack $\mathcal{Z}$, and $x = wx'$, $y = w'y'$ with $w, w' \in L^{(i)}$, $\mathcal{Z} \xrightarrow{x'} \mathcal{Y}$ and $\mathcal{Z} \xrightarrow{y'} \mathcal{Y}$.*

Proof. Only a top E_i with $i \geq 1$ admits two different letters, namely $\mathtt{b}$ and $\mathtt{c}$. Say $x = \mathtt{c}x_0$ and $y = \mathtt{b}y_0$, so $E_{i-1}\hat{\mathtt{c}}_i\mathcal{Z} \xrightarrow{x_0} \mathcal{Y}$ and $E_{i-1}E_{i-1}\hat{\mathtt{b}}_i\mathcal{Z} \xrightarrow{y_0} \mathcal{Y}$. By (D3) both runs are in the same case. Case (b) is impossible: it would give $\mathcal{Y} = \mathcal{Y}_1\mathcal{Z} = \mathcal{Y}_2\mathcal{Z}$ with $\mathcal{Y}_1$ ending in $\hat{\mathtt{c}}_i$ and $\mathcal{Y}_2$ in $\hat{\mathtt{b}}_i$. In case (a), $x_0 = x_1x'$ with $E_{i-1}\hat{\mathtt{c}}_i \xrightarrow{x_1} \varepsilon$ and $\mathcal{Z} \xrightarrow{x'} \mathcal{Y}$, so $w := \mathtt{c}x_1$ satisfies $E_i \xrightarrow{w} \varepsilon$ and $w \in L^{(i)}$ by Lemma F.1(i). The same argument gives $y = w'y'$. □

Proposition F.4. *If $\mathcal{X} \xrightarrow{x} \mathcal{Y}$ and $\mathcal{X} \xrightarrow{y} \mathcal{Y}$, then $x \equiv y$. Consequently L_n is Clark–Eyraud substitutable.*

Proof. Induct on $|x| + |y|$. If x or y is empty, then $\mathcal{X} = \mathcal{Y}$, and the other is empty by (D2). If $x = cx'$ and $y = cy'$ for a letter c, (D1) gives $\mathcal{X} \xrightarrow{c} \mathcal{X}'$ with $\mathcal{X}' \xrightarrow{x'} \mathcal{Y}$ and $\mathcal{X}' \xrightarrow{y'} \mathcal{Y}$, and induction applies. Otherwise Lemma F.3 gives $x = wx'$, $y = w'y'$; induction gives $x' \equiv y'$, Lemma F.2 gives $w \equiv w'$, and $\equiv$ is a congruence.

For the consequence, let $x, y \in \Sigma^+$ with $uxv, uyv \in L_n$. By (D1) and Lemma F.1(i) there are runs $E_n \xrightarrow{u} \mathcal{X} \xrightarrow{x} \mathcal{Y} \xrightarrow{v} \varepsilon$ and $E_n \xrightarrow{u} \mathcal{X} \xrightarrow{y} \mathcal{Y}' \xrightarrow{v} \varepsilon$; $\mathcal{Y}$ and $\mathcal{Y}'$ are reachable, so Lemma F.1(ii) gives $\mathcal{Y} = \mathcal{Y}'$, and $x \equiv y$. Letters outside $\{\mathtt{a}, \mathtt{b}, \mathtt{c}\}$ only have empty distributions. □

Proof of Proposition 7.1. (i) Every template is nonempty and uses each variable once; at fan-out one, nonpermutation and nonmerging are automatic. The grammar is reduced since $E_i \Rightarrow^* \mathtt{c}^i\mathtt{a}\mathtt{c}^i$. It has $n+2$ nonterminals and rules of sizes 3 (start), 2 (base), 7 and 5 (the two rules of each level) (Definition 2.5), whence $\|G_n\| = 13n+7$. A shortest yield of E_i has length $2i+1$, and the levels above E_i add at least $2(n-i)$ letters, attained by the rules $E_j \to \mathtt{c}E_{j-1}\mathtt{c}$. So the start, base and unary rules have rule thickness $2n+1$, and the binary rule of level i has $2(n-i)+2+2(2i-1) = 2n+2i$; hence $\bar{\theta}_{G_n} = 4n$.

(ii) The first claim is Proposition F.4. It is the condition of Definition 2.8 at $f = 1$ for the trivial typing, and the finite language L_n is a 1-MCFL; item (a) after Definition 2.8 gives $L_n \in \mathcal{C}^{\mathrm{mcf}}_{1,s}$.

(iii) A derivation yields a word without $\mathtt{c}$ iff it uses no rule $E_i \to \mathtt{c}E_{i-1}\mathtt{c}$. Such a derivation from E_i is the complete binary tree of height i and yields W_i, and $|W_i| = 2|W_{i-1}| + 2$ with $|W_0| = 1$.

(iv) By (ii), $L_n \in \mathcal{C}^{\mathrm{mcf}}_{1,s_U}$, and by (iii), $1 \in \mathsf{S}_{L_n}$ with $\mu_1(L_n) = |W_n|$. Proposition 4.7 with $m = 1$ forces a word of type 1, which can only be W_n. □

G The Linear Factor at Every Fan-Out

This appendix proves Proposition 6.1. Ψ denotes the Parikh map.

Proof of Proposition 6.1. Put $t := 2f$ and $L := L^{(f)}_{q,n}$.

(i) The language $R := \mathtt{a}_1^*\mathtt{d}_1\mathtt{a}_2^*\mathtt{d}_2 \cdots \mathtt{a}_t^*\mathtt{d}_t\mathtt{z}^*$ is the path language from 0 to t of the graph on $\{0, \dots, t\}$ with edges $\mathtt{a}_h : h-1 \to h-1$, $\mathtt{d}_h : h-1 \to h$ and $\mathtt{z} : t \to t$, and a word of R is determined by its Parikh vector. In w_i the marker $\mathtt{d}_h$ occurs once, at position $h(i+1)$, so the position of one marker, and the distance $(h'-h)(i+1)$ of two, determine i; all words have length W.

Let $J \subseteq L$ and $1 \le d \le f$, let $\vec{x}, \vec{y}$ be nonempty d-tuples and $E = (v_0, \dots, v_d)$ a d-context with $E[\vec{x}] = w_i \in J$ and $E[\vec{y}] = w_j \in J$, and let F be a d-context with $F[\vec{x}] \in J$. We show $F[\vec{y}] \in J$; the converse is symmetric. By Lemma 3.17, $F[\vec{y}] \in R$. If $i = j$, then $\sum_h \Psi(x_h) = \sum_h \Psi(y_h)$, so $F[\vec{y}]$ has the Parikh vector of $F[\vec{x}]$ and equals it. Let $i \neq j$. A marker in v_0 would have the same position in w_i and w_j, a marker in v_d the same distance from the end, and two markers in one v_h the same distance; each forces $i = j$. So the fixed part of E carries at most $d-1$ markers, and $\vec{x}$ carries at least $t - d + 1 \ge d + 1$ of them. Some component x_h carries two, and every word of L with factor x_h is w_i. Hence $F[\vec{x}] = w_i = E[\vec{x}]$, the fixed parts of F and E have equal Parikh vectors, $\Psi(F[\vec{y}]) = \Psi(E[\vec{y}])$, and $F[\vec{y}] = w_j \in J$. Finite languages are context-free, and the inclusion is item (a) after Definition 2.8.

(ii) For each i, $L \setminus \{w_i\} \in \mathcal{C}^{\mathrm{mcf}}_{f,s}$ by (i), so every characteristic sample contains w_i by Lemma 4.6. Hence $F = L$. Conversely, $K = L$ is the only finite K with $L \subseteq K \subseteq L$, and the canonical operators return L on it by sample consistency and soundness.

(iii) Take nonterminals $P_0, \dots, P_q$ of fan-out f and $S_0, \dots, S_q$, $C_0, \dots, C_n$ of fan-out one, start symbol S_q, and the rules

$$P_0 \to (\mathtt{d}_1, \mathtt{d}_3, \dots, \mathtt{d}_{2f-1}), \qquad C_0 \to (\mathtt{z}), \qquad C_h \to (\xi_1^1\xi_2^1)(C_{h-1}, C_{h-1})\ (1 \le h \le n),$$
$$P_i \to (\mathtt{a}_1\xi_1^1\mathtt{a}_2,\ \mathtt{a}_3\xi_1^2\mathtt{a}_4,\ \dots,\ \mathtt{a}_{2f-1}\xi_1^f\mathtt{a}_{2f})(P_{i-1})\ (1 \le i \le q),$$
$$S_j \to (\xi_1^1\mathtt{d}_2\xi_1^2\mathtt{d}_4 \cdots \xi_1^f\mathtt{d}_{2f}\xi_2^1)(P_j, C_n)\ (0 \le j \le q), \qquad S_j \to (\xi_1^1\mathtt{z}^{2f})(S_{j-1})\ (1 \le j \le q).$$

By induction $L(C_h) = \{\mathtt{z}^{2^h}\}$,

$$L(P_i) = \{(\mathtt{a}_1^i\mathtt{d}_1\mathtt{a}_2^i, \dots, \mathtt{a}_{2f-1}^i\mathtt{d}_{2f-1}\mathtt{a}_{2f}^i)\}, \qquad L(S_j) = \{\mathtt{a}_1^i\mathtt{d}_1 \cdots \mathtt{a}_{2f}^i\mathtt{d}_{2f}\mathtt{z}^{2f(j-i)+2^n} : 0 \le i \le j\},$$

so $L(S_q) = L$. Every variable occurs once and in order, consecutive components of P_j are separated by $\mathtt{d}_2, \mathtt{d}_4, \dots$, and no component is empty; so the grammar is good, and it is reduced. It has $2q+n+3$ nonterminals and $1, q, q+1, q, 1, n$ rules of sizes $f+1$, $3f+2$, $2f+4$, $2f+3$, 2, 5 for P_0, P_i, the first and second rules of S_j, C_0 and C_h (Definition 2.5); this gives the stated size, and $n \le q$ gives $(7f+17)(q+1)$. In a reduced presentation every rule occurs in a successful derivation and all words have length W, so $\bar\theta_G = W$. Finally let G be any reduced good fan-out-$\le f$ presentation of L. For the trivial typing, Theorem 3.13 makes $\mathsf{W}(\widetilde{G}_0)$ characteristic for $\mathcal{B}_{f,s_{\mathrm{triv}}}$, and it has at most $|V| + |P| \le \|G\|$ words (Definition 3.9 and the remark after Theorem 4.3); by (ii) for s_{triv}, it has $q+1$ words. The final estimate follows from (ii), (iii) and $|w_i| = W$.

(iv) The identification changes no length and no template shape, so the grammar stays reduced and good with the same size and thickness, and the arguments for (ii) and (iii) apply unchanged once (i) is shown. For (i), let $J \subseteq L$, and let $u\,x\,v = w_i \in J$ and $u\,y\,v = w_j \in J$ with $x, y \in \Sigma^+$; if $i = j$ then $x = y$. Let $i \ne j$. All words have length W. If u contained $\mathtt{d}$, the first $\mathtt{d}$ would have position $i+1$ in w_i and $j+1$ in w_j; if v contained $\mathtt{d}$, the last $\mathtt{d}$ would be followed by $2(q-i)+2^n$ and by $2(q-j)+2^n$ letters. So x contains both letters $\mathtt{d}$ of w_i and has the factor $\mathtt{da}^i\mathtt{d}$, which occurs in no other word of L and only once in w_i. Hence $p\,x\,q' \in J$ forces $(p, q') = (u, v)$, and then $p\,y\,q' = w_j \in J$; the converse is symmetric. □

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During preparation of this work, the author used ChatGPT (OpenAI) and Claude (Anthropic) to assist with organization of the manuscript, expression and translation of text, cross-checks of references and notation, and editing of LaTeX source. After using these tools, the author reviewed and edited the material as needed and takes full responsibility for the content of the publication.

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